

Enabling Organizational Strategy through Information Systems

Module 1 asked what an information system is. This one asks the harder question a manager actually gets paid to answer: **which** system, **where**, and **why that one**. A company can spend heavily on technology and end up merely faster at what it already did. The difference between spending and strategy is a way of thinking — about where the pressure in your industry comes from, where in your own operation value is created, and what a competitor would have to do to copy you.

1

OBJECTIVE 2.1

Discuss how information systems are used for automation, organizational learning, and strategic advantage

2

OBJECTIVE 2.2

Describe how information systems support the business models companies use in the digital world

3

OBJECTIVE 2.3

Explain why and how companies keep looking for innovative ways to use information systems for competitive advantage

4

APPLICATION 2.4

Decide where artificial intelligence belongs in strategy work, and where buying it is not the same as gaining an advantage

5

APPLICATION 2.5

Produce the evidence yourself with a spreadsheet formula and a database query that actually run

6

APPLICATION 2.6

Write the analysis steps as functions and answer harder questions across several database tables

How to use this page

1. **Read the short setup first.** Every term is explained in ordinary language before an activity uses it, so nothing assumes a background you have not been given.

2. **Answer as you go.** The activities sit inside the reading. When you answer a question, **every option explains itself** — including the three you did not pick, because knowing why a wrong answer is wrong is the part that transfers.
3. **Write real formulas and real queries.** The last section is not a description of a spreadsheet and a database. You type into them and they run, on the page, against real rows — and they tell you plainly when something will not work.
4. **Retry anything.** Restart one activity, one section, or the whole page. Your best result stays in this browser until you clear it; answers reset on reload so every attempt is fresh.
5. **Finish with the final challenge.** Thirty situations, scored by objective, so a weak area shows up instead of being averaged away.

LEVEL

Who decides what

Operational, managerial and executive work need different systems because they face different kinds of decision.

AMBITION

Faster, better, smarter

Three different reasons to build something, with three different payoffs — and only one of them is strategy.

POSITION

Where and how to compete

The five forces say what is squeezing you; the value chain says where you could answer it.

EVIDENCE

Getting the numbers

A claim without evidence is a preference. Two tools close most of the gap between question and answer.

ONE IDEA CONNECTS THE WHOLE MODULE

Technology that everyone can buy cannot, by itself, make you different from everyone who bought it. Advantage comes from what you put around the technology — the processes, the data and the people — which is why this module spends its time on analysis rather than on products.

SECTION 2-1A

How an organization is arranged for deciding

Before anyone can say which information system a company needs, they have to say who is asking. A cashier, a store manager, and a chief executive all work for the same

firm and all want information — but different information, on different clocks, for different reasons. This section maps the organization first, so every system has somewhere to sit.

Business processes: what a company actually does all day

Companies do not exist to own technology; they exist to do things, and the things they do have a name.

Business processes are the activities an organization performs to reach its business goals, and they come in two kinds.

Core activities — the activities that transform inputs and produce outputs, because that transformation is the thing a customer eventually pays for: flour and labor become a loaf, and an order becomes a delivered package.

Supporting activities — the activities that enable the core activities to take place, because nobody bakes the loaf if payroll never runs, the oven is never repaired, and no baker was ever hired.

A broad range of information is needed to support those processes, which is why structure comes first: it shows which type of system supports which process, and how much value each one delivers.

Three levels, three different questions

Every organization is composed of different decision-making levels. The usual drawing is a pyramid: a wide **operational level** at the bottom, a narrower **managerial level** in the middle, and a narrow **executive level** at the top.

The shape matters less than what follows from it. Each level has different responsibilities and, therefore, different informational needs — which is why each level is commonly supported by a different type of information system.

The operational level: running the day

At the operational level of a firm, the routine, day-to-day business processes and the interactions with customers occur. This is the checkout lane, the loading dock, the order screen. The chapter's figure states three facts about it.

Who — foremen and supervisors, the managers closest to the work, who make day-to-day decisions that are highly structured and recurring.

What the systems do — they automate routine and repetitive activities and events, such as processing sales transactions, and improve the efficiency of business processes at the customer interface.

Why — to improve organizational efficiency, meaning the extent to which goals are accomplished faster, at lower cost, or with relatively little time and effort.

The unit of work here is the transaction. A **transaction** is anything that occurs as part of a firm's daily business of which it must keep a record: a sale rung up, an item received, a shift clocked. The clock is short too, since operational planning typically has a time frame of a few hours or days.

Because activities here are clearly delineated and well focused, optimizing them can offer quick returns on the investment, and the same systems help managers find the underlying causes of performance problems.

The managerial level: running a function

At the managerial level — also called the **tactical level** — functional managers such as marketing, finance, manufacturing, and human resource managers monitor and control operational-level activities and pass information up to higher levels. The chapter calls them **midlevel managers**.

Who — midlevel managers and functional managers, who typically focus on problems inside one business function such as marketing or finance.

What the systems do — they automate the monitoring and controlling of operational activities, so a manager can see what the level below is producing.

Why — to improve organizational effectiveness, meaning the extent to which goals or tasks are accomplished well, in service of the organization's strategic objectives.

A managerial decision is usually contained within the business function, is moderately complex, and has a time horizon of a few days to a few months — a horizon with its own name, **tactical planning**. The chapter's example is a marketing manager at Nike allocating the advertising budget for the next business quarter.

Managers here watch a few numbers rather than every transaction. The metrics deemed most critical to assessing progress toward a certain goal are called **key performance indicators**, or KPIs, and they are displayed on **digital dashboards**.

The chapter's illustration is a production manager at Nike whose business intelligence system supplies analytics and forecasts on multiple product lines, inventory levels, and overall production capacity. The manager builds several hypothetical production schedules, then runs predictive analyses on each to compare inventory levels and potential sales profitability.

The executive level: running the company

At the executive level — also called the **strategic level** — managers face the long-term questions: which products to produce, which countries to compete in, and what strategy to follow. The people

here are the president and chief executive officer, the chief information officer, vice presidents, and possibly the board.

Who — executive-level managers, who must weigh the ramifications of any decision for the overall organization rather than for one department.

What the systems do — they supply aggregate summaries of past organizational data and projections of the future, rather than individual records.

Why — to improve organizational strategy and planning, which is a longer and broader job than either efficiency or effectiveness.

Executive-level decisions deal with complex problems whose consequences are broad and long-term. Deciding to develop a new product, or discontinue an existing one, can have vast effects on the organization’s levels of employment and profitability. Three things follow for the systems.

Balancing KPIs — some indicators here focus on balancing performance across the organization, staggering product launches so spikes in demand do not hit the supply chain all at once.

Benchmarking KPIs — others benchmark the organization’s performance against its competitors, because a rising number means little until you know what rivals did.

Unstructured data — the systems must take in global economic factors, demographic changes, and changing customer tastes and preferences, none of which arrive as tidy rows of transactions.

Executives also engage in **environmental scanning**, assessing how new information systems could help the organization offer new products or services and support its strategy.

Reading the three levels side by side

Set out together, the differences stop looking like job titles and start reading like a design brief.

LEVEL	WHO WORKS THERE	WHAT THE SYSTEMS DO	WHY	DECISION
Operational	Foremen and supervisors	Automate routine and repetitive activities and events	Improve organizational efficiency	Structured

LEVEL	WHO WORKS THERE	WHAT THE SYSTEMS DO	WHY	DECISION
Managerial tactical	Midlevel and functional managers	Automate the monitoring and controlling of operational activities	Improve organizational effectiveness	Semistructured
Executive strategic	Executives: the CEO, CIO, vice presidents, possibly the board	Provide aggregate summaries of past data and projections of the future	Improve organizational strategy and planning	Unstructured

Efficiency is not effectiveness. Efficiency is the extent to which goals are accomplished faster, at lower cost, or with relatively little time and effort, and it belongs to the operational level. **Effectiveness** is the extent to which goals or tasks are accomplished *well*, and it belongs to the managerial level. A warehouse shipping the wrong item in record time is efficient, not effective.

The diagram redraws each level on its own, so you can recall who, what, why, and the decision type without the table in front of you.

INTERACTIVE DIAGRAM

The three decision-making levels, one at a time

Pick a level to see who works there, what its information systems do, why the organization wants them, and what kind of decision gets made.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Operational level — The widest part of the pyramid, where the day happens one recorded event at a time

Who (Foremen and supervisors) → What (Automate routine and repetitive activities and events) → Why (To improve organizational efficiency) → Decision type (Structured and recurring)

- This is where the routine, day-to-day business processes and the interactions with customers occur, so the systems sit at the customer interface: registers, order screens, receiving scanners.
- The unit of work is the **transaction** — anything that occurs as part of a firm's daily business of which it must keep a record.
- Decisions here are **structured**: the procedures to follow can be specified in advance, so they can be programmed directly into the system and made with little or no human intervention.

- Operational planning typically has a time frame of a few hours or days, and because the activities are clearly delineated and well focused, optimizing them can offer quick returns on the investment.

Managerial level, also called tactical — The middle of the pyramid, where one business function is monitored and steered

Who (Midlevel managers and functional managers) → What (Automate the monitoring and controlling of operational activities) → Why (To improve organizational effectiveness) → Decision type (Semistructured)

- Functional managers — marketing, finance, manufacturing, human resource managers — monitor and control the level below them and provide information to the levels above them.
- Their aim is **effectiveness**, the extent to which goals or tasks are accomplished well, achieved by effectively utilizing and deploying organizational resources toward the strategic objectives.
- Decisions here are **semistructured**: some procedures can be specified in advance, but not to the point of producing a specific recommendation, so judgment and expertise are still required.
- The scope is usually contained within the business function, is moderately complex, and has a horizon of a few days to a few months, which the chapter calls **tactical planning**.

Executive level, also called strategic — The narrow top of the pyramid, where commitments are made for the whole organization

Who (Executives: CEO, CIO, vice presidents, possibly the board) → What (Aggregate summaries of past data and projections of the future) → Why (To improve organizational strategy and planning) → Decision type (Unstructured)

- The questions are long-term: which products to produce, which countries to compete in, and what organizational strategy to follow.
- Decisions here are **unstructured**, because few or no procedures can be specified in advance for problems that are relatively complex, nonroutine, and consequential for the overall organization.
- KPIs at this level balance performance across the organization — staggering product launches so demand spikes do not hit the supply chain at once — and benchmark the firm against its competitors.
- These systems must handle unstructured data such as global economic factors, demographic changes, and shifting customer tastes, and executives use **environmental scanning** to see how new systems could support strategy.

Test the three levels against a few situations before turning to the decisions.

CHECK YOURSELF**Placing work at the right level**

Four options, one best answer; read every explanation, including the ones for options you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A supervisor on a distribution floor needs to know whether tonight's shift can clear the orders already sitting in the building. Which level is she working at, and what are its systems for?

- **A.** The managerial level, because a supervisor is a manager and the question covers a whole shift rather than one order — Job titles are not levels. The chapter places foremen and supervisors at the operational level precisely because their decisions are day-to-day, highly structured, and recurring, whatever the word on the badge.
- **B.** The operational level, where systems automate routine and repetitive activities and events in order to improve efficiency — **correct**. This is routine day-to-day work with a horizon of a few hours, and operational systems exist to automate routine and repetitive activities and events so that goals are accomplished faster, at lower cost, or with relatively little time and effort.
- **C.** The executive level, because whether orders ship tonight affects whether customers stay with the firm at all — Executive-level decisions deal with complex problems whose ramifications are broad and long-term, such as which countries to compete in. Tonight's backlog matters, but it is settled before the executives next meet.
- **D.** The managerial level, because the systems there automate the monitoring and controlling of operational activities — The description of managerial systems is accurate, and it would fit a functional manager comparing this week's throughput across several shifts. Here the supervisor is inside the shift doing the work, not standing above it monitoring a function.

2. A company reports that one process now finishes faster and at lower cost than it did last year. Which term names that result, and which level does the chapter associate it with?

- **A.** Effectiveness, at the managerial level, since the goal is now being accomplished well — Effectiveness is the extent to which goals or tasks are accomplished *well*, and it is the managerial-level aim. Speed and cost alone say nothing about whether the right thing was produced.
- **B.** Efficiency, at the managerial level, since midlevel managers deploy the organization's resources — Half right is still wrong here. Efficiency is the correct term, but the chapter ties it to the operational level, where systems optimize clearly delineated processes and return the investment quickly.
- **C.** Efficiency, at the operational level, since goals are accomplished faster, at lower cost, or with relatively little time and effort — **correct**. Efficiency is defined as the extent to which goals are accomplished faster, at lower cost, or with relatively little time and effort, and improving it is the stated purpose of operational-level systems.

- **D.** Effectiveness, at the executive level, since a faster process supports the organization's long-term strategy — Executive systems aim at strategy and planning through aggregate summaries and projections, not at the speed of a single process. Naming this effectiveness also misses that nothing was said about how well the work was done.

3. A marketing manager is deciding how to allocate the advertising budget for the next business quarter. How does the chapter describe the scope and time horizon of a decision like this one?

- **A.** The scope is usually contained within one business function, is moderately complex, and runs a few days to a few months — **correct**. That is the chapter's description of managerial decisions, and it is why this horizon of a few days to a few months carries the name tactical planning.
- **B.** The scope reaches across the whole organization with long-term ramifications, since real money is being committed — Broad scope and long-term ramifications describe executive-level decisions, such as developing or discontinuing a product. A quarter of advertising spend is a large sum inside one function, not a commitment for the whole organization.
- **C.** The scope is a single business function, but the horizon is a few hours or days, matching operational planning — The scope is right and the horizon is not. A few hours or days is operational planning, the pace of the register and the loading dock, whereas a business quarter is months away.
- **D.** The scope covers several functions at once, which is exactly why the chapter calls this level tactical planning — Tactical planning is named for its time horizon, not for spanning functions, and midlevel managers typically focus on problems within one business function such as marketing or finance.

Structured, semistructured, and unstructured decisions

The levels differ in more than seniority. The decisions themselves have different shapes, and the shape determines how much of the decision a machine can take over.

Structured decisions — those in which the procedures to follow for a given situation can be specified in advance, so they can be programmed directly into operational information systems and made with little or no human intervention.

Semistructured decisions — those in which some procedures can be specified in advance, but not to the extent where a specific recommendation can be made, because the problems and solutions are not clear-cut and often require judgment and expertise.

Unstructured decisions — those in which few or no procedures can be specified in advance, because the problems are relatively complex and nonroutine and their consequences reach the whole organization.

The chapter's smallest example is the clearest. An inventory system in a mall shoe store tracks stock and orders more when levels drop below a specified point, and the store's operational managers simply confirm the order was needed. The rule existed before the situation did, so software can carry it.

The less a procedure can be written down in advance, the higher in the organization the decision travels, and the more the system shifts from deciding to informing.

SORT

How structured is this decision?

Place each decision in the category the chapter defines for it. Situations that are not the chapter's own examples are practice situations, and each one states the conditions you need.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Structured — the procedures to follow can be specified in advance, so the system can carry the decision itself

- An inventory system in a mall shoe store tracks stock and issues an order for more shoes whenever the count falls below a set level, and the operational managers in the store only confirm it — This is the chapter's own example of a rule written before the situation arose, which is why it can be programmed directly into an operational system and run with almost no human intervention.
- A payroll system multiplies the hours a system recorded by a stored pay rate to produce gross pay for each employee — The procedure is fixed in advance and the same every period, so it is highly structured and recurring — exactly the work operational systems automate.
- A register records each completed sale so the firm keeps a record of its daily business — Recording a transaction is the definition of routine operational work, and no judgment enters it: the steps are identical for every customer.
- In a practice situation, a supervisor assigns a returning part-time worker to a shift using a posted seniority rule that already covers this exact case — Because the rule was specified in advance and fits the situation without interpretation, the decision is structured even though a person happens to be the one applying it.

Semistructured — some procedures can be specified in advance, but not far enough to produce a specific recommendation

- A marketing manager at Nike decides how to allocate the advertising budget for the next business quarter — The chapter gives this as managerial work: past results and forecasts narrow the choice, but no procedure returns the specific allocation, so judgment finishes the job.

- A production manager uses a business intelligence system to build several hypothetical production schedules and then runs predictive analyses on each one — The system supplies analytics, forecasts, and comparisons, but it stops short of naming the schedule to use, which is precisely the semistructured boundary.
- In a practice situation, a functional manager watching a dashboard of quarterly KPIs must choose which of two underperforming processes to repair first, with the metrics known but no rule ranking them — Measurement is specified in advance and the choice is not, so the manager brings judgment and expertise to a moderately complex problem inside one function.
- In a practice situation, a midlevel manager decides how many temporary staff to bring into one department for a seasonal peak, using a forecast that narrows the range without naming a number — The scope stays inside a single business function on a horizon of a few weeks, and the forecast informs the decision rather than making it.

Unstructured — few or no procedures can be specified in advance, and the ramifications reach the whole organization

- Top managers decide whether to develop a new product or discontinue an existing one, knowing employment levels and profitability may be affected for years — The chapter uses this decision to show ramifications that are vast and long-term, and no procedure written in advance can settle it.
- Executives decide which countries the organization will compete in — This is one of the long-term strategic questions the chapter puts at the executive level, and it turns on complex, nonroutine judgments about the whole organization.
- Executives weigh global economic factors, demographic changes, and shifting customer tastes in choosing what organizational strategy to follow — These are the unstructured data types the chapter says executive systems must consider, and no advance procedure converts them into a single answer.

Functional areas and the systems that serve them

Decision-making levels cut the organization horizontally. Functional areas cut it vertically. A **functional area** is a discrete area of an organization that focuses on a specific set of activities.

Two examples show how different those sets of activities can be. People in marketing focus on activities that promote the organization and its products in a way that attracts and retains customers, while people in accounting and finance manage and control its capital assets and financial resources.

Because the activities differ, the systems differ. **Functional area information systems** are designed to support the unique business processes of specific functional areas, and the chapter's Table 2.1 names four of them.

FUNCTIONAL AREA	INFORMATION SYSTEM	SAMPLE APPLICATIONS
Accounting and finance	Systems used for managing, controlling, and auditing the financial resources of the organization	Accounts payable; expense accounts; cash management; payroll processing
Human resources	Systems used for managing, controlling, and auditing the human resources of the organization	Recruiting and hiring; education and training; benefits management; employee termination; workforce planning
Marketing	Systems used for managing new product development, distribution, pricing, promotional effectiveness, and sales forecasting of the products and services offered by the organization	Market research and analysis; new product development; promotion and advertising; pricing and sales analysis; product location analysis
Production and operations	Systems used for managing, controlling, and auditing the production and operations resources of the organization	Inventory management; cost and quality tracking; materials and resource planning; job costing; resource utilization

The two cuts meet in one picture. The chapter's Figure 2.6 puts five functional area systems along the base — accounting, financial, human resource, marketing, and operational — and stacks applications above them by how demanding the decision is.

Transaction processing and basic reporting — payroll processing in accounting, cash management in finance, an employee skill inventory in human resources, pricing and sales analysis in marketing, and manufacturing resource utilization in operations.

Tactical and managerial control — order processing in accounting, financial forecasting in finance, compensation analysis in human resources, sales force automation in marketing, and inventory management in operations.

More complex planning and coordination — portfolio management and ratio analysis in finance, personnel forecasting in human resources, advertising and promotion planning in marketing, and manufacturing resource planning in operations.

Read the two cuts together and you get a grid: the function names the column and the level names the row. Naming both narrows a vague request such as “we need better reporting” down to a single cell.

Ask which function owns the process and which level makes the call; those two answers describe the system before a single vendor is contacted.

MATCH

Functional areas and the systems that serve them

Pair each function or functional area system with the work it does. Four pairs come from Table 2.1 and name a functional area; the other four come from the chapter’s figure of applications stacked by decision level.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Accounting and finance

Systems used for managing, controlling, and auditing the financial resources of the organization Accounts payable, expense accounts, cash management, and payroll processing are the sample applications, and all four track where money sits and where it moves.

Human resources

Systems used for managing, controlling, and auditing the human resources of the organization Recruiting and hiring, education and training, benefits management, employee termination, and workforce planning cover a person’s whole span with the organization.

Marketing

Systems used for managing new product development, distribution, pricing, promotional effectiveness, and sales forecasting This function focuses on activities that promote the organization and its products in a way that attracts and retains customers, which is why forecasting and pricing sit here.

Production and operations

Systems used for managing, controlling, and auditing the production and operations resources of the organization Inventory management, cost and quality tracking, materials and resource planning, job costing, and resource utilization all concern making the thing rather than selling it.

Financial information systems

Cash management, then financial forecasting, then portfolio management and ratio analysis One functional system reaches across all three application layers, from a basic transaction record up to complex planning and coordination.

Human resource information systems

An employee skill inventory, then compensation analysis, then personnel forecasting The same function is served at three heights: recording who has which skill, controlling pay, and planning the workforce the organization will need.

Marketing information systems

Pricing and sales analysis, then sales force automation, then advertising and promotion planning Analysis of what sold supports tactical control of the sales force, which in turn feeds

the longer planning of campaigns.

Operational information systems

Manufacturing resource utilization, then inventory management, then manufacturing resource planning. Measuring how resources were used supports managing stock, and both feed the complex coordination of manufacturing resource planning.

One last pass over this section's vocabulary, from KPIs to functional area systems.

CHECK YOURSELF

Indicators, executives, and functions

Three questions on the vocabulary this section introduced; the explanations say what each wrong option genuinely describes.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What are key performance indicators, and where does the chapter say they are displayed?

- **A.** The complete set of transactions a firm records each day, displayed inside a transaction processing system — That describes raw transaction data, which is the record of everything that happened rather than the few numbers chosen to show progress. Indicators exist because nobody can watch every transaction.
- **B.** The metrics deemed most critical to assessing progress toward a certain goal, displayed on digital dashboards — **correct.** KPIs are the metrics deemed most critical to assessing progress toward a certain goal, and they are displayed on digital dashboards for managers who need the state of things at a glance.
- **C.** Targets set once a year by executives for every function, displayed in an annual report prepared for the board — KPIs are watched continuously rather than published once a year, and both managers and executives use them. This would be closer to a description of annual financial reporting.
- **D.** The procedures specified in advance for a recurring decision, displayed to supervisors as they do the work — Procedures specified in advance define a structured decision, not an indicator. A rule tells you what to do; an indicator tells you how the goal is going.

2. How does the chapter describe the information executives need from their systems?

- **A.** Detailed records of individual transactions, so that nothing is lost between the register and the boardroom — Individual transaction records are the output of operational systems. Executives receive them summarized, because the question at that level is the shape of the trend and not the content of one sale.

- **B.** Summaries built from the organization's own internal transaction data, because outside signals such as demographic change belong to the marketing function — The chapter says the opposite about the sources: executive-level systems need to consider various types of unstructured data, such as global economic factors, demographic changes, and changing customer tastes and preferences, none of which start inside the firm.
- **C.** Reports confined to one business function at a time, because each executive is responsible for a single function — That is the managerial level, where a midlevel manager typically focuses on problems within a specific business function. Executives must consider the ramifications of a decision for the overall organization.
- **D.** Aggregate summaries of past organizational data and projections of the future, with KPIs that balance performance and benchmark competitors — **correct.** Executive systems supply aggregate summaries of trends and projections of the future, with KPIs that balance performance across the organization and benchmark it against competitors.

3. What is a functional area, and what makes a functional area information system different from an information system in general?

- **A.** A functional area is one of the decision-making levels, and its systems serve everyone standing on that rung — Levels and functions are the two different cuts through an organization. Operational, managerial, and executive are the levels; marketing, human resources, and the rest are the functions.
- **B.** A functional area is a discrete area focused on a specific set of activities, and its systems support that area's unique business processes — **correct.** A functional area is a discrete area of an organization that focuses on a specific set of activities, and functional area information systems are designed to support the unique business processes of specific functional areas.
- **C.** A functional area is a physical site such as a store or a plant, and its systems are installed and operated locally — Location is not what defines a function. A single marketing function can operate from several sites, and a single site usually contains people from several functional areas.
- **D.** A functional area is a single business process such as order fulfillment, and its systems automate the steps of that one process — Business processes are the activities an organization performs to reach its goals, and a functional area is the standing part of the organization that owns a whole set of them: marketing runs market research, pricing, and promotion, not one process.

Automating, learning, and strategizing

Three companies can buy identical software for completely different reasons. One wants the same work done faster. One wants to learn whether the work is worth doing at all. One wants to change what it competes on. The chapter treats these as a ladder, where each rung adds more value than the one below.

Deployed across an organization's decision-making levels and functional areas, an information system can add value in three general ways: by enabling **automating** activities, by enabling **learning**, and by enabling the execution of **organizational strategy**. These are not necessarily mutually exclusive, but the chapter holds that each is progressively more useful to the firm and adds more value to the business.

The shorthand worth memorizing is faster, better, smarter. These are ambitions rather than product categories: one system can be defended and measured on any of the three, and people make that choice, not the software.

EXPLORE

Faster, better, smarter

Open each card and read all four panels; the fourth card sets the three against each other, which is where the distinction usually breaks down.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Automating · Doing things faster

- **The mindset in one line:** Technology used to help complete a task within an organization faster, more cheaply, and perhaps with greater accuracy or consistency — in other words, to increase efficiency.
- **What it looks like in practice:** Loan applications go into a computer database so the people deciding on them work faster, more easily, and with fewer errors, and customers can complete the application online instead of carrying paper home.
- **The payoff it produces:** Time and cost fall. In the fully automated loan process a small loan can be finished in as little as 15 minutes, and moving off the manual process may enable the organization to deploy its employees more efficiently.
- **Where it stops:** It changes what a process costs, never what the process is. If the underlying process is flawed, automating use of technology may mask the problem while multiplying its effects.

Organizational learning · Doing things better

- **The mindset in one line:** The ability of an organization to use past behavior and data to improve its business processes, built on top of the automating mentality rather than in place of it.
- **What it looks like in practice:** The loan processing system tracks types of application by date, month, or season, so a manager can see trends, plan timely staffing and training, and manage the funds used to fulfill the loans.
- **The payoff it produces:** Effectiveness rises, meaning goals get accomplished well rather than merely quickly, because new data about the underlying process can be used to monitor, control, and change that process.
- **Where it stops:** Learning tells you how well the current process is working. It does not by itself say what the organization ought to be competing on, which is a question decided a level higher up.

Supporting strategy · Doing things smarter

- **The mindset in one line:** Going beyond automating and learning to find ways of using information systems to achieve the organization's chosen strategy — by innovating, streamlining operations, optimizing the supply chain, or better understanding customers.
- **What it looks like in practice:** Uber's use of information systems for personalization, anticipation, and coordination sits at the heart of its organizational strategy, is fundamental to its business model, and is essential to the long-term survival of the organization.
- **The payoff it produces:** Strategic value, meaning help in improving the business so that it can compete better — and the chapter is blunt that a system delivering none of that, even while saving money, is not likely to be funded.
- **Where it stops:** It depends on the organization already having a strategy worth serving. Strategic planning happens at the executive level, and no system can supply the vision it is being built to carry out.

The three compared · Why the order matters

- **The mindset in one line:** Three general ways an information system can provide value, which the chapter presents as progressively more useful to the firm and therefore as adding progressively more business value.
- **What it looks like in practice:** One product can occupy all three at once: Uber automatically matches riders and drivers, learns preferences from past behavior, and builds its strategy on exactly those capabilities.
- **The payoff it produces:** The chapter draws the comparison as three bars of business value added, where automating is the shortest bar, learning is taller, and strategizing is the tallest of the three.
- **Where it stops:** They are not mutually exclusive, so the real question is never which single rung a proposal belongs to but how far up it reaches, since a strategic system almost always automates and learns as well.

Automating: doing things faster

Someone with an **automating** perspective thinks of technology as a way to complete a task inside an organization faster, more cheaply, and perhaps with greater accuracy or consistency. What they are buying is **efficiency** — the extent to which goals are accomplished faster, at lower cost, or with relatively little time and effort. The work itself is never questioned; only what it costs.

The chapter’s worked example is a bank screening loan applications. A person with an automating mentality makes three moves, and not one of them changes who decides or on what grounds.

Put the applications into a computer database — so the people deciding on the loans process them faster, more easily, and with fewer errors than they could from a stack of paper.

Let customers apply online — because an application completed from home arrives already entered, which removes the trip home and back from the schedule.

Redeploy the people the change frees — moving from a manual to an automated loan process may enable the organization to deploy employees more efficiently, which is where the further cost savings come from.

Automation is not confined to paperwork. With increasing digital density, almost every sector can gain efficiency: sensors monitoring machinery or product flows throughout production can tremendously improve a manufacturing company’s performance. Operational-level systems automate repetitive activities and also gather valuable data for higher decision-making levels, which is the hinge into the next rung.

The same process, three times over

The table takes one loan application through three versions of the same process. Read down a column to see one whole process; read across a row to watch a single activity change.

PRIMARY ACTIVITY	MANUAL LOAN PROCESS	TECHNOLOGY-SUPPORTED PROCESS	FULLY AUTOMATED PROCESS
Complete and submit loan application	Customer takes the application home, completes it, and returns it (1.5 days)	Customer takes the application home, completes it, and returns it (1.5 days)	Customer completes application from mobile device (1.5 days)
Check application for errors	Employee does this in batches (2.5 days)	Employee does this in batches (2.5 days)	Computer checks application (0.5 days)

PRIMARY ACTIVITY	MANUAL LOAN PROCESS	TECHNOLOGY-SUPPORTED PROCESS	FULLY AUTOMATED PROCESS
Input data from application into the information system	Applications are kept in paper form, although there is handling time involved (1 hour)	Employee does this in batches (2.5 days)	Done in 15 minutes
Assess loan applications under \$250,000 to determine whether to fund them	Employee does this completely by hand (15 days)	Employee does this with the help of the computer (1 hour)	Computer automated
Committee decides on any loan over \$250,000	15 days	15 days	15 days
Applicant notified	Employee generates letters manually in batches (1 week)	Employee generates letters with the help of a computer (1 day)	System email
Total time	Anywhere from 25–40 days, depending on size of loan	Anywhere from 5–20 days, depending on size of loan	Anywhere from 15 minutes to 1 day, depending on size of loan

SEQUENCE

The loan application process, activity by activity

Put the six primary activities into the order the loan table lists them, then read why each one sits where it does.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The table you just read runs the same six activities through three different processes. Getting the sequence straight is what makes the time savings legible, because each activity can only be shortened by the technology that touches it.

1. **Complete and submit the loan application. Manually the customer takes the application home, completes it, and returns it in 1.5 days; in the fully automated process the customer fills it out from home via the web in 15 minutes.** — Nothing else can start until an application exists, and this is the only activity where the work sits with the customer rather than the bank, which is why moving it to the web removes a day and a half at a stroke.

2. **Check the application for errors. An employee does this in batches over 2.5 days in both the manual and the technology-supported processes; the computer does it in 1 second as the form is being completed.** — Errors are caught before the data travel any further into the bank. Batching is what makes this take days rather than minutes, because a form waits for the batch it belongs to rather than for a person.
3. **Input the data from the application into the information system. Paper handling takes about 1 hour in the manual process, an employee keys batches over 2.5 days in the technology-supported process, and no extra time is needed once the customer fills the form in online.** — This is the activity that gets worse before it gets better: the middle column costs 2.5 days where the manual process cost an hour, because a half-computerized process still needs somebody to type everything in.
4. **Assess loan applications under \$250,000 to determine whether to fund them. An employee does this completely by hand in 15 days, does it with the help of the computer in 1 hour, or the computer does it automatically in 1 second.** — This is the largest single saving in the table, falling from 15 days to 1 second, and it is the one activity where an actual funding decision was handed over to a machine rather than merely assisted by one.
5. **The committee decides on any loan over \$250,000, which takes 15 days in the manual, technology-supported, and fully automated processes alike.** — The table lists this immediately after the smaller-loan assessment because the two are alternative paths chosen by the size of the loan, which is why every total is given as a range. This is also the only row technology never touches.
6. **The applicant is notified. Manually generated letters go out in batches over a week, computer-assisted letters take a day, and the fully automated system notifies the applicant by email in 1 second.** — A loan process is not finished when the decision is made but when the customer knows the decision, which is why a week of letter writing belonged in the manual total all along.

What the totals actually show

The headline is a fall from weeks to minutes, but three quieter facts in that table teach more.

One row never moves — the committee deciding any loan over \$250,000 takes 15 days in all three processes, because none of the three columns changed who makes that decision.

That row is the ceiling — the fully automated process runs from 15 minutes to 15 days depending on loan size, and 15 days is exactly the committee's figure, so the slowest automated loan is slow for a human reason.

One step gets worse first — entering data takes about an hour in the manual process, where applications stay on paper, but 2.5 days in the technology-supported one, where an employee keys in the batches. Half-computerizing can create work where none existed.

Automation compresses the parts of a process a machine is permitted to touch and leaves everything else where it was, which is why the fully automated column reports a range instead of a number.

CHECK YOURSELF

Reading an automation table honestly

Four options, one best answer; read every explanation, including the ones for the answers you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Which activity in the loan table takes the same amount of time in the manual, technology-supported, and fully automated processes, and why does that matter?

- **A.** Checking the application for errors, because a person still has to read every form no matter how it arrives — Error checking is one of the activities that changes the most across the table: an employee does it in batches over 2.5 days in the first two processes, while the computer does it in 1 second as the form is being completed. This would be right only if a person still had to read each form.
- **B.** The committee decision on loans over \$250,000, at 15 days in all three, because none of the three processes changed who makes that decision — **correct**. That row reads 15 days in every column, so the ceiling on the fully automated process — anywhere from 15 minutes to 15 days — is set by a decision reserved for people rather than by any limit of the technology.
- **C.** Notifying the applicant, because a decision still has to reach a human being at the other end — Notification changes in all three columns: letters generated manually in batches take a week, computer-assisted letters take a day, and the fully automated system emails the applicant in 1 second. The letter is precisely the part that disappeared.
- **D.** Inputting the data into the information system, because entering an application takes as long as it takes — This activity does change, and unusually it moves in both directions: about an hour of paper handling in the manual process, 2.5 days of batch keying in the technology-supported process, and no extra time at all once the customer completes the form online.

2. Entering application data takes 2.5 days in the technology-supported process but only about an hour in the manual one. What is the accurate reading of that?

- **A.** The columns must be reversed, since a process that gained a computer should not take longer at that step than the paper one did — It is reasonable to expect every row to improve, but the table reports what it reports: applications are kept in paper form with handling time in the manual process, and an employee keys batches for 2.5 days in the technology-supported one. Partly computerizing a process can create work where none existed.
- **B.** In the manual process nothing is keyed in at all, since applications stay on paper, while the technology-supported process needs an employee to key batches into the system — **correct.** The hour in the manual column is handling paper, not data entry. Once a system exists but the customer is not filling it in directly, somebody has to type every application into it, and that work is done in batches.
- **C.** The technology-supported process is checking applications more thoroughly, and the extra scrutiny is what takes the extra time — Checking the application for errors is a separate row of the table and takes 2.5 days in both the manual and the technology-supported processes, so the extra time here is not extra scrutiny. It is data entry that simply did not exist before.
- **D.** The two processes handle different sizes of loan, so their times for this activity cannot fairly be compared — Loan size splits the assessment activity and the committee activity, not data entry, and the three columns describe three versions of one process rather than three kinds of loan. Every row is meant to be read straight across.

3. A manager says: "Our new system lets each reviewer handle three times as many files a day at a lower cost per file. That is all I need to know." Which mentality is that, and what does the chapter say it leaves out?

- **A.** An automating mentality, and it leaves out whether the process being sped up is a sound one — **correct.** Automating aims at efficiency — accomplishing goals faster, at lower cost, or with relatively little time and effort — and the chapter warns that automating a flawed process with no learning built in can mask the problem while multiplying its effects.
- **B.** A learning mentality, and it leaves out how much the system costs the bank to run — A learning mentality uses past behavior and data to improve the business process itself, and this manager has asked about neither. Cost is also the one thing he did mention, so it cannot be what is missing from his account.
- **C.** A strategic mentality, and it leaves out the return the investment will produce — A strategic mentality asks how a system helps achieve the organization's chosen strategy and produces competitive advantage. Throughput and cost per file are efficiency measures, which the chapter places on the lowest rung rather than the strategic one.
- **D.** An automating mentality, and it leaves out how many employees the change will let the bank redeploy — The first half is right and the second half is not. Redeploying employees more efficiently is part of the automating case, since the chapter names it as where the further cost savings come from, so adding that number makes the efficiency story fuller rather than answering whether the process being sped up is sound.

Organizational learning: doing things better

Organizational learning is the ability of an organization to use past behavior and data to improve its business processes. It builds on the automating mentality rather than replacing it: by analyzing the data created when a process was automated, you develop an improved understanding of the work process itself.

Return to the loan department. A computer-based loan processing system can track types of loan application by date, month, or season, and that single capability changes what a manager can do.

Trends become visible — the manager can see more auto loan applications arriving in the fall, for instance, and plan the timely staffing and training of loan department personnel around that shape.

Money can be planned — the manager can more efficiently manage the funds used to fulfill the loans, because demand stops being a surprise arriving one application at a time.

Patterns in the decisions appear — the manager can study which types of people apply at which times of year, the patterns of the decisions made, and the subsequent performance of those loans.

The test for learning is not how much data a system holds. It is whether the system creates data about the underlying business process that can be used to monitor, control, and change that process. Speed was efficiency; this is **effectiveness**, the extent to which goals or tasks are accomplished well. Used this way, a loan processing system sits at the managerial level of the organization.

What automating alone can hide. If the underlying process is flawed, a learning use of technology may help you detect the problem and change it: a pattern among accepted loans can separate low-performing from high-performing ones and prompt a change to the acceptance criteria. Automate the same process with no learning built in, and you never uncover the data that would show the process is bad.

The chapter puts numbers on it. Suppose the criteria are poor: they approve someone with a high level of debt as long as no payment has been late recently.

- Working by hand, a reviewer might get through four applications a day and, because of those criteria, inadvertently accept about two bad applications a week.
- Automate that same faulty process with no learning built in, and the reviewer might handle 12 applications a day, resulting in up to six bad applications accepted per week on average.
- The technology served only to magnify the existing business problem, so automating a loan process requires sound underlying business processes or the errors rapidly increase.

Learning also scales outward. Increasing digital density, together with the ability to learn from data gathered through many interactions, can increase personalization, anticipation, and coordination, and the chapter's example is Uber.

Personalization — Uber uses a rider's past behavior to learn about her preferences and provide personalized ride recommendations.

Anticipation — the app uses data about riders' and drivers' locations and a host of other factors to anticipate surges in demand and change pricing accordingly.

Coordination — that same data is then used to coordinate riders and drivers so that the matching of supply and demand is optimized.

Plain efficiency is on that list too: the app automatically matches riders with drivers. One product doing all four at once shows why the three ways of adding value are not mutually exclusive.

Supporting strategy: doing things smarter

Automating and improving processes both have real advantages, but in most cases the chapter argues that the best way to use an information system is to support the organization's strategy.

Organizational strategy is a firm's plan to accomplish its mission and goals and to gain or sustain competitive advantage over rivals.

That plan is made at the executive level through **strategic planning**: senior managers form a vision of where the organization needs to head, convert it into measurable objectives and performance targets, and craft a strategy to reach them. Someone with a strategic mentality then looks for ways to use information systems to achieve that strategy, and the chapter names four.

- Innovating, because a genuinely new offering changes what the company sells rather than what it costs.
- Streamlining operations, so that the savings serve the chosen plan instead of being an end in themselves.
- Optimizing the supply chain, since what a firm can promise a customer depends on what its suppliers deliver.
- Understanding customers better, because deep knowledge of a customer is not something a rival can simply buy.

This person still wants the benefits of automating and learning, and also looks for some strategic, competitive advantage from the system. The funding rule follows. In today's business environment, if a proposed information system will not clearly deliver strategic value — help improve the business so it can compete better — while also helping people work smarter and save money, it is not likely to be funded.

Uber shows the top rung too. Using information systems for personalization, anticipation, and coordination is at the heart of its organizational strategy, is fundamental to its business model, and is essential to the organization's long-term survival. What counted as learning one rung down counts as strategy here.

Why the top rung matters more than it used to. Rather than solely improving an organization's efficiency, information systems now enable technology- and data-driven business model disruption: incumbents see their ways of working challenged by technology startups that approach the business in fundamentally different ways, grabbing market share and ultimately making existing business models obsolete.

A firm parked on the automating rung is getting steadily cheaper at doing something a newcomer may be about to make unnecessary.

DECIDE

Which ambition does this proposal actually serve?

Work the four decisions in order, and after each one compare every outcome before moving on.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A hypothetical practice situation. You manage the loan department at a community bank, and four proposals land on your desk. For each one the question is not whether it is a good idea, but which of the three ways of adding value it actually serves — because that decides what you measure and who has to approve it.

Decision 1. A vendor offers to replace the paper application with a web form customers complete at home, with the computer checking for errors as they type and the data flowing straight into the loan system with no separate keying.

- Automating. It compresses activities the bank already performs, and the payoff is time and cost. — **the stronger call.** Right. Completing the application from home via the web, error checking as the form is filled in, and no separate data entry step are exactly the moves in the fully automated column of the loan table. Judge it on efficiency: faster, cheaper, fewer errors.
- Organizational learning, because the applications will now sit in a database where they can be analyzed. — Storing data and learning from data are different things. Nothing in this proposal creates information about the underlying process — who applies when, or how the approved loans later perform — and nobody has been given a report. A database is a precondition for learning, not learning itself.
- Supporting strategy, because customers will prefer a bank they can apply to from the sofa. — Convenience may well matter to customers, but nothing here ties it to a plan to gain or sustain advantage over rivals, and any competitor can buy a comparable web form. Labelling an efficiency project strategic is how a project ends up measured against the wrong target.

Decision 2. The same vendor offers a reporting add-on. It charts application types by date, month, and season, and tracks how the loans that were approved perform afterwards. It will not speed up a single approval by one minute.

- Nothing worth funding, because a report does not process an application. — This is the mistake the chapter warns about most directly. Without learning it is more difficult to uncover a bad process underneath the system, so the bank would simply run the existing process faster. Reports are how a process gets examined at all.
- Automating, because reporting is one more clerical task the computer takes over. — Efficiency is measured in time and cost, and this proposal saves neither by its own description. Judging it on approval speed would kill a proposal whose whole purpose is to tell you something you do not currently know.
- Organizational learning, because it creates data about the process that can be used to change the process. — **the stronger call.** Right. The manager can now plan timely staffing and training, manage the funds used to fulfill loans, and study the patterns in the decisions made and the later performance of those loans. That is using past behavior and data to improve a business process.

Decision 3. Now the pressure. A consultant reports a finding in this hypothetical bank: the department approves applicants carrying a high level of debt as long as they have not been late on any payments recently, and those loans are performing badly. The vendor's throughput system would take each reviewer from four applications a day to twelve.

- Deploy the throughput system now and revisit the acceptance criteria later, once the backlog is cleared. — This is the chapter's cautionary arithmetic. A reviewer working by hand under those criteria might accept two bad applications a week; automated to twelve applications a day, the same faulty process yields up to six bad ones a week. The technology serves only to magnify the existing business problem.
- Fix the acceptance criteria first and install the reporting that would have exposed them, then raise throughput. — **the stronger call.** Right. Automating a loan process requires sound underlying business processes, and the learning layer is what surfaces the pattern — distinguishing low-performing from high-performing loans over their lives and changing the acceptance criteria accordingly.
- Keep technology out of this department altogether, since automation is what produced the bad loans. — Automation did not write the criteria; it would only multiply their effect. Removing the technology leaves a flawed process in place and slow, and leaves nobody any way to see the pattern that would identify the flaw.

Decision 4. The executive committee will fund one write-up of the same project. Three drafts of the case sit in front of you.

- "It will cut our processing cost per application." — True, and on its own insufficient. This is a pure efficiency case, and the chapter states that a proposed system that does not clearly deliver strategic value as well is not likely to be funded in today's business environment.
- "It gives us same-day decisions on small loans, built on application and performance data we already hold, and it lowers cost per application while doing it." — **the stronger call.** Right. That reaches strategic value — helping the business compete better — while also helping people work smarter and saving money in the process, which is the combination the chapter says a proposal needs to be likely to be funded.
- "Every competitor is digitizing their loan process, so we cannot afford to be the last one." — This argues for parity, never for advantage, and it hands the bank's technology plan to its rivals. It also tells the committee nothing about what the system would do here or how anyone would know whether it worked.

Reading the ladder correctly

Three ambitions, three payoffs, and three different questions to ask about any proposal.

- 1 **Faster** — ask what this work costs today and what it will cost afterwards, because the payoff is efficiency, measured in time and money.
- 2 **Better** — ask what the system will teach you about the process itself, because the payoff is effectiveness, measured in decisions that turn out well.
- 3 **Smarter** — ask what the system lets you do that a rival cannot, because the payoff is competitive advantage, measured against competitors rather than last year.

Most good systems reach all three rungs, so the useful question is not which one a proposal is but how far up it reaches — and a project that answers only the first is an efficiency story told in a room that funds strategy.

CHECK YOURSELF

Learning, strategy, and the funding test

Three questions on the upper two rungs; the wrong answers are the ones people actually give in meetings.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What makes an information system produce organizational learning rather than mere speed?

- **A.** It stores considerably more data than the system it replaced, since learning is a matter of volume — Volume by itself changes nothing, because an automated process also generates data and the question is what the data are about and whether anyone analyzes them. A bigger database nobody examines teaches an organization exactly as much as a smaller one.
- **B.** It creates data about the underlying business process that can be used to monitor, control, and change that process — **correct**. That is the chapter's own test, and the loan example shows it working: trends by season, the patterns of the decisions that were made, and the later performance of those loans, all of which can change how applications are evaluated.
- **C.** It removes human judgment from the process, so that the decisions come out consistent — Consistency is one of the benefits claimed for automating, not for learning, and the chapter's learning example depends on a manager reading trends and acting on them. A system that removed the judgment would leave nothing to improve.
- **D.** It operates at the executive level of the organization rather than the managerial level — The chapter places the learning-focused loan processing system at the managerial level, where midlevel managers monitor and control operational activity. Which level a system serves is a separate question from whether it produces learning.

2. In the chapter's illustration, a reviewer working by hand under flawed acceptance criteria handles four applications a day and inadvertently accepts about two bad ones a week. The same faulty process is automated so that the reviewer handles twelve a day. What follows?

- **A.** Up to six bad applications a week are accepted, because the technology serves only to magnify the existing business problem — **correct**. Three times the throughput under the same bad criteria produces up to three times the bad loans, which is why automating a process requires sound underlying business processes or the errors rapidly increase.
- **B.** Bad acceptances stay near two a week, because the criteria were the problem and the criteria did not change — The criteria did not change, which is why the failure rate per application does not improve — but the number of applications reviewed tripled, so the count of bad acceptances rises with it. Holding the total steady would require the system to catch what the criteria miss.

- **C.** Bad acceptances fall, because a computer applies the criteria more consistently than a tired reviewer does — Consistency is a real benefit of automation, and it is beside the point when the rules being applied consistently are the wrong rules. Applying a flawed criterion identically every time simply produces flawed results faster.
- **D.** It cannot be worked out from the illustration, because tripling the number of files reviewed says nothing about how many of the extra files were bad ones — The illustration deliberately holds the criteria and the applicant mix constant so that only throughput changes, which is how it arrives at up to six bad acceptances a week — three times the volume run through the same flawed rules, drawn as cumulative errors climbing as automation increases.

3. A project team proposes a system that will clearly cut cost per transaction but has no answer to how it helps the company compete. What does the chapter say about its prospects?

- **A.** It will be funded, because cost reduction is the most reliable justification an information system can offer — Cost reduction is genuine value and it is the automating rung, which the chapter treats as the least of the three ways of adding value. It is not presented as the strongest case a proposal can make in today's business environment.
- **B.** It is not likely to be funded, because a proposal is expected to deliver strategic value as well as helping people work smarter and save money — **correct.** The chapter states the test directly: a system that will not clearly deliver strategic value, meaning help to improve the business so that it can compete better, while also helping people work smarter and save money, is not likely to be funded.
- **C.** It is not likely to be funded, because a proposal is expected to show competitive advantage instead of cost savings — The conclusion is right and the reasoning is not. The chapter's test asks for strategic value while also helping people to work smarter and save money, so cost savings are half of what a proposal is expected to show rather than the wrong sort of argument.
- **D.** It will be funded, because the strategic value of a system only becomes clear once the system is running — Strategy is set in advance through strategic planning, in which executives form a vision, convert it into measurable objectives and performance targets, and craft a strategy to reach them. Discovering the strategic value afterwards is what the funding test exists to prevent.

Where to compete: analyzing competitive forces

Every company has more ideas than money. Before anyone can sensibly say which system to build, somebody has to say what is actually pressing on the business — and this section teaches the framework the chapter uses to say it.

Organizations struggle with identifying the best uses of their resources to execute their strategy. Some companies have a competitive advantage simply by being the first to enter a market, which is what the chapter calls a **first-mover advantage**. Most organizations never get that opening. They must compete within established industries, against firms that were already there and customers who already have habits.

Because every industry is different, organizations need to analyze the competitive forces within their industry to better understand where to focus their resources. An **industry** is simply the set of firms selling the same kind of product to the same kind of customer, and the analysis asks one narrow question: of everything squeezing this industry, which pressure is doing the most damage to us right now?

The five primary competitive forces

The framework the chapter uses is Porter's (1979) notion of the five primary competitive forces. Read them as five separate directions that pressure can arrive from, rather than as five names to memorize.

Traditional rivals within your industry — the rivalry among competing sellers already inside your industry, which shows up as competition in price, product distribution, and service, because all of you are chasing the same buyers.

Threat of new entrants into your market — the chance that firms not competing with you today will start tomorrow, which brings increased capacity in the industry, reduced prices, and decreased market share for everyone already there.

Customers' bargaining power — the ability of the people who buy from you to push back on your terms, which arrives as reduced prices, a need for increased quality, and demand for more services.

Suppliers' bargaining power — the ability of the firms you buy from to set the terms of the relationship, which reaches you as increased costs and reduced quality in the very things you resell.

Threat of substitute products or services from other industries — a different kind of product meeting the same underlying need, which brings product returns from customers, decreased market share, and losing customers for life.

The chapter draws this as one circle and four boxes: rivalry among existing firms sits at the center, and bargaining power of buyers, threat of new entrants, threat of substitute products or services, and bargaining power of suppliers each press inward on it. The picture carries the model's claim — five forces influence the profit potential of an industry, so a firm can be competently run and still earn very little if the forces around it are severe.

INTERACTIVE DIAGRAM

The five competitive forces, one at a time

Pick a force to redraw it: what the pressure is, what the chapter says it implies, what the internet did to it, and the kind of system that relieves it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Traditional rivals within your industry — Pressure from the firms already selling what you sell

Your firm (One of several established sellers) → Established rivals (Same goods, same suppliers, same buyers) → One pool of customers (Every sale won is a sale someone else lost) → The usual weapon (Price cuts that thin every margin at once)

- **What it is** — the rivalry among competing sellers in your industry, meaning the firms that already sell what you sell to the customers you already have.
- **What the chapter says it implies** — competition in price, product distribution, and service, so the fight is fought on what you charge, how the product reaches the buyer, and how well the buyer is treated.
- **What the internet did to it** — an increase of competitors due to wider geographic reach, and customers who can more easily compare products, so competition focuses more on price than it used to.
- **Signal and relief** — you know rivalry is the strong force when every promotion is matched within days and margin falls while volume holds, and the systems that help are the ones moving the contest off price, such as reliable stock information so the advertised item is actually on the shelf.

Threat of new entrants into your market — Pressure from firms that are not in your industry yet

Outsiders (Have money, want your customers) → Barriers to entry (Capital, scale, expertise, loyalty) → Entry happens (Increased capacity in the industry) → The result (Reduced prices, decreased market share)

- **What it is** — the threat of potential new entrants into your industry, which is about companies that do not compete with you today and could begin to tomorrow.

- **What the chapter says it implies** — increased capacity in the industry, reduced prices, and decreased market share, because more sellers are now chasing a pool of customers that did not grow to match.
- **What the internet did to it** — reduced barriers to entry, since the internet reduces the difficulty of obtaining critical resources or entering new markets, so the money and infrastructure that once kept outsiders out no longer do.
- **Signal and relief** — you know entrants are the strong force when the firms taking your customers were not in this business last year, and the systems that help are the ones building what a newcomer cannot buy quickly, such as years of customer history and reordering that runs on its own.

Customers' bargaining power — Pressure from the people who hand you money

Your customers (Comparing alternatives in seconds) → Switching cost (Lower than it once was) → What they ask for (Lower price, higher quality, more service) → Your side (Terms set partly by the buyer)

- **What it is** — the bargaining power that customers have within your industry, which is their ability to insist on better terms because walking away costs them little.
- **What the chapter says it implies** — reduced prices, a need for increased quality, and demand for more services, so the buyer is effectively setting part of what you charge and what you must include.
- **What the internet did to it** — wider choices for customers lead to lower switching costs and higher bargaining power of customers, so the comparison a buyer once had to drive across town to make now happens in a moment.
- **Signal and relief** — you know buyer power is the strong force when shoppers leave over small price differences or a few large accounts squeeze you at renewal, and the systems that help give the customer something to lose by leaving, such as a loyalty record and offers built from that customer's own history.

Suppliers' bargaining power — Pressure from the firms you buy from

Few suppliers (Little real choice about whom to buy from) → Terms set upstream (Prices and lead times move on their schedule) → What reaches you (Increased costs and reduced quality) → The internet twist (Equal access, but suppliers reach more buyers too)

- **What it is** — the bargaining power that suppliers have within your industry, which is their ability to raise prices or dictate terms because you have few alternatives.
- **What the chapter says it implies** — increased costs and reduced quality, and both of those land in the product your own customer eventually sees, so supplier pressure is never only a purchasing problem.
- **What the internet did to it** — companies have equal access to suppliers and it is easier to find new ones, but suppliers also have access to more potential buyers, so this is the one row where the influence genuinely runs both ways.

- **Signal and relief** — you know supplier power is the strong force when few firms can supply what you need and nobody in your company can state what the firm buys in total, and the systems that help consolidate purchasing into one record and let qualified alternatives be compared side by side.

Threat of substitute products or services from other industries — Pressure from a different answer to the same need

The underlying need (The job the customer wants done) → Your industry's answer (The product you happen to sell) → Another industry's answer (Meets the same need a different way) → The damage (Returns, lost share, customers lost for life)

- **What it is** — the potential for substitute products or services from other industries, where the customer satisfies the same need without ever entering your industry.
- **What the chapter says it implies** — product returns from customers, decreased market share, and losing customers for life, which is the harshest wording in the table and the reason this force deserves attention it rarely gets.
- **What the internet did to it** — new substitutes are created by the internet and other information technologies, so an industry can acquire a competitor that did not exist in any industry a few years earlier.
- **Signal and relief** — you know substitutes are the strong force when discounts stop working and the whole category shrinks rather than your share of it, and the systems that help extend your offer into the substitute's territory instead of defending the old one.

What each force does to a firm, and what the internet did to each force

Table 2.3 sets every force beside two things: what living with it implies for a firm, and how the internet has influenced it. Read the middle column as the damage and the right column as the reason the damage grew.

COMPETITIVE FORCE	IMPLICATION FOR FIRM	INFLUENCE OF THE INTERNET
Traditional rivals within your industry	Competition in price, product distribution, and service	Increase of competitors due to wider geographic reach; customers can more easily compare products, so competition focuses more on price.
Threat of new entrants into your market	Increased capacity in the industry, reduced prices, and decreased market share	Reduced barriers to entry, as the internet reduces the difficulty of obtaining critical resources or entering new markets.

COMPETITIVE FORCE	IMPLICATION FOR FIRM	INFLUENCE OF THE INTERNET
Customers' bargaining power	Reduced prices, need for increased quality, and demand for more services	Wider choices for customers lead to lower switching costs and higher bargaining power of customers.
Suppliers' bargaining power	Increased costs and reduced quality	Companies have equal access to suppliers; easier to find new suppliers; suppliers have access to more potential buyers.
Threat of substitute products or services from other industries	Product returns from customers, decreased market share, and losing customers for life	New substitutes are created by the internet and other information technologies.

One pattern runs straight down the right-hand column: almost every entry describes the internet making a force stronger, not weaker. Wider reach adds competitors, easier comparison pushes the fight toward price, barriers to entry fall, switching costs drop, suppliers reach more potential buyers, and information technologies create substitutes that did not exist before.

The same technology that hands a firm new ways to compete hands its rivals, its customers, its suppliers, and firms in other industries exactly the same gift, which is why this analysis has to be redone rather than filed.

CHECK YOURSELF

Naming the force

Four options, one best answer; the company situations are generic practice examples rather than reports about real firms, and every explanation is worth reading, including the ones for options you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A regional retailer has lost share to two companies that have sold the same goods in the same region for decades, and every promotion it runs is matched within a week. Which competitive force is at work?

- **A.** Threat of new entrants into its market, because competitors are taking customers it used to keep — New entrants are firms that were not competing in this industry before, and the distinguishing fact here is that both competitors have been selling here for decades. It would be the right answer if the companies taking share had just arrived.

- **B.** Traditional rivals within its industry, because the pressure comes from established sellers competing on price and service — **correct.** Rivalry among competing sellers in your industry shows up as competition in price, product distribution, and service, and matched promotions between long-established sellers is precisely that pressure.
- **C.** Threat of substitute products or services, because customers are choosing something other than this retailer — A substitute meets the same need from another industry, so the customer never shops your industry at all. These shoppers are still buying the same goods, simply from a different seller of the same kind.
- **D.** Customers' bargaining power, because shoppers are the ones deciding to buy elsewhere — Buyer power is about customers forcing better terms because leaving costs them little, which usually shows up as pressure on price, quality, and services across all sellers. Here the visible actor is a competitor matching offers, not a buyer negotiating.

2. According to Table 2.3, what does the threat of new entrants imply for the firms that are already in an industry?

- **A.** Increased costs and reduced quality in what those firms buy and resell — Increased costs and reduced quality is the implication the table gives for suppliers' bargaining power, because supplier terms reach you through what you purchase rather than through who else is selling.
- **B.** Product returns from customers, decreased market share, and losing customers for life — Product returns, decreased market share, and losing customers for life is the table's implication for substitute products or services, and the phrase about customers lost for life is what makes that row distinct.
- **C.** Increased capacity in the industry, reduced prices, and decreased market share — **correct.** A new entrant adds capacity to an industry whose customer pool did not grow, so the table lists increased capacity, reduced prices, and decreased market share.
- **D.** Competition in price, product distribution, and service among the established sellers — Competition in price, product distribution, and service is the implication listed for traditional rivals, which is the force describing sellers already established inside the industry rather than newcomers.

3. Why does the chapter say organizations need to analyze the competitive forces within their industry at all?

- **A.** Because a first-mover advantage protects most organizations, so the analysis simply confirms the position they already hold — A first-mover advantage does exist, and it belongs to companies that are first to enter a market. The chapter's point is the opposite of protection for most firms: most organizations must compete within established industries.
- **B.** Because every industry is different, so the analysis shows where a firm should focus its limited resources — **correct.** Organizations struggle with identifying the best uses of their resources, and because every industry is different, analyzing its forces is how a firm decides where to concentrate what it has.

- **C.** Because the analysis produces a ranked shortlist of vendors, which is what a technology investment decision needs — The model helps determine which forces matter most and which kinds of technology could address them, and that knowledge then becomes the basis for identifying investments. Naming products is a later step, and doing it first inverts the sequence.
- **D.** Because the analysis forecasts how much profit this particular firm will earn next year, which is the number a technology budget has to be built from — The chapter presents the five forces as influences on the profit potential of an industry, not as a forecast of one firm's earnings. What the model produces is a judgment about which forces matter most and which technologies could address them, which guides where money goes rather than saying how much there will be.

Telling which force is the strong one

A single lost sale can be told as a story about almost any of the five, so the real skill is not reciting the list but choosing between them from what you can actually observe. Each force leaves a different fingerprint.

Rivalry is the strong force when — the companies taking your customers have sold in this industry for years, discounting is the only move anyone reaches for, and each promotion is matched within days while volume stays flat and margin falls.

New entrants are the strong force when — the companies taking your customers were not in this business last year, and nothing you hold — capital, scale, expertise, customer loyalty — would take a newcomer long to reproduce.

Customers' bargaining power is the strong force when — buyers compare your price against alternatives in seconds and leave over small differences, or a few large accounts hold most of your revenue and use that leverage at every renewal.

Suppliers' bargaining power is the strong force when — few firms can supply what you need, your costs rise or your quality slips on their schedule rather than yours, and nobody in your company can state what the firm buys in total.

Substitutes are the strong force when — price cuts stop working because the customers you lost are not shopping in your industry at all, and the whole category shrinks rather than just your share of it.

The confusion that costs the most is rivalry mistaken for new entrants. Both look like a competitor taking share, but rivalry says the industry is crowded and you must give buyers a reason beyond price,

while an entrant says your barriers to entry are low and the answer is to build something a newcomer cannot assemble in a quarter.

The second confusion is a rival mistaken for a substitute. A rival sells what you sell, so a better price or a better shelf can win the customer back; a substitute meets the same need from another industry, so the departing customer never compares the two prices and your discount reaches nobody.

SORT

Which force does this situation signal?

Each situation below is a generic, hypothetical example written for practice, with no real company named; place it under the competitive force it points to, then read why it belongs there. Every force is used twice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Traditional rivals within your industry — Pressure from sellers already established in your industry, fought on price, distribution, and service

- Three chains that have sold the same products in the same city for thirty years now copy each other's advertised prices within days, and none of them gains volume. — The pressure comes from sellers already established inside the industry, and it is being fought on price, which is exactly the implication the table lists for traditional rivals.
- A delivery firm finds that its two oldest competitors have both extended free next-day service into the same counties it serves, and margins fall for all three. — Established competitors matching one another on service and distribution is rivalry within the industry, not the arrival of anyone new.

Threat of new entrants into your market — Firms that were not competing here before, adding capacity and cutting prices and share

- A company that has never sold in this category announces it will begin next quarter, and the permits, equipment, and web storefront it needs turn out to be inexpensive and quick to obtain. — A firm from outside the industry is entering because barriers to entry are low, which adds capacity and pushes prices down for everyone already there.
- Two startups launch this year selling what an established manufacturer sells, renting their computing and warehousing instead of building either one. — Renting critical resources is the reduced difficulty of obtaining them that the table attributes to the internet, and it is what makes these firms entrants rather than long-standing rivals.

Customers' bargaining power — Buyers forcing lower prices, higher quality, and more services because leaving costs them little

- Buyers arrive at the counter having already compared the price against four alternatives on their phones, and they say so before asking for a discount. — Wider choice and easy comparison lower switching costs and raise customers' bargaining power, which shows up first as pressure on price.
- Four large accounts provide most of a supplier's revenue, and each renewal brings a demand for a lower rate plus added services at no charge. — Concentrated buyers can insist on reduced prices, increased quality, and more services, which is the implication the table lists for customers' bargaining power.

Suppliers' bargaining power — The firms you buy from setting terms, which reaches you as higher costs and lower quality

- Only two firms in the world make the component a product depends on, and both raised prices this year while lengthening delivery times. — Few alternatives upstream let suppliers dictate terms, and the result reaches the firm as increased costs and reduced quality in what it sells.
- Every branch of a chain phones in its own orders, so nobody at headquarters can say what the company buys in total or what it pays. — A firm that cannot see its own purchasing has no comparison and no leverage, which leaves supplier bargaining power unopposed even when alternatives exist.

Threat of substitute products or services from other industries — The same need met from outside your industry entirely, so the category itself shrinks

- A tool retailer's customers have stopped buying and now rent what they need from a service that belongs to no part of the retail tool industry. — The same need is met from another industry, so the customer never enters the tool retailer's market and a lower shelf price would reach nobody.
- A photo-printing firm sees returns rise and share fall as customers keep their pictures on screens instead, an option no printing company sells. — New substitutes created by information technologies produce exactly the table's implications: product returns, decreased market share, and customers lost for life.

From the force to the technology, in that order

The chapter is explicit about what the model is for: Porter's five-forces model of competition can help you determine which forces may be most important and which specific technologies could be used to address those forces, and you then use that knowledge as the basis for identifying particular investments. The sequence is the lesson — force first, technology second, investment third.

Each force therefore has a family of systems that answers it, so naming the force narrows the shopping list before anyone opens a vendor website.

Against traditional rivals — systems that move the contest off price, such as accurate stock information so the advertised item is genuinely on the shelf, because when competition focuses more on price, a firm that competes only on price keeps nothing it wins.

Against new entrants — systems that build what a newcomer cannot simply buy, such as accumulated customer history, saved orders, and reordering that runs on its own, because the internet already lowered the barriers that permits and capital used to provide.

Against customers' bargaining power — systems that give a customer something to lose by leaving, such as a loyalty record and offers built from that household's own purchases, because wider choice has already driven switching costs down.

Against suppliers' bargaining power — systems that let a firm see and consolidate its own buying and compare qualified alternatives, such as one purchasing record covering every location, because equal access to suppliers only helps a company able to act on it.

Against substitutes — systems that extend your offer into the substitute's territory, such as online ordering, dependable delivery windows, and subscriptions serving the same need the new way, because the table describes a customer lost this way as a customer lost for life.

DECIDE

Naming the force before naming the technology

Work through four decisions for this hypothetical company, and after each one read every outcome before moving on.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

This is a hypothetical practice situation, built only to give the analysis something to work on. A regional chain of bicycle shops is losing ground, leadership wants to spend on technology, and nobody has yet said what problem the spending is meant to solve. You are the analyst in the room, and your job in each decision is to name the force before anyone names a product.

Decision 1. Two chains that have sold bicycles in this region for thirty years now match every promotion the company runs within a week. Unit sales are flat across all three, and margins are falling for all three.

- Traditional rivals within the industry, because established sellers are competing on price and service for the same customers — **the stronger call**. Right. Rivalry among competing sellers shows up as competition in price, product distribution, and service, and matched promotions with flat volume is the clean signal. Naming it this way rules out any initiative that would only cut price further.
- Threat of new entrants, because competitors are taking sales the company used to make — Losing sales to a competitor does not by itself make that competitor an entrant. These two chains have been in the industry for thirty years, and an entrant is a firm that was not competing here before. The distinction changes the response: rivalry pushes you to compete on something other than price, while an entrant tells you your barriers to entry are low.
- Customers' bargaining power, because customers are choosing whichever store is cheapest that week — Buyer power is real in almost every retail market, so this is a reasonable instinct. What makes it the wrong call here is that the observable behavior belongs to the competitors: they are matching each promotion. Buyer power would be the answer if the evidence were customers negotiating or leaving over small price differences with no competitor action described.

Decision 2. Separately, a national online retailer that has never sold bicycles announces regional delivery. It owns no stores, rents its warehouse space, and needed only a contract and a website to begin.

- Traditional rivals, because it is one more company selling bicycles in this market — It will indeed become a rival, but naming it that way loses the fact worth acting on. What the situation reveals is how little it cost this company to start, and that is a statement about barriers to entry rather than about crowded competition.
- Threat of new entrants, because a firm from outside the industry started selling here with very little standing in its way — **the stronger call**. Right, and the evidence is the giveaway: no stores, rented warehousing, a website. The chapter attributes exactly this to the internet, which reduces the difficulty of obtaining critical resources or entering new markets, and the consequence for incumbents is increased capacity, reduced prices, and decreased market share.
- Suppliers' bargaining power, because the new retailer will compete for the same manufacturers' output — Supplier power is about the terms the firms you buy from can impose on you, such as higher prices or longer lead times. A second buyer in the market can eventually tighten supply, but nothing in this situation describes a supplier changing its terms, so the force being signaled is entry.

Decision 3. Only two manufacturers make the frames the chain sells most. Both raised prices this year and lengthened lead times, and each shop still phones in its own orders, so no one can say what the chain buys in total.

- Customers' bargaining power, because the price increase will have to be passed to shoppers — Passing on a cost is a consequence, not a diagnosis. Buyer power would be the finding if customers were forcing terms on you; here the terms are being set on the other side of the business, before a customer is involved at all.
- Suppliers' bargaining power, because few firms supply what the chain needs and the chain cannot see its own buying — **the stronger call**. Right on both halves. Few alternatives upstream produce the table's implication of increased costs and reduced quality, and a chain that cannot state its own volumes has no comparison to negotiate with. The initiative follows from the finding: one purchasing record across every shop, plus qualified alternate manufacturers.
- Traditional rivals, because competitors buy from the same two manufacturers and face the same increase — Shared exposure is a genuine observation and a useful one, since a rival with the same problem will not undercut you easily. It still does not name the pressure, which is coming from upstream rather than across the street.

Decision 4. Finally, a pattern nobody expected: households that once bought a second bicycle are buying electric scooters and using ride-hailing apps instead. Discounts on bicycles have stopped producing visits, and the customers who leave do not come back for a sale.

- Threat of substitute products or services from other industries, because the same need is being met outside the bicycle industry entirely — **the stronger call**. Right, and the two symptoms confirm it: discounts stopped working, and departing customers do not return. The chapter's implication for this force is product returns, decreased market share, and losing customers for life, and it also notes that new substitutes are created by the internet and other information technologies. The response has to extend the offer rather than defend the old one.
- Traditional rivals, because some other company is clearly getting the money these households used to spend — Money is leaving, but not to a bicycle seller. A rival sells what you sell, which is why a better price can win the customer back. That is precisely what stopped working here, and that failure is the clue.
- Customers' bargaining power, because customers are refusing to buy at the current price — It looks like price resistance, and price is where the symptom appears. The tell against it is that lower prices produce no visits at all: a buyer with strong bargaining power is still buying the product, just on better terms, while these households have left the category.

Turning a finding into something a decision maker can fund

A finding is not yet a proposal. Turning one into a recommendation an executive can approve takes four parts, and they have to arrive in this order.

- 1 **Name the force** exactly, using one of the chapter's five names, because a recommendation that opens with a technology has skipped the diagnosis and cannot be checked against anything.

- 2 **Give the observable evidence** that this force is the strong one, such as promotions matched within a week or a sole supplier lengthening lead times, because evidence is what keeps a meeting from being two opinions.
- 3 **State the initiative concretely** — not “improve our digital capabilities” but “one purchasing record covering all fourteen locations, with three qualified alternate suppliers” — because only a concrete initiative can be costed and built.
- 4 **Commit to a measure and a date**, naming the number you expect to move and when you will check it, because a system with no defined result can be defended forever and judged never.

Say it in one sentence. Because [force] is the strongest force on us, shown by [evidence], we should build [initiative], and we will judge it by [measure] on [date]. If any slot is empty, what you are holding is a preference rather than a recommendation.

EXPLORE

Vague recommendation, fundable recommendation

Open each card and read the two versions of the same proposal side by side, one force at a time; both versions are invented practice wording rather than quotations from any real company.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Answering established rivals · Traditional rivals within your industry

- **The vague version somebody actually says:** “We need to be more competitive, so we should invest in better technology for our stores.”
- **The same idea written to be funded:** “Rivalry is our strongest force: both long-standing competitors match our promotions within a week and volume is flat. We will put every location on one live stock record so advertised items are actually available, and we will judge it by out-of-stock rate on promoted items at the next quarterly review.”
- **What the rewrite added:** A named force, the evidence behind it, one concrete system rather than a category of spending, and a number with a date attached to it.
- **Why a decision maker cares about the difference:** The first sentence would justify almost any purchase, so nobody can later say whether the money worked. The second can be argued with before it is funded and judged after.

Answering a new entrant · Threat of new entrants into your market

- **The vague version somebody actually says:** “An online seller has entered our market, so we need a digital transformation strategy.”

- **The same idea written to be funded:** "New entrants are our strongest force: a company with no stores began selling here this quarter, which means our barriers to entry are low. We will build purchase history and automatic reordering for regular customers, judged by the share of revenue from repeat accounts at the end of two quarters."
- **What the rewrite added:** The reason the entrant matters, which is low barriers to entry rather than one more competitor, plus a system that accumulates something a newcomer cannot rent.
- **Why a decision maker cares about the difference:** Digital transformation names no force and no measure, so it cannot be compared against any other proposal competing for the same money.

Answering customer bargaining power · Customers' bargaining power

- **The vague version somebody actually says:** "Customers keep asking for discounts, so we should launch a loyalty app like everyone else has."
- **The same idea written to be funded:** "Customer bargaining power is our strongest force: switching costs are near zero and we cannot identify a repeat buyer. We will record purchases by household and make offers from that history, judged by repeat-visit rate and average basket at the six-month review."
- **What the rewrite added:** The evidence that buyer power is high, the reason the system helps, which is raising the cost of leaving, and the two numbers that will show whether it did.
- **Why a decision maker cares about the difference:** Copying a competitor argues for parity and never for advantage, and it hands the technology plan to whichever rival moved first.

Answering supplier bargaining power · Suppliers' bargaining power

- **The vague version somebody actually says:** "Our costs are up. We should renegotiate with suppliers and maybe get some new software for purchasing."
- **The same idea written to be funded:** "Supplier bargaining power is our strongest force: two manufacturers supply most of what we sell, prices rose, and no one can state our total volume. We will consolidate purchasing into one record across all locations and qualify three alternates, judged by average unit cost and lead time at the next quarterly review."
- **What the rewrite added:** The specific weakness giving suppliers their leverage, the two-part initiative that removes it, and a measure covering both cost and delivery time.
- **Why a decision maker cares about the difference:** You cannot renegotiate from a position you cannot describe, so the vague version asks for a better outcome without changing anything that produced the current one.

Answering a substitute · Threat of substitute products or services from other industries

- **The vague version somebody actually says:** "Sales are down across the category, so we should cut prices and advertise harder until it recovers."

- **The same idea written to be funded:** “Substitutes are our strongest force: discounts have stopped producing visits and departing customers do not return, because they are meeting the same need outside our industry. We will offer subscription servicing and same-week delivery so the need can be met through us, judged by subscription sign-ups and repeat service visits after two quarters.”
- **What the rewrite added:** The recognition that the lost customer is not comparing prices at all, and an initiative aimed at the need rather than at the old product.
- **Why a decision maker cares about the difference:** Discounting against a substitute spends margin on people who have already left, which is why this force is the one most often answered with the wrong instrument.

Two limits worth saying out loud. The analysis tells you where to aim, and it does not do more than that:

It names conditions, not products — the model identifies which forces matter most and which kinds of technology could address them, so a vendor shortlist assembled before the force is named is a set of products hunting for a problem.

The answer expires — the internet altered all five forces at once, so any diagnosis is a photograph of an industry on one particular day, and a firm that files it away ends up defending a position that has already moved.

CHECK YOURSELF

The internet, the order of analysis, and the proposal

Three questions on what the internet did to the forces and on how the analysis is meant to be used; the recommendations quoted in the last question are invented practice examples, not real proposals.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Table 2.3 records how the internet influenced each competitive force. Which entry belongs in the row for customers’ bargaining power?

- **A.** Reduced barriers to entry, as the internet reduces the difficulty of obtaining critical resources or entering new markets — Reduced barriers to entry is the entry recorded for the threat of new entrants, because it describes how easily an outsider can now assemble what it needs to start competing.
- **B.** Wider choices for customers lead to lower switching costs and higher bargaining power of customers — **correct.** More choice lowers what it costs a customer to switch, and a customer who can leave cheaply can insist on lower prices, higher quality, and more services.

- **C.** Companies have equal access to suppliers, it is easier to find new suppliers, and suppliers reach more potential buyers — Equal access to suppliers and suppliers reaching more buyers is the entry for suppliers' bargaining power, and it is the one row where the table describes influence flowing in both directions.
- **D.** New substitutes are created by the internet and other information technologies — New substitutes created by information technologies is the entry for the threat of substitute products or services from other industries, which is about needs met outside your industry rather than terms demanded inside it.

2. A project team opens a meeting by proposing a specific new system, then looks for reasons the company needs it. What does the chapter's use of the five-forces model say about that order?

- **A.** It is efficient, because starting from a concrete system keeps the discussion practical rather than theoretical — Concreteness is worth something, but a concrete answer to an unnamed question cannot be evaluated. The practical version of this meeting states the force and the evidence first, and it stays just as concrete afterwards.
- **B.** It is acceptable when the proposed system is one competitors already run, because a system already working in this industry has effectively been diagnosed for you — Matching competitors argues for parity rather than advantage, and it says nothing about which force is actually pressing on this firm. A rival's purchase is evidence about the rival's situation, not about yours.
- **C.** It reverses the sequence: the model determines which forces matter most and which technologies address them, and that knowledge then identifies the investment — **correct.** The model helps determine which forces may be most important and which specific technologies could be used to address those forces, and that knowledge is then the basis for identifying particular investments.
- **D.** It does not matter much, because the five forces describe conditions outside the firm while a technology decision is made inside it — The forces describe pressure from outside the firm, and that is exactly why they drive internal decisions: analyzing them is how an organization decides where to focus its limited resources.

3. Which of these is a recommendation a decision maker could actually fund and later judge?

- **A.** "Buyer power is our strongest force, shown by shoppers who have compared our price before they reach the counter, so we should modernize the customer experience across every channel and review our progress regularly through the coming year" — This one names a force and offers evidence for it, which already puts it ahead of most proposals. Modernizing the customer experience across every channel is a category of spending rather than an initiative, and reviewing progress regularly names neither a number nor a date.

- **B.** "Our three largest competitors have all installed new store systems in the past year and customers have noticed the difference, so we should adopt comparable systems this year rather than keep falling behind them on service" — This names no force at all, and its evidence is about what rivals have bought rather than about what is actually pressing on this firm. It justifies parity with competitors, and a plan built from rivals' purchases can never do better than match them.
- **C.** "Supplier power is our strongest force, shown by two manufacturers raising prices while no one here can state our total volume; we will consolidate purchasing into one record and qualify three alternate suppliers, judged by unit cost and lead time at the next quarterly review" — **correct.** It names the force, gives observable evidence, states a concrete two-part initiative, and commits to specific measures with a review date, so it can be argued with before funding and judged afterwards.
- **D.** "Substitutes are our strongest force, shown by discounts that no longer bring departed customers back to us, so we need to become a more innovative and data-driven organization able to respond to whatever the market does next" — The force and the evidence are right, and failed discounts really are the signal for substitutes, so the diagnosis half of this proposal is sound. Innovative and data-driven then describes a wish rather than a system, so nothing here could be costed, built, or checked afterwards.

How to compete: strategy, resources, and the value chain

The five forces tell a company where the pressure in its industry comes from. They never say what to do about it. This section is the other half: once a company knows what it is up against, how does it choose what to be good at, and which systems are worth the money?

Choosing a generic strategy

To achieve superior returns, a company has to position itself within the industry it operates in so that it can counter the effects of the five forces. Figure 2.12 lays the common positions out as a grid built on two questions: is the differentiating attribute **low cost** or **uniqueness**, and is the market focus **narrow** or **broad**?

Three named strategies sit inside that grid, and each is a promise the company's later spending has to keep.

Low-cost leadership strategy — the organization offers the best prices in its industry on its goods and/or services, which is how the chapter describes Walmart.

Differentiation strategy — the organization tries to provide better products or services than its competitors, as Porsche, Nordstrom, and IBM do, so the customer is paying for something rivals have not matched.

Best-cost provider strategy — a middle-of-the-road position offering products or services of reasonably good quality at competitive prices, as Dell does, placed at the center of the figure where the quadrants overlap.

Differentiation comes in two widths: a company might aim it broadly at many types of consumers, or focus on one segment, as Apple did for years with high-quality computers for home and educational markets. Crossing that choice with the two axes produces the five general types the figure names.

TYPE OF STRATEGY	WHERE IT SITS IN FIGURE 2.12	WHAT THE CUSTOMER IS PROMISED
Overall low-cost leadership	Broad market, low cost	The best prices in the industry, for everyone
Broad differentiation	Broad market, uniqueness	Better products than competitors, for many types of consumers

TYPE OF STRATEGY	WHERE IT SITS IN FIGURE 2.12	WHAT THE CUSTOMER IS PROMISED
Focused low-cost leadership	Narrow market, low cost	The lowest price inside one slice of the market
Focused differentiation	Narrow market, uniqueness	Something better, built for one segment
Best-cost provider	The center, where the quadrants overlap	Reasonably good quality at competitive prices

MATCH

Generic strategies and the situations that suit them

Each situation below is a generic, hypothetical one written for practice, with no real company named. Pair it with the general type of strategy in Figure 2.12 that fits it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Overall low-cost leadership

A company selling everyday goods to nearly every kind of buyer, whose customers choose a store mainly on price A low-cost leadership strategy means offering the best prices in the industry on goods and/or services, and the chapter names Walmart as the example. The market is broad and the differentiating attribute is cost.

Broad differentiation

A company whose buyers come from many different segments and who pay for products or services that are better than the alternatives A differentiation strategy means trying to provide better products or services than competitors, as Porsche, Nordstrom, and IBM do, and here the differentiation is aimed broadly at many different types of consumers.

Focused differentiation

A company that deliberately serves one segment of buyers and gives that segment higher quality than anyone else offers it The chapter describes exactly this width of differentiation: Apple focused for many years on high-quality computers for the home and educational markets rather than aiming at every buyer.

Focused low-cost leadership

A company competing on price, but only inside one narrow slice of the market rather than across the whole industry Figure 2.12 names this as one of the five general types, in the narrow-market and low-cost quadrant. The chapter names no company for it, so you have to recognize it by its position rather than by an example.

Best-cost provider strategy

A company offering reasonably good quality at competitive prices, sitting on purpose between the price players and the premium ones This is the middle-of-the-road strategy the

chapter describes, with Dell as the example, and the figure places it at the center where all the quadrants overlap.

A chosen middle is not the same as no middle. The best-cost provider position is a real strategy with a real promise: reasonably good quality at competitive prices, held steady.

A company that has never said whether it competes on price or on uniqueness, and for whom, has no test to apply to the next proposal, so the budget goes to whoever argues best.

Resources and capabilities: the second lens

Position tells the market what to expect; it does not make the company able to deliver. Whichever generic strategy an organization pursues, it must have resources and/or capabilities superior to those of its competitors in order to gain or sustain a competitive advantage.

Resources — the organization's specific assets used to achieve cost or product differentiation from competitors, such as proprietary technology, brand equity, or a loyal and established customer base.

Capabilities — the organization's ability to leverage those resources in the marketplace, such as design quality or efficient operations, so a capability is something a company does rather than something it owns.

Distinctive competencies — what resources and capabilities produce together, such as innovation, agility, quality, or low cost, and they are what makes the product valuable to customers relative to competitors.

Superior value creation — what the chain is aiming at, which occurs when an organization can provide products at a lower cost or with superior, differentiated benefits to the customer.

Figure 2.13 draws that as a chain: resources and capabilities feed distinctive competencies, competencies produce a cost or differentiation advantage, and that advantage earns the name sustained competitive advantage. An asset nobody uses well never becomes a competency.

What separates a lasting advantage from a temporary one is comparative: the resources and capabilities have to be superior to those of competitors, not merely present. Anything a rival can order this afternoon fails that test.

Figure 2.14 gives eight sources of competitive advantage, one company each.

SOURCE OF ADVANTAGE	COMPANY THE CHAPTER NAMES
Quality	Toyota
Service	IBM
User base	Facebook
Proprietary	Coca-Cola
Innovation	Apple
Brand	Nike
Value	Nintendo
Data	Uber

Several of those sources are informational rather than physical. For many organizations, the chapter says, information systems are at the heart of the different resources, capabilities, or distinctive competencies.

CHECK YOURSELF

Strategy, resources, and capabilities

Four options, one best answer; read every explanation, including the ones for the answers you did not pick. Any company described in a question is hypothetical unless the chapter names it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Which description matches the best-cost provider strategy as the chapter presents it?

- **A.** Offering the best prices in the industry on goods and/or services, the way the chapter describes Walmart — That is the low-cost leadership strategy, whose whole promise is price. It occupies the low-cost end of the horizontal axis rather than the center of the figure, and the chapter names Walmart for it.
- **B.** Providing products or services of reasonably good quality at competitive prices, the way the chapter describes Dell — **correct.** The chapter calls this a middle-of-the-road strategy, names Dell as the example, and Figure 2.12 places the best-cost provider at the center where all four quadrants overlap.

- **C.** Providing better products or services than competitors and aiming them at one segment of consumers, as Apple did in home and educational computers — That is focused differentiation: the differentiating attribute is uniqueness and the market focus is narrow. Quality is aimed at one segment rather than balanced against price for everybody.
- **D.** Providing better products or services than competitors and aiming them broadly at many different types of consumers rather than at one segment — That is broad differentiation. It shares the uniqueness end of the axis with focused differentiation and differs only in how wide the market is, so neither one describes a company balancing quality against price. The chapter names Porsche, Nordstrom, and IBM for differentiation without saying which width each pursues.

2. Which of these is a capability rather than a resource, as the chapter defines the two words?

- **A.** Proprietary technology the company owns and no competitor can use — Proprietary technology is one of the chapter's own examples of a resource: a specific asset the organization holds and uses to achieve cost or product differentiation.
- **B.** Brand equity built up with customers over many years — Brand equity is also listed as a resource. It is something the organization possesses, and possessing it says nothing yet about how well the organization puts it to work.
- **C.** Efficient operations, meaning the company runs its processes better than others do — **correct.** Capabilities reflect the organization's ability to leverage its resources in the marketplace, and the chapter offers design quality and efficient operations as the examples.
- **D.** A loyal and established customer base that keeps returning — A loyal and established customer base is the third resource the chapter lists. Turning that base into repeat revenue would be a capability; having the base is an asset.

3. A company holds years of purchase data and a well-known brand, but no department can agree on how to use either, and nothing the company does with them is better than what rivals do. What does the chapter's framework say about its position?

- **A.** It already has a sustained competitive advantage, because owning data and a brand is what the chapter means by resources — Resources alone do not finish the chain. The organization must have resources and/or capabilities superior to those of its competitors, and superiority is judged by what reaches the market, not by what sits in the building.
- **B.** It has resources but not the capabilities to leverage them, so no distinctive competency and no advantage follows yet — **correct.** Capabilities are the ability to leverage resources in the marketplace; without them the resources never become distinctive competencies, and Figure 2.13 runs from competencies to a cost or differentiation advantage to a sustained advantage.
- **C.** It has capabilities but no resources, since knowing about customers counts as ability rather than as an asset — This reverses the two definitions. Data and a brand are assets the organization holds, which makes them resources; the ability to use them well would be the capability that is missing here.

- **D.** It has achieved superior value creation, because customers can recognize the brand more easily than they recognize rivals — Superior value creation occurs when an organization can provide products at a lower cost or with superior, differentiated benefits to the customer. Recognition without a better cost or a better benefit does not meet that definition.

Analyzing the value chain

To find where inside the company a system would actually pay, managers use value chain analysis. Think of an organization as a big input/output process: supplies are purchased and brought in, integrated into products and services it markets, sells, and distributes, with customer service after the sale.

Throughout that process, opportunities arise for employees to use information systems to add value by acquiring supplies more effectively, improving products, and selling more products. That set of activities is the **value chain**. Value has a plain meaning here: the analysis asks what each activity adds for the customer and what it costs to do.

Value chain analysis is the process of analyzing an organization's activities to determine where value is added to products and/or services and what costs are incurred for doing so. It runs in three steps.

- 1 **Draw the chain first** — flesh out each activity, function, and process where value is or should be added and performance can be improved, because nobody improves a step that was never written down.
- 2 **Then find the costs** — determine the costs in each area and the factors that drive them or make them fluctuate, because a cost with no known driver can be complained about but not managed.
- 3 **Then decide what to optimize** — determine which activities need optimizing to improve performance, cut costs, and gain or sustain competitive advantage, which is the first point at which naming a system is reasonable.

The chain has two halves. The **primary activities** run left to right as a flow, because a product passes through them on its way to a customer. The **support activities** are bands drawn underneath, because they serve every stage rather than one.

ACTIVITY	HALF OF THE CHAIN	WHAT THE NAME COVERS
Inbound logistics	Primary	Receiving what was purchased and getting it inside, before any of it is sellable
Operations	Primary	Integrating supplies into the products and services the customer buys

ACTIVITY	HALF OF THE CHAIN	WHAT THE NAME COVERS
Outbound logistics	Primary	Distributing the finished product or service out to the customer
Sales and marketing	Primary	Making customers aware and giving them a reason to choose it
Service	Primary	Looking after the customer after the sale
Administration and firm infrastructure	Support	Running the organization as a whole, billable to no single product
Human resources	Support	Who is hired, trained, and scheduled to do the work above
Product research and development, technology, and systems development	Support	Improving the product and building the systems the chain runs on
Procurement	Support	The buying decision itself, not the physical receiving of goods

The model was built for manufacturing but applies to service industries too, sometimes with different activities. The chapter walks through a hotel, and that mapping repays slow reading.

Inbound logistics — receiving supplies, and handling reservations, because a booking is an input that arrives and must be held until it is used.

Operations — checking guests in or out, cleaning rooms, and preparing breakfast, since a clean, ready, served room is the product itself.

Outbound logistics — work the chapter says might be performed by the hotel's catering or event service, because that is where the finished offering reaches the guest.

Sales and marketing — promoting the hotel and attracting business meetings and conventions, so rooms are booked before they can be cleaned.

Service — the activities performed after the guest has left, such as registering or resolving guest complaints.

Other organizations may lack inbound and/or outbound logistics altogether and may classify activities differently, while the supporting activities are likely to resemble those of other organizations. The chain is a tool for analysis, not a law.

INTERACTIVE DIAGRAM

The value chain, one half at a time

Step through the primary activities, then the support activities, then the places information systems add value along the chain.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Primary activities — The flow a product travels, from arriving supplies to the customer after the sale

Supplies come in (Inbound logistics) → Inputs become the product (Operations) → The product goes out and is sold (Outbound logistics, sales and marketing) → The buyer is looked after (Service)

- Think of the organization as a big input/output process: supplies are purchased and brought in, integrated into products and services, marketed, sold, and distributed to customers, with customer service provided after the sale.
- The five primary activities the chapter names in Figure 2.15 are **inbound logistics**, **operations**, **outbound logistics**, **sales and marketing**, and **service**, drawn in that order and connected by arrows.
- The model was initially created to focus on manufacturing, but it also applies to service industries, sometimes with different activities being performed — in a hotel, inbound logistics covers receiving supplies and handling reservations, while operations covers checking guests in or out, cleaning rooms, and preparing breakfast.
- Some organizations may lack inbound and/or outbound logistics entirely, and different activities may be classified differently, so the chain is a tool for identifying and analyzing what a particular organization actually does.

Support activities — The bands drawn underneath the whole flow, because they serve every stage rather than one

Administration and firm infrastructure (Running the organization as a whole) → Human resources (Hiring, training, and scheduling people) → Product R and D, technology, systems development (Improving the product and building the systems) → Procurement (Choosing suppliers and terms)

- Figure 2.15 draws these as layers beneath the primary flow rather than as steps inside it — administration and firm infrastructure at the top, then human resources, then product research and development, technology, and systems development, then procurement — which is the visual way of saying that their payoff appears spread across all five primary activities instead of in one of them.
- Whereas the primary activities may differ from one organization to another, the chapter notes that the supporting activities are likely to resemble those of other organizations.
- **Procurement** is the decision about whom to buy from and on what terms; **inbound logistics** is what happens to the goods after that decision is made. Filing one under the other sends a recommendation to the wrong department.
- The figure shows these bands supported by a private cloud and computer-aided design systems, and names a financial decision support system, an employee self-service portal, and a supply chain management system alongside them.

Where information systems add value — The analysis that decides which system to fund, and the systems the figure places along the chain

Draw the chain (Every activity where value is or should be added) → Find the costs (And the factors that drive or move them) → Choose what to optimize (Performance, cost, advantage) → Name the system by its activity (Not by the technology it uses)

- Because information systems can automate and optimize many activities along the value chain, using them has become one of the primary ways that organizations improve their value chains.
- Many organizations use the internet to connect businesses with one another electronically so that they can exchange orders, invoices, and receipts online in real time, which removes waiting from the seams between activities.
- Organizations use various social media such as blogs, X, or Instagram to connect with their customers, and Figure 2.15 shows supply chain management and customer relationship management systems, along with the website, Facebook, and X, laid across the primary flow.
- Value chain analysis is the process of analyzing an organization's activities to determine where value is added to products and/or services and what costs are incurred for doing so — which is why the costs come before the shopping.

The role of information systems in value chain analysis

Because information systems can automate and optimize many activities along the chain, using them has become one of the primary ways organizations improve their value chains. Figure 2.15 places ordinary systems on a sample chain.

Systems laid across the primary flow — supply chain management and customer relationship management systems run along the whole flow, with the website, Facebook, and X, because those touch goods coming in and customers going out.

Connections between businesses — many organizations use the internet to connect electronically, exchanging orders, invoices, and receipts in real time, which takes waiting out of the seams between activities.

Connections with customers — organizations use social media such as blogs, X, or Instagram to connect with customers, so marketing and service stop being one-way announcements.

Systems beneath the support bands — the figure shows a private cloud and computer-aided design systems supporting those bands, and names a financial decision support system, an employee self-service portal, and a supply chain management system alongside them.

Name a proposed system by the activity it improves. “A better app” cannot be argued about; “a system that cuts the cost driver in inbound logistics” can be costed, funded, or refused.

SORT

Which primary activity does this system improve?

Each proposed system below is hypothetical; place it in the primary activity it would strengthen, using the chapter’s own mapping of the value chain onto a hotel.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Inbound logistics — in a hotel, receiving supplies and handling reservations

- A reservation system that records each booking as it is made and holds the right room for the right night — The chapter puts handling reservations in inbound logistics for a hotel, alongside receiving supplies, because a booking is an input that arrives and has to be held until it is used.
- A delivery portal where linen and food suppliers confirm what will reach the loading dock tomorrow morning — Receiving supplies is the other half of inbound logistics in the chapter’s hotel example, and this system manages what arrives rather than who was chosen to send it.

Operations — in a hotel, checking guests in or out, cleaning rooms, preparing breakfast

- A housekeeping tablet that routes cleaners room by room and marks each room ready for the next guest — Cleaning rooms is named as a day-to-day operations activity, because a clean and ready room is part of the product the guest is buying.
- A check-in screen that issues a key card without the guest waiting in line at the front desk — Checking guests in or out is listed under operations for a hotel, since it is part of turning inputs into the service the customer receives.

Outbound logistics — in a hotel, work that might be performed by the catering or event service

- A catering scheduler that assigns staff, food, and equipment to each function the hotel's event service delivers — The chapter says outbound logistics activities in a hotel might be performed by the hotel's catering or event service, so a system that plans those functions belongs here rather than under room operations.
- A tracking system that follows each event order from the kitchen to the room where the function is being held — Outbound logistics is the half of the flow that gets the finished offering to the customer, and in the chapter's hotel example that work sits with the catering or event service.

Sales and marketing — in a hotel, promoting it and attracting business meetings and conventions

- A campaign tool that measures which advertisement brought more convention organizers to the booking page — Attracting business meetings and conventions is listed under sales and marketing for a hotel, and measuring what attracts them is part of the same activity.
- A pipeline tracker showing which large organizations are choosing a host city next year so the sales team can approach them — Promoting the hotel and pursuing group business is sales and marketing, because the work is creating awareness and a reason to choose this hotel.

Service — in a hotel, the activities after the guest has left, such as resolving complaints

- A complaint log that registers each issue a departed guest reports and tracks it until it is resolved — The chapter names registering or resolving guest complaints as customer service activities performed after the guest has left.
- A follow-up system that flags a guest whose last stay went badly so the next stay is handled with extra care — Looking after the customer after the sale is the service activity, and this system acts on a past stay rather than on the current one.

The technology/strategy fit

If a system does things faster and saves money, who cares whether it matches the strategy? Money and time are limited. If money grew on trees an organization would build every system it could imagine; in reality it acquires only those adding the most value, meaning the ones that help it automate and learn *and* carry strategic value.

Managers once treated information systems as a necessary service, a necessary evil, and a distasteful expense to be minimized. Successful managers now treat them as a competitive asset to be nurtured and invested in, an enabler of opportunities and a mechanism for executing the business model.

The test has a name, **business/IT alignment**, and organizations try to maximize it: in most cases they do not want systems that fail to match the strategy, even when those systems offer real automating and learning benefits.

The chapter's illustration is blunt. A firm whose strategy is overall low-cost leadership probably does not want a system that differentiates its products on high quality, because a low-cost leader values the investments that drive costs down over those that do not.

Two qualifications stop this from becoming a rule applied without thinking.

Strategic necessity — organizations sometimes have no choice about investments that may or may not coincide with their overall strategy, and the chapter calls such an investment a strategic necessity: something the organization must do to survive.

The change comes with the system — implementing a new or innovative system is not sufficient by itself, because any significant implementation needs commensurate, significant organizational change in processes, roles, culture, and mindset.

Underneath it all is a floor: an information system can be only as effective as the business model it serves, and bad business models cannot be overcome by good information systems.

SEQUENCE

From competitive forces to a funded system

Put the eight steps of a full analysis into the order the chapter builds them, from the industry outside to the money inside.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Each step narrows the question the next one answers. Run them out of order and you end up with a technology looking for a problem.

1. **Analyze the competitive forces in the industry to see where the pressure on profit is actually coming from.** — Given that every industry is different, organizations need to analyze the competitive forces within their industry to better understand where to focus their resources, so this comes before any choice about how to compete.
2. **Choose the position the company will compete from: low-cost leadership, differentiation, or the best-cost provider middle, broad or focused.** — To achieve superior returns, a company positions itself within its industry to counter the effects of the five forces, and that position is what everything afterwards has to serve.
3. **Take stock of the resources and capabilities the company holds, and ask whether they are superior to those of competitors.** — No matter which generic strategy is chosen, the organization must have resources and/or capabilities superior to those of its competitors in order to gain or sustain a competitive advantage.
4. **Draw the value chain, fleshing out each activity, function, and process where value is or should be added and where performance can be improved.** — The chapter makes drawing the chain the first move of value chain analysis, because an activity nobody has written down cannot be measured or improved.

5. **Determine the costs within each area of the diagram, along with the factors that drive those costs or cause them to fluctuate.** — Costing comes second in value chain analysis, and knowing the driver behind a cost is what separates a number you can act on from a number you can only report.
6. **Decide which activities need to be optimized to improve performance, cut costs, and ultimately gain or sustain competitive advantage.** — This is the third step of value chain analysis and the first moment at which naming a system is reasonable, because now the target activity is known.
7. **Test the candidate system against the strategy, keeping the ones that match and setting aside the ones that do not, unless the investment is a strategic necessity.** — Organizations try to maximize business/IT alignment and in most cases do not want systems that fail to match the strategy, though a strategic necessity is something the organization must do to survive regardless.
8. **Plan the organizational change that has to accompany the system, then take the proposal for funding with its value stated plainly.** — Merely implementing a system is not sufficient; there must be commensurate, significant organizational change, and most companies are willing to spend only when they can see clear, significant value.

Position first, then the resources and capabilities that make it deliverable, then the value chain activity where a system would pay, then the alignment test a proposal must survive to be funded.

CHECK YOURSELF

The value chain and the technology/strategy fit

Three questions on locating work in the value chain and on deciding whether a system deserves the money. The firms described are hypothetical; only the chapter's own examples name real companies.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A hotel is deciding where an employee self-service portal for scheduling and time off belongs on its value chain. Where does it sit?

- **A.** Operations, because scheduling decides who is available to clean rooms and serve breakfast — Operations covers the day-to-day activities of checking guests in or out, cleaning rooms, and preparing breakfast. Scheduling touches operations, but it touches the other primary activities just as much, which is the signature of a support activity.
- **B.** Human resources, a support activity, because it serves the staffing of every stage rather than one stage — **correct.** Human resources is one of the support bands Figure 2.15 draws beneath the whole flow, and the figure lists an employee self-service portal among the systems it places on that chain.

- **C.** Inbound logistics, because staff time is one of the inputs the hotel receives and consumes — Inbound logistics is about goods and, in a hotel, reservations arriving and being handled. Treating employees as arriving inventory stretches the definition past the one the chapter uses.
- **D.** Service, because the portal exists to serve the people who work at the hotel — Service in the value chain means looking after the paying customer after the sale, such as registering or resolving guest complaints. The word service in ordinary speech is broader than the activity the model names.

2. A firm pursuing overall low-cost leadership is offered a system that would help it differentiate its products on high quality, and the system genuinely automates work and produces useful learning. What does the chapter say the firm should conclude?

- **A.** Fund it, because a system that automates work and produces learning has already justified itself — Automating and learning benefits are real but not sufficient. Organizations are constrained by time and money to acquire only the systems that add the most value, meaning those that automate and learn and also carry strategic value.
- **B.** Fund it, because a company that can differentiate on quality should abandon a low-cost position — Nothing in the situation says the low-cost position was wrong. A single system offer is a weak reason to abandon a position the company has aligned its costs, suppliers, and customer promise around.
- **C.** Probably decline it, because organizations do not want systems that fail to match the strategy even when they offer automating and learning benefits — **correct.** The chapter says an organization probably does not want a system that differentiates products on high quality when its strategy is overall low-cost leadership, because a low-cost leader values the investments that drive costs down.
- **D.** Declare it a strategic necessity, because any system a competitor might also buy is something the firm must have — A strategic necessity is something the organization must do to survive, not a label for any purchase a competitor might also make. Applying it this loosely would let it excuse every proposal on the table.

3. Which statement about implementing a new information system matches the chapter?

- **A.** A well-chosen system can carry a weak business model, which is why the system decision matters more than the model — The chapter states the reverse: an information system can be only as effective as the business model that it serves, and bad business models cannot be overcome by good information systems.
- **B.** Implementing the system is the hard part, so processes, roles, and culture can be left as they are afterwards — This inverts the caution. Merely choosing and implementing new or innovative systems is not sufficient to gain or sustain competitive advantage, and the change in processes, roles, culture, and mindset is part of the work rather than an afterthought.
- **C.** Any significant implementation needs commensurate organizational change, and the system can be only as effective as the business model it serves — **correct.** Both cautions are stated: significant implementations require commensurate, significant organizational change, and a system can be only as effective as the business model it serves.

- **D.** Change the processes, roles, and culture only after the system has run long enough to show where it falls short, since reorganizing first would be premature — The chapter puts the change alongside the implementation rather than after it: in any significant implementation there must be commensurate, significant organizational change, and waiting leaves the system dropped into an organization that still works the old way.

Business models in the digital world

Before anyone can argue about which technology a company should buy, the company has to be able to say plainly how it makes its living: who it serves, what it offers them, how the offer reaches them, and where the money comes from and where it goes. That plain statement is the **business model**, and technology investments have to be aligned with it.

What a business model actually is

A **business model** is a summary of a business's strategic direction that outlines how the objectives will be achieved. It specifies the **value proposition** as well as how a company will create, deliver, and capture value, and it identifies nine things: customer segments, value propositions, channels, customer relationships, revenue streams, key resources, key activities, key partners, and cost structure.

Three verbs sit inside that definition, and it is worth slowing down on them, because a company can be good at one and fail at another.

Create value — make something somebody genuinely wants, because a repaired bicycle, an hour of streaming, or a ride across town is worth something to the person receiving it and nothing at all if no one wants it.

Deliver value — get that something into the customer's hands, because a repair nobody can reach and an app nobody can install create value that never arrives.

Capture value — keep a share of it as income, because a company that creates and delivers value but never captures any of it runs out of money while its customers are still delighted.

The chapter compresses the whole idea into five questions. Walk them in this order, because each answer narrows the next one.

- 1 What does a company do?** Name the product or service in words a stranger would understand, because everything below is meaningless until this is settled.
- 2 How does a company uniquely do it?** Say what makes this offer different from the one next door, because a model that could describe any competitor is not a strategy.
- 3 What are the key resources and activities needed?** List the assets you must have and the work you must perform, because promises you cannot staff or equip are not promises.

- 4 **In what way, or ways, does the company get paid for doing it?** Say who hands over money and when, because the customer and the payer are not always the same person.
- 5 **What are the costs involved?** Add up what running all of the above consumes, because a model that earns well and spends more is still a failing model.

Those answers are not an academic exercise. How a company answers these questions dictates how and where information systems investments can be utilized to execute a competitive strategy and sustain an advantage over competitors. Generally speaking, anything that has to do with the day-to-day functioning of the organization is part of its business model, and information systems can be used to support and execute many aspects of it.

CHECK YOURSELF

What a business model is for

Four options, one best answer; read every explanation, including the ones for options you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Which statement best describes what a business model is?

- **A.** A summary of a business's strategic direction that outlines how objectives will be achieved, specifying the value proposition and how the company will create, deliver, and capture value — **correct.** That is the chapter's definition, and the three verbs matter: creating value, delivering it, and capturing some of it are separate achievements that a company can succeed and fail at independently.
- **B.** The set of information systems a company has purchased or built, listed with the vendor, the annual cost, and the department that uses each one — That is an inventory of technology, and it answers none of the five questions. The chapter's point runs the other way: how a company answers the business model questions is what dictates where information systems investments can usefully go.
- **C.** A forecast of the revenue and profit the company expects to report over the coming financial year, broken down by product line — A revenue forecast is a number the model produces, not the model itself. Two companies could forecast identical revenue while serving different customers through different channels at wildly different costs.
- **D.** A chart showing who reports to whom, so that responsibility for each part of the business can be traced to a named manager — An organization chart shows reporting lines and responsibility, which is genuine and useful information about a company's structure. It says nothing about who the customers are, why they would choose this firm, or how the firm gets paid.

2. Someone says: "Our business model is simple — we sell coffee." What is missing from that answer?

- **A.** Nothing is missing, because naming the product answers the question of what the company does, and the other components follow from the product itself — Naming the product does answer the first question — what does the company do — and that is a real start. The rest does not follow from the product, though: two shops selling the identical cup can differ on segments, channels, relationships, and costs.
- **B.** Only the price, since a product and a price together already state what is sold and how the firm gets paid for it — Price is part of the revenue streams component, so this is closer than it looks, but a price is not the same as knowing in what ways the firm gets paid. It also leaves the segments, channels, relationships, resources, activities, partners, and costs entirely unstated.
- **C.** Everything after the first question: how the company uniquely does it, what key resources and activities are needed, in what ways it gets paid, and what the costs are — **correct.** The five questions ask what the company does, how it uniquely does it, what key resources and activities are needed, in what ways it gets paid, and what the costs are, and a one-line product description answers only the first.
- **D.** Only the technology, because a business model in the digital world is mainly a description of the systems the company runs on — Technology is not itself a component of the business model; it appears through the resources, activities, and channels that the model needs. A model written as a list of systems has skipped the question of who is being served and why.

3. Why does the chapter insist that a manager answer the business model questions before deciding on information systems investments?

- **A.** Because how a company answers them dictates how and where information systems investments can be used to execute a competitive strategy and sustain an advantage — **correct.** The model is what tells you which investment executes the strategy, which is why the chapter presents technology investment as something to be aligned with the model rather than chosen on its own merits.
- **B.** Because the answers reveal which technology vendor is offering the lowest price for the systems the company needs to buy this year — Vendor pricing is a procurement question that arrives much later. The business model tells you what the system has to accomplish, and a cheap system aimed at the wrong component is not a saving.
- **C.** Because the systems a company already owns are what determine which business model it is able to adopt in its industry — This reverses the chapter's order of reasoning: how a company answers the business model questions is what dictates where systems investments go, not the other way round. Treating the installed systems as the constraint locks a firm into whichever model it happens to have inherited.
- **D.** Because information systems are only relevant to the revenue streams component and can safely be ignored when writing the other eight — Revenue streams are one component of nine, and systems support many of the others directly, from the channels that reach customers to the key activities the firm performs every day.

The nine components

The chapter breaks a proper business model into nine components. Each one plays a critical role in shaping all aspects of the business, including the expenses, revenues, operating strategies, corporate structure, and sales and marketing procedures. The table gives each component, what it means, and the question it answers.

COMPONENT	DESCRIPTION	QUESTIONS TO ASK
Customer segments	The customers targeted with the product/service offering	Who will be our target customers? Who are the most important customers?
Value proposition	The utility that the product/service has to offer to customers	Why do customers need our product/service? What problems will our product/service solve? Why would customers choose our product/service over our competitors' products/services?
Channels	The ways in which the product/service offerings reach the target customers	How will our customers be reached? Which channels are best in terms of cost and convenience for the customers?
Customer relationships	The relationships formed with the target customers	What types of relationships do we build with our customers, such as one-off versus long-term? How do we maintain these relationships?
Revenue streams	The way a firm generates income	How do we generate income? What are we selling? What are customers willing to pay for?
Key resources	The most important assets needed to make the business model work	What key resources are needed to enable our value proposition, channels, customer relationships, and revenue streams?
Key activities	The most important activities needed to make the business model work	What key activities are needed to enable our value proposition, channels, customer relationships, and revenue streams?
Key partners	The network of partners and suppliers needed to make the business model work	Who are our key partners and suppliers? What resources do they offer, and what activities do they perform?

COMPONENT	DESCRIPTION	QUESTIONS TO ASK
Cost structure	The costs incurred when operating the business model	What are the costs incurred when operating the business model? Which resources and activities are most expensive?

It helps to read the nine in two halves. The first five face the customer and describe the promise the company is making, and each one has an example you already know. The split into halves is a reading aid rather than something the chapter numbers, but it keeps the outward promise clearly separate from the machinery that has to keep it.

Customer segments — the customers you are targeting, named rather than assumed, because a model that serves two different groups has to say so: Uber connects people in need of rides with drivers willing to give one, and both groups have to appear in its model.

Value proposition — the utility your offering has for those customers, which is your answer to why they would choose you: Walmart answers with price, pursuing low-cost leadership by offering the best prices in its industry, while Porsche, Nordstrom, and IBM answer by trying to provide better products or services than their competitors.

Channels — the ways the offering actually reaches those customers, chosen on cost and on convenience for the customer, which is why Amazon uses the web as an economic medium to reach a large customer base rather than opening a counter in every town.

Customer relationships — what kind of connection you form and how you keep it, because a one-off sale and a long-term relationship demand different work: Uber uses a rider's past behavior to learn her preferences and provide personalized ride recommendations, which only pays off across many rides.

Revenue streams — the way the firm generates income and what customers will actually pay for, and the answers differ sharply: Netflix charges a monthly or annual fee for use of the service, while Google's services are free to the user and paid for by a third party.

The remaining four face inward. They describe what the company must have, do, borrow, and spend in order to keep the promise the first five just made.

Key resources — the most important assets the model needs, which the chapter illustrates elsewhere with proprietary technology, brand equity, and a loyal and established customer base, because a promise resting on an asset you do not have is a wish.

Key activities — the most important things the company must actually do, such as the work Uber's app performs when it automatically matches riders with drivers and anticipates surges in demand so it can change pricing accordingly, because owning an asset and never using it changes nothing.

Key partners — the network of partners and suppliers you depend on, together with what they supply and what they do, which is the arrangement the chapter calls affiliate marketing: Amazon's Associates program pays businesses that bring or refer customers to it, typically through revenue sharing, so that reach is real but borrowed.

Cost structure — what operating the whole arrangement costs and which resources and activities are the most expensive, because this is the component that decides whether the other eight can survive contact with a bank statement.

EXPLORE

The five component pairs beginners mix up

Open each pair and read all four panels; the third panel gives you a test you can apply in seconds.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Value proposition, not revenue stream · What they get vs. what they pay

- **What it really is:** The value proposition is the utility the product or service has to offer to customers: why they need it, what problem it solves, and why they would choose it over a competitor's offering.
- **The look-alike beside it:** The revenue stream sits right next to it and answers something else entirely — the way the firm generates income, what it is selling, and what customers are willing to pay for.
- **The question that separates them:** Ask "what does the customer receive?" for the value proposition, and "who hands over money, and when?" for the revenue stream. The two answers are frequently about different people.
- **What the mix-up costs:** The same value can be charged for in completely different ways: Netflix charges the viewer a monthly or annual fee, while Google's services are free to the user and paid for by a third party. If you fuse the two, you will believe that changing the price means changing the product.

Channels, not customer relationships · The route vs. the rapport

- **What it really is:** Channels are the ways in which the offering reaches the target customers, and the chapter judges them on two things at once: cost to the firm and convenience for the customer.

- **The look-alike beside it:** Customer relationships are the relationships formed with those customers — one-off or long-term — along with the work of maintaining them once they exist.
- **The question that separates them:** A channel is a path the offering travels down; a relationship is what remains after the offering has arrived. One app can do both jobs, which is exactly why the two get collapsed into one.
- **What the mix-up costs:** Uber uses a rider's past behavior to provide personalized ride recommendations, which is relationship work happening on top of the delivery channel. A firm that counts its app as its relationship keeps spending on delivery and never understands why customers do not return.

Key resources, not key activities · What you have vs. what you do

- **What it really is:** Key resources are the most important assets needed to make the model work, which the chapter illustrates with proprietary technology, brand equity, and a loyal and established customer base.
- **The look-alike beside it:** Key activities are the most important things the firm must actually perform, such as the coordination of riders and drivers that Uber's app carries out to optimize the matching of supply and demand.
- **The question that separates them:** Resources are nouns you own or control; activities are verbs you perform. A delivery van is a resource, and driving the collection route every morning is an activity.
- **What the mix-up costs:** Both are defined as what is needed to enable the value proposition, channels, customer relationships, and revenue streams, so leaving one out breaks the same promise. Buying the asset and never staffing the work is a quiet and expensive way to fail.

Key partners, not key resources · Owned vs. borrowed

- **What it really is:** Key partners are the network of partners and suppliers needed to make the model work, described by what resources they offer and what activities they perform on your behalf.
- **The look-alike beside it:** Key resources are the assets the firm itself must have. The confusion is genuine, because the same capability can often be either owned or borrowed without the customer noticing.
- **The question that separates them:** Ask who would still have the capability if the relationship ended tomorrow. If the answer is not you, it belongs under partners, no matter how essential it feels.
- **What the mix-up costs:** The chapter's example of affiliate marketing is Amazon's Associates program, which pays businesses that bring or refer customers to it, typically through revenue sharing, so that reach is real but borrowed. Filing a partner under resources hides the risk that the partner leaves and takes the capability with them.

Customer segments, not "everyone" · Naming who you are for

- **What it really is:** Customer segments are the customers targeted with the offering, and the component asks two questions: who our target customers will be, and who among them are the most important.
- **The look-alike beside it:** The look-alike here is not another component but a non-answer. "Anyone who needs this" feels generous and safe, and it is a refusal to choose rather than a segment.
- **The question that separates them:** Try to name a person the offering is deliberately not for. If no such person exists, you have not segmented the market; you have only described it.
- **What the mix-up costs:** Every other component points back at the segment, because a channel can only be called convenient for somebody in particular and a value proposition can only be compelling to somebody in particular. An unnamed customer makes the remaining eight answers unfalsifiable.

Once the nine are clear, the fastest way to test yourself is to work backwards: read the question the chapter attaches to a component, then name the component from the question alone. That direction is harder than reciting the list, and it is the direction a real conversation runs in, because people describe their difficulties as questions long before anyone knows which box the answer belongs in.

MATCH

Component to the question it answers

Pair each business model component with the question the chapter uses to draw it out.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Customer segments

Who will be our target customers, and who are the most important ones? This component names the customers targeted with the offering, and the second half of the question matters as much as the first, because most firms serve several groups unequally.

Value proposition

Why do customers need this, and why choose it over a competitor's? The value proposition is the utility the offering has to customers, so the test is always comparative: a benefit a rival also provides is not yet a reason to choose you.

Channels

How will customers be reached, and which routes are best on cost and convenience? Channels are the ways the offering reaches the target customers, and the chapter weighs cost to the firm against convenience for the customer rather than either one alone.

Customer relationships

Is this one-off or long-term, and how will we maintain it? This component covers the relationships formed with target customers, and the maintenance half is where the ongoing work and the ongoing expense actually live.

Revenue streams

How do we generate income, and what are customers willing to pay for? Revenue streams describe the way a firm generates income, and the second clause is the honest one, because what a firm wants to sell and what customers will pay for often differ.

Key resources

Which assets must we have to enable the promise we just made? Key resources are the most important assets needed to make the model work, and the chapter ties them explicitly to enabling the value proposition, channels, relationships, and revenue.

Key activities

Which things must we actually perform to enable that same promise? Key activities are the most important activities needed to make the model work, and they are the verbs to the resources' nouns, tied to the same four components.

Key partners

Who supplies us, what do they offer, and what do they perform? Key partners are the network of partners and suppliers needed to make the model work, and naming what each one contributes is what turns a contact list into a component.

Cost structure

What does operating all of this cost, and which parts are most expensive? Cost structure covers the costs incurred when operating the business model, and the chapter asks specifically which resources and activities are most expensive, because that is where a model is fixed or broken.

Why the pieces interlock

Beginners treat the nine components as nine independent boxes to fill in. They are not independent, and the chapter's own questions show the wiring.

Look again at the questions for key resources and key activities. Both ask what is needed *to enable our value proposition, channels, customer relationships, and revenue streams* — so those two components are defined by the first five and cannot be settled before them. Then look at cost structure, which asks which resources and activities are most expensive. Cost is the shadow cast by the choices already made.

That wiring means a change in one place travels. Five common moves show how far.

Change the customer segment — and the channel usually has to change with it, because a route that is convenient for a walk-in neighbour may be unreachable for a customer thirty miles away.

Change the channel — and key resources and key activities change immediately, because reaching people a new way requires assets and daily work that the old way never needed.

Change the revenue stream — and the customer relationship changes with it, because charging once at the counter and charging every month create very different expectations about what happens after the sale.

Change a key partner — and you have quietly changed your resources and your risks, because a capability you borrow can be withdrawn on terms you do not control.

Try to change only the cost structure — and you will find you cannot, because costs are produced by the resources and activities you chose, so cutting the number without changing the choices only breaks the promise.

A business model is not nine separate answers; it is nine answers that all have to be true at the same time, which is why a single change is never really a single change.

DECIDE

Change one component and follow the consequences

Work through the five decisions in order, choosing the option that keeps the business model coherent.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The situation below is hypothetical and exists only for practice; no real business is being described. You advise the owner of a small neighborhood bicycle-repair shop. Today the model is simple: local riders walk their bikes in, two mechanics repair them, customers pay per repair, and the only significant assets are a rented storefront and a wall of tools. The owner wants to change one component. Follow where that change travels.

Decision 1. The owner decides to add a new customer segment: office workers ten miles away who commute by bicycle and cannot bring one in during shop hours. Which component is forced to change next?

- Channels, because the new segment cannot reach the storefront and the offering has to travel to them instead — **the stronger call**. Right. Channels are the ways in which the offering reaches the target customers, chosen on cost and on convenience for the customer. Naming a segment that cannot get to your door has not served anyone yet; the route has to change before anything else does.
- Cost structure, because the shop should cut its expenses first so it can afford the expansion — Cost structure is the costs incurred when operating the business model, and it is the sum of the other choices rather than the first lever to pull. Cutting costs before you know what the new segment requires just makes the shop smaller and no closer to the commuters.

- Nothing, because the value proposition is still good repairs at a fair price and that has not changed — The value proposition is the utility the offering has to customers, and “good repairs you cannot physically get to” has very little utility to a commuter. Choosing a new segment always puts pressure on the components that reach and serve it.

Decision 2. The owner picks a mobile channel: a van that collects bicycles from office parks in the morning and returns them in the evening. Which components does that choice immediately force?

- Key resources and key activities, because the van is a new asset and the daily collection round is new work — **the stronger call.** Right. Key resources are the most important assets needed to make the model work and key activities are the most important things it must do, and both are defined as what is needed to enable the channels, value proposition, relationships, and revenue. A new channel almost always demands both.
- Key partners only, because the shop can simply have a courier company carry the bicycles instead — A partner is a genuine option — key partners are the network of partners and suppliers needed to make the model work. But it does not escape the point: whether the vehicle is owned or borrowed, the model now requires a resource and an activity it did not require before. Borrowing changes which component holds them, not whether they exist.
- Customer segments, because the shop should now drop its walk-in customers to keep things simple — Nothing about adding a channel requires abandoning the original segment, and dropping profitable walk-in customers would shrink the revenue that pays for the van. Segments change when you decide who you are for, not automatically whenever a route changes.

Decision 3. The van costs money every single day it runs, but the shop still charges only when something breaks, so income arrives at unpredictable intervals. What change to revenue streams keeps the model coherent?

- Offer a maintenance plan the commuter pays for monthly, so income arrives on the same rhythm as the cost — **the stronger call.** Right. Revenue streams answer how the firm generates income and what customers are willing to pay for, and a commuter who depends on the bike daily plausibly will pay to keep it running rather than to fix it after it fails. A recurring cost met only by occasional income is one of the most common ways a well-meant model runs out of money.
- Leave the pricing alone and assume the extra volume from the new segment will cover the van — This is a hope rather than a component. Revenue streams are supposed to state how income is generated, and volume that has not been sold to anyone yet is not a revenue stream. The van’s costs, meanwhile, arrive whether the volume does or not.
- Add a one-time pickup fee charged at the moment of each repair — This is closer than doing nothing, and for an occasional customer it might be enough, since it does at least connect the new cost to the new service. It still ties all income to breakdowns, and the van is paid for by the day rather than by the breakdown.

Decision 4. The monthly plan means the shop has made an ongoing promise instead of completing a one-time sale. Which component now has to be rewritten?

- Customer relationships, because a subscription is a long-term relationship that has to be maintained rather than a single transaction — **the stronger call**. Right. This component asks what types of relationships the firm builds — one-off versus long-term — and how it maintains them. A subscriber who hears nothing for two months and then finds a mechanic unavailable will cancel, so maintenance is now part of the offering.
- Key partners, because the shop should sign a contract with a parts supplier to guarantee stock — A reliable parts supplier is a sensible partner and would genuinely help, since key partners are described by the resources they offer and the activities they perform. It just does not address the change in front of you, which is that the shop now owes something to the customer continuously.
- Cost structure, because the owner should recalculate what the plan costs before doing anything else — Recalculating is worth doing, but cost structure records the costs incurred in operating the model and therefore follows from the choices made. The promise to the customer is the choice; the cost is its consequence.

Decision 5. A year on, the owner wants to know whether the whole change was worth making. Which component tells him what it actually took to run the new model?

- Cost structure, because it records the costs of operating the model and which resources and activities are the most expensive — **the stronger call**. Right. Cost structure asks exactly that pair of questions, so it is where the van, the fuel, the extra hours, and the software subscription finally show up next to each other. It is also the component that tells him which single expensive activity to attack first.
- Revenue streams, because the income from the monthly plans is the measure of whether the change worked — Revenue streams tell you how income is generated and are half of the answer, which is what makes this tempting. Income alone cannot say whether the change was worth it, because a plan that sells beautifully and costs more to deliver than it earns is a loss with good marketing.
- Key resources, because listing the assets shows the value of what the shop now owns — Key resources name the most important assets needed to make the model work, which is useful for spotting what the model depends on. A list of assets is not a running total of what operating them consumed, and that total is what the owner is asking for.

How to write one down. Take a business you know and answer the nine questions in a single page, in the order the table gives them. Wherever an answer is “everyone,” “the usual,” or “we will figure that out,” you have found the part of the model that has not been decided yet — and almost always the part where an information system either helps or is wasted.

CHECK YOURSELF**Naming components and tracing consequences**

Three questions on identifying the right component and on what happens when one of them changes.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Which component is defined as the most important assets needed to make the business model work?

- **A.** Key activities — Key activities are the most important activities needed to make the model work — the things the firm performs, such as matching riders with drivers, rather than the things it holds.
- **B.** Key resources — **correct.** Key resources are the most important assets needed to make the business model work, and the chapter's examples of a firm's resources include proprietary technology, brand equity, and a loyal and established customer base.
- **C.** Key partners — Key partners are the network of partners and suppliers needed to make the model work, described by what resources they offer and what activities they perform. The capability is real but it belongs to somebody else.
- **D.** Cost structure — Cost structure is the costs incurred when operating the business model, including which resources and activities are most expensive. It measures the assets rather than naming them.

2. A firm decides that its offering will now reach customers through a subscription app instead of a retail counter. Which component has it changed, and what must be revisited straight away?

- **A.** It changed its channels, so key resources and key activities must be revisited, because both are defined as whatever enables the channels and the rest of the promise — **correct.** Channels are the ways the offering reaches the target customers, and because resources and activities are defined as whatever enables the channels and the rest of the promise, changing the route changes what the firm must own and do.
- **B.** It changed its value proposition, and nothing else needs revisiting, because the product being sold to the customer is identical either way — The product may indeed be identical, which is what makes this tempting, but the value proposition is the utility to the customer and a change of route is a change of channel. Even if the offer were untouched, the components downstream of the channel would still move.
- **C.** It changed its cost structure, and the customer segments must be renamed to match the new pattern of spending the app creates — Cost structure will certainly move, since an app has to be built and run, but it moves as a consequence rather than as the change itself. Segments are named by deciding who you serve, not by observing what you spent.

- **D.** It changed its key partners, and only the partner contracts need to be rewritten, since the app will be built and hosted by someone else — Partners may well be involved in building or hosting the app, so this is a plausible reading. It still misidentifies the change, which is how the offering reaches the customer, and it wrongly implies that resources and activities can stay as they were.

3. Which statement about cost structure is accurate?

- **A.** It is another name for revenue streams, viewed from the paying side of the same transaction rather than the earning side — Revenue streams describe how the firm generates income and what customers will pay for, which is money coming in. Cost structure describes money going out, and the two can be healthy or unhealthy quite independently of each other.
- **B.** It is the total price the customer pays, since the costs a firm incurs are ultimately passed through to whoever buys — This confuses cost with price. Cost structure records the costs incurred when operating the model, which is money leaving the firm; what a customer hands over belongs to revenue streams, and a firm can perfectly well charge less than a thing costs it or many times more.
- **C.** It lists what the company spends on technology, which is why it is the one component information systems have any effect on — Technology spending is part of the cost structure but nowhere near all of it, and information systems affect far more than costs. Systems reach the channels, the relationships, the key activities, and the revenue streams as well.
- **D.** It records the costs incurred when operating the model and asks which of the resources and activities are the most expensive — **correct.** The component covers the costs incurred when operating the business model and asks which resources and activities are most expensive, which makes it downstream of the resource and activity choices rather than an independent decision.

Revenue models, platforms, and network effects

A business model says what a company does. This half of the story answers the blunt part: in what way does the company get paid, and who stands on the other side of that payment? Several of the answers below were impractical until digital delivery made one more customer nearly free to serve.

The revenue model: how the money arrives

A **revenue model** describes how the firm will earn revenue, generate profits, and produce a superior return on invested capital. Even nonprofits need one, because something has to pay for the work. Sales, transaction fees, and advertising existed offline; the internet enabled or enhanced others, such as affiliate marketing.

The table lists the typical revenue models in the digital world and who uses each.

REVENUE TYPE	DESCRIPTION	WHO IS DOING THIS?
Affiliate marketing	Paying businesses that bring or refer customers to another business; revenue sharing is typically used.	Amazon's Associates program
Advertising	Free services are provided to customers and paid for by a third party.	Google, Instagram, X, TikTok
Subscription	Users pay a monthly or yearly recurring fee for the use of the product or service.	Netflix, Microsoft 365, Dropbox, Spotify, Adobe Creative Cloud
Transaction fees / brokerage	A commission is paid to the business for aiding in the transaction.	PayPal, eBay, Airbnb, Uber
Traditional sales	A consumer buys a product or service from the website, or makes in-app purchases.	Amazon, Zappos, Nordstrom, TikTok
Freemium	Basic services are offered for free, but a premium is charged for special features.	Flickr, iCloud, Dropbox

Read each row by asking whose hand the money leaves. Five of the rows sort into four groups.

The customer pays for the thing — traditional sales, because a shopper buys the product from the website or in the app.

The customer pays for the time — subscription, because the fee recurs monthly or yearly for continued use, as at Netflix.

Somebody outside the deal pays — advertising, because a third party covers the cost, which is why Google charges users nothing.

The firm is paid for arranging someone else's deal — brokerage commissions the transaction, as at PayPal, and affiliate marketing pays for the referral.

Freemium is the sixth and the odd one out, because it splits one product into a free version and a paid one.

SORT

Which revenue model is this?

Place each offer under the revenue model that describes how the money actually reaches the firm; every model in the table appears here twice, and the one item that names no company is a generic example written for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Affiliate marketing — paying a business that brings or refers customers to another business, usually by sharing revenue

- A website earns a share of the revenue every time the shoppers it sends to a large online retailer go on to buy something — the arrangement behind Amazon's Associates program — Affiliate marketing pays businesses that bring or refer customers to another business, and revenue sharing is the typical form that payment takes.
- A price-comparison site is paid by a merchant for each customer it refers, rather than by the customers it serves — The payment is for the referral itself, which is what separates affiliate marketing from a commission earned for aiding a transaction between two other parties.

Advertising — free services provided to customers and paid for by a third party

- Google, Instagram, X, and TikTok let people use their services without charging them, and bill someone else instead — Under an advertising model, free services are provided to customers and paid for by a third party, and these four are the chapter's named examples.
- YouTube charges neither the producers who upload videos nor the consumers who watch them, and takes its money from advertisers — The chapter uses YouTube as the contrast to Airbnb: the platform monetizes a third party rather than either side of its own market.

Subscription — a recurring monthly or yearly fee for continued use of the product or service

- Netflix customers pay a recurring monthly or yearly fee, and the service stops when they stop paying — A subscription model charges a recurring fee for the use of the product or service, which is exactly what a lapsed account demonstrates.
- Microsoft 365, Spotify, and Adobe Creative Cloud each bill a recurring fee for continued use of the software — All three are named subscription examples, and the giveaway is that the fee buys a period of use rather than a permanent copy.

Transaction fees / brokerage — a commission paid to the business for aiding in someone else's transaction

- Airbnb collects a commission from both the host and the guest for aiding the booking, without owning the room — Transaction fees and brokerage mean a commission paid to the business for aiding in the transaction, and the chapter names Airbnb charging both parties.
- PayPal and eBay are paid a commission for helping a transaction between two other parties go through — Neither firm is the buyer or the seller; each is paid for making the exchange between them possible, which is the definition of brokerage.

Traditional sales — a consumer buys the product or service from the website, or makes in-app purchases

- A shopper buys a pair of shoes outright from the Zappos website, and Nordstrom sells the same way from its own site — Traditional sales means a consumer buys a product or service from the website, and both companies are named examples of it.
- Amazon uses the web to reach a very large customer base, turning inventory over quickly so it can offer low prices and still make a profit on each item sold — This is the chapter's description of using the web as an economic medium to sell goods, so the revenue still arrives as the price of a product.

Freemium — basic service free, a premium charged for special or unrestricted features

- Dropbox and iCloud give every user an account with limited storage capacity and charge those who want more space — Freemium offers basic services free and charges a premium for unrestricted versions, and storage is one of the chapter's listed restrictions.
- Flickr offers a basic service at no cost and charges only for the special features on top of it — The defining feature is one product existing in a free limited version and a paid premium one, which is how the chapter defines freemium.

Why giving it away became a real strategy

In a competitive marketplace, price is set by **marginal cost** — the cost of producing one additional unit of output. The building blocks of a digital product are processing, storage, and bandwidth; those costs have fallen, and cloud computing turned many into variable costs.

When serving one more user costs almost nothing, an organization may give limited versions away free to build a large customer base and charge a premium for unrestricted ones, typically by subscription. That is **freemium**, and what the free version withholds varies.

Features — the free version does less, so whoever needs the missing capability pays.

Bandwidth — data moves more slowly, which bites hardest on the heaviest users.

Storage — Dropbox and iCloud limit free capacity and sell subscriptions to larger accounts.

Number of users — the free version covers one person or a small team, so a growing group outgrows it.

Class of user — the product or service is restricted to a certain class of users, such as educational users, so anyone outside that group has to pay for it.

Choosing that line is the hard part. Free offerings must be compelling enough to attract users, yet if too many features are free, few will pay for the premium version. Often a process of fine tuning and adjustment is needed, and it is hardest when the change reduces the service level of the free offering — as when the calendar app Fantastical moved previously free features into its premium offering.

Free games are the extreme case. Mobile games such as Candy Crush or Honor of Kings download free and sell **in-app purchases** of extra features, content, or in-game currency. In 2024 alone, Honor of Kings took in more than US\$1.8 billion globally that way, and players gain too, since developers depend on that revenue and keep upgrading the game.

CHECK YOURSELF

Revenue models and the economics of free

Pick the best answer, then read all four explanations — the wrong ones name what that option genuinely describes. Any situation here that names no company is a generic example written for practice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A cloud storage company gives every user an account with limited capacity at no charge and asks those who want more space to pay a monthly fee. Which revenue model best describes this?

- **A. Advertising**, because the free accounts are being paid for by a third party — Advertising means free services are provided to customers and paid for by a third party, as at Google or TikTok. Nobody outside this relationship is paying, so the answer would need advertisers funding the free accounts.

- **B.** Freemium, because a basic version is free and a premium is charged for the unrestricted version — **correct**. The chapter names cloud storage providers such as Dropbox and iCloud as the standard freemium case: free accounts with limited storage capacity, and a subscription for larger space.
- **C.** Transaction fees and brokerage, because the company collects a fee each time a user upgrades to a larger account — Transaction fees and brokerage mean a commission paid to a business for aiding in a transaction between two other parties, as with PayPal, eBay, or Airbnb. Here the storage company is one of the two parties, so what it charges its own customer is a price rather than a commission, however it is timed.
- **D.** Traditional sales, because the customer buys the larger account from the company's website — Traditional sales is a consumer buying a product or service from the website, and money genuinely does change hands here. It misses what is distinctive, though: the free limited tier exists on purpose to build the base that the paid tier is sold into.

2. Why did falling costs make free versions of digital products practical in the first place?

- **A.** Because in a competitive marketplace price is set by marginal cost, and processing, storage, and bandwidth have become cheap enough that serving one more user costs very little — **correct**. The chapter's argument runs through marginal cost — the cost of producing an additional unit of output — and the falling costs of the fundamental building blocks of digital products and services.
- **B.** Because cloud computing turned variable costs into fixed costs, so additional users are absorbed by money the firm has already spent — This reverses the direction the chapter describes: cloud computing turned many of these costs into variable costs, not fixed ones. Variable costs are what make each additional user cheap rather than free.
- **C.** Because advertising revenue rises with every free user a service adds, so free users are not really free — Some free services genuinely are funded by advertisers, which is the advertising model. But freemium firms such as Dropbox and iCloud charge their own premium users, and the chapter grounds the shift in costs rather than in ad revenue.
- **D.** Because the cost of writing the software is paid once up front, so every user after the first one is pure profit — This is close enough to be tempting, and it is part of why free versions are thinkable at all, but it overstates the case. Delivering the service to one more user still consumes processing, storage, and bandwidth; the chapter says those costs have fallen and become variable, not that they vanished.

3. A company running a freemium product is deciding which features to leave in the free version. What guidance does the chapter give, and what does the Fantastical example add?

- **A.** Give away as much as possible, because a larger base of free users turns into a larger number of paying subscribers — Scale does matter, and the chapter says app developers benefit from the ability to scale their user base quickly. It also states the opposite pressure: if too many features are offered for free, few people will be willing to pay for the more feature-rich premium version.

- **B.** Keep the free version as thin as possible, because users decide on price alone and will upgrade once they need the product — This fails the other half of the test: the free offering must be compelling enough to attract new users in the first place. A free version nobody wants builds no base for the premium version to sell into.
- **C.** The free version must be compelling enough to attract users but not so generous that few will pay, and pulling a free feature back later is especially hard, as Fantastical found — **correct.** The chapter describes both sides of the balance, says a process of fine tuning is often needed, and notes the special difficulty of reducing the service level of the free offering, as when Fantastical moved previously free features into its premium offering.
- **D.** The balance is settled at launch, because changing which features are free afterwards is a routine pricing update — The opposite is true. Fine tuning and adjustment are described as ongoing, and the Fantastical example exists precisely because a later change to the free tier is the hardest kind to make.

Pipes and platforms

Traditionally, many business models functioned like **pipes**: the firm produced a product and pushed it out to customers, so value flowed in one direction and the firm, not the customer, was the one producing it.

Increasing digital density enabled another shape. A **platform-based business model** does not provide the product or service itself; it coordinates others — businesses and users — to co-create value, building a digital ecosystem where some users produce and others consume. People uploading to Instagram or Pinterest produce what others consume, and Uber and Airbnb depend on users to offer and take the rides and rooms.

What gets exchanged on a platform is wider than most people assume.

VALUE CREATED OR EXCHANGED	EXAMPLES
Products	Amazon Marketplace, eBay
Services	Airbnb, Uber
Payments	Square, PayPal, Venmo, Zelle
Investments and funding	Kickstarter, Lending Club
Content	Wikipedia, X, YouTube, TikTok
Communication	WhatsApp, Zoom
Collaboration	Dropbox

VALUE CREATED OR EXCHANGED	EXAMPLES
Social relationships	Facebook, LinkedIn

Serving producers and consumers at once creates a **two-sided market**, whose success depends on the network effect. Who pays is a separate decision: Airbnb charges a transaction fee to both parties, while YouTube charges a third party, the advertisers.

The two shapes are not rivals on a shelf — many pipeline businesses already carry platform elements such as user reviews, so the question is how much platform a business model should contain.

INTERACTIVE DIAGRAM

Pipe, platform, and the two network effects

Pick a model to redraw it: the first two contrast the two shapes of business, and the last two show which side each network effect acts on.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The pipe business — Value produced by the firm and pushed out one direction

Suppliers (Inputs bought by the firm) → The firm (Produces the product itself) → The product (Pushed out to the market) → Customers (Buy, consume, and that is the end)

- Traditionally, many business models functioned like **pipes**, where products were produced and pushed to the customers, so the firm creates the value and the customer only receives it.
- A jet-engine manufacturer selling engines and spare parts to airlines is the pure form: everything of value is made inside the company and sold outward.
- Growth in a pipe means producing more or selling to more people, because there is no second group whose participation the firm could recruit instead.
- Pipes are not obsolete. Today many pipeline-based businesses include platform elements, such as reviews created by users, so the two shapes mix inside one company.

The platform business — Value created by others whom the firm coordinates

Producers (Upload, list, drive, host) → The platform (Coordinates, matches, sets rules) → Consumers (Watch, buy, ride, stay) → Who pays (One side, both sides, or a third party)

- A **platform-based business model** coordinates others — both other businesses and users — to co-create value, rather than providing the product or service itself.
- Platform businesses create digital ecosystems where some users create value and other users consume it: people uploading to Instagram or Pinterest produce what others consume, and Uber and Airbnb depend on users to offer and consume the services.

- Serving both groups at once creates a **two-sided market**, and its success depends on the network effect — a platform has value only if users participate in production or consumption.
- Monetizing is a separate decision from building the market: Airbnb charges a transaction fee to both parties, while YouTube charges neither producers nor consumers but a third party, the advertisers.

Same-side network effects — Value changes because more users join your own side

More users on one side (Your neighbours, doing what you do) → Value can rise (Wider iOS use, wider Keynote compatibility) → Value can fall (More riders competing for the same rides) → The management tool (Surge pricing handles excessive demand)

- **Same-side network effects** relate to the increase or decrease in value as additional users join on the same side of the network, so the effect comes from people in the same role as you.
- The positive example: as more users switch to iOS, everyone benefits from wider compatibility of its Keynote presentation format, because the format is understood in more places.
- The negative example: as more riders use Uber, competition for available rides increases, and surge pricing exists to help manage that excessive demand.
- This is why a platform cannot simply chase user counts — platform businesses must carefully evaluate the various same-side and cross-side network effects instead of assuming every new user adds value.

Cross-side network effects — Value changes because more users join the other side

More apps arrive (Producers join the developer side) → Users gain (The app store becomes more useful) → More users arrive (Consumers join the other side) → Developers gain (More reason to build for iOS)

- **Cross-side network effects** relate to the change in value for a user if users join on the other side, so the benefit arrives from the group you are matched with rather than the group you belong to.
- The chapter's pair points both ways: the app store becomes more useful for users as more apps are added, and app developers are more likely to develop iOS apps as more users use iOS.
- Because each side is the reason the other side shows up, any platform-based business model must be attractive to both producers and consumers at the same time.
- This is also the hard part of starting a platform, since a side with nobody on it gives the other side no reason to arrive — value exists only where users participate in production or consumption.

Network effects, and which side they come from

A network effect is what happens to a platform's value when the number of people on it changes. The chapter separates two kinds, and the only difference is which side the newcomers join.

Same-side network effects — the increase or decrease in value as users join on the same side of the network, so people doing what you do change what the platform is worth to you.

Cross-side network effects — the change in value for a user if users join on the other side, so the people you are matched with change what it is worth.

Notice the phrase increase or decrease. Same-side effects run both ways, and the chapter supplies an example of each.

- Positive: as more users switch to iOS, everyone benefits from wider compatibility of its Keynote presentation format.
- Negative: as more riders use Uber, competition for available rides increases, which is what surge pricing manages.

Cross-side effects run between the two sides, and the chapter's example arrives as a pair pointing opposite ways: the app store becomes more useful for users as more apps are added, and app developers are more likely to develop iOS apps as more users use iOS. Each side is why the other shows up.

So a platform has value only where users participate in production or consumption, and it must attract producers and consumers alike. That is why these firms evaluate the various same-side and cross-side effects carefully rather than chase a user count.

Reasoning from the mechanism, not a claim the chapter makes. The chapter defines the two effects but never describes a market tipping to one winner, so check this rather than memorize it. If each consumer who joins makes the platform more attractive to producers, and each producer makes it more attractive to consumers, the platform slightly ahead keeps getting easier to join, and an early lead compounds. Expect concentration, not a guarantee: a negative same-side effect, a regulator, or a stronger local rival can interrupt the loop.

MATCH

Platform vocabulary, and which side the effect came from

Pair each term or example with what it actually describes; three of the pairs are network effects you must place on the correct side.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Same-side network effect

The increase or decrease in value as additional users join on the same side of the network
The definition allows movement in both directions, which is why a crowded side can be worth less rather than more.

Cross-side network effect

The change in value for a user when users join on the other side of the network The value arrives from the group you are matched with, which is why a platform must recruit both sides at once.

More riders join a ride-hailing service, so competition for available rides increases

A same-side effect that reduces value, managed with surge pricing The newcomers are riders and the people affected are riders, so the effect is same-side even though drivers also trade in this market.

The app store becomes more useful as more apps are added to it

A cross-side effect, since users gain from arrivals on the developer side Developers and users occupy opposite sides of the platform, so a gain that crosses between them is cross-side by definition.

More people switch to iOS, so the Keynote presentation format is compatible with more machines

A same-side effect that increases value The benefit reaches the same group that grew, which is what makes this the positive form of a same-side effect.

Two-sided market

The market a platform creates by serving producers and consumers at once, with success depending on the network effect Because both sides must be present, the platform has value only if users participate in production or consumption.

Platform-based business model

Coordinating other businesses and users to co-create value instead of providing the product or service itself Uber and Airbnb are the chapter's cases: successful platforms that do not offer the rides or the rooms themselves.

Disintermediation

Matching providers and customers directly, without the traditional middlemen Removing the intermediary is what lets a platform connect the two sides cheaply, and the chapter flags it for later study.

Gig economy

An arrangement in which workers are not employed by the company but hired only for short-term, temporary jobs The chapter raises it as the human cost of platform disruption, because protections such as social security and job security often go missing.

The sharing economy

Many platforms match providers and customers directly, with no traditional middleman — **disintermediation**. As individuals began sharing physical goods and services this way, not only content, many concluded we are moving toward a **sharing economy**, sometimes called a collaborative economy. The chapter takes its definition from a dictionary: an economic system in which assets or services are shared between private individuals, either free or for a fee, typically by means of the internet.

The claim underneath is that ownership is not what counts; access to the asset is. Three consequences follow.

Assets get used harder — research cited in the chapter found that one vehicle on a car-sharing platform such as Free2Move, a subsidiary of the multinational automaker Stellantis, can replace up to 13 privately owned vehicles, and for some people Uber replaces the need to own one.

Goods last longer — platforms such as Craigslist or eBay encourage selling used goods, extending the usable life of many products.

The definition blurs — the Lyft scooters show the boundary between true sharing and refined renting becoming hard to draw.

Early sharing-economy thinking included social goals such as waste reduction and greater human interaction, which matter less now. The benefits usually cited today are practical ones: more flexible work arrangements, and the chance to earn extra income by renting out an apartment through Airbnb or driving for Uber.

These business models are not without criticism. In the **gig economy**, workers are not employed by the company but hired only for short-term, temporary jobs, and rights fought for since the Industrial Revolution are often neglected: no protection against discrimination, no social security, little job security, limited income opportunity. Uber and Airbnb have also sought advantage over traditional businesses by breaking, or at least bypassing, existing laws and regulations, which can often be regarded as unfair competition.

EXPLORE

What breaks a platform

Open each card and read all four facets; every failure here comes from something the chapter says a platform business has to manage.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

One side stops showing up · The empty half of a two-sided market

- **The failure:** A platform does not produce the value it sells; it coordinates others to co-create it. So a platform only has value if users participate in the production or the consumption, and a side with nobody on it leaves nothing to coordinate.
- **How it shows up on the platform:** Listings go stale because no producers post, or listings sit unseen because no consumers arrive. Each side looks at the empty other side and concludes the platform is not worth joining.
- **What it costs if nobody acts:** The cross-side effects run the wrong way: fewer producers make the platform less useful to consumers, and the consumers who then leave make it less useful to the producers still there.

- **What the chapter says about it:** Any platform-based business model must be attractive to both producers and consumers, which makes the two sides one design problem rather than two marketing campaigns.

Crowding on one side · A negative same-side effect left unmanaged

- **The failure:** Same-side network effects can decrease value as well as increase it. The chapter's example is direct: as more riders use Uber, competition for available rides increases, so the newest arrivals make the platform worse for the people already there.
- **How it shows up on the platform:** Everyone on the crowded side waits longer, is matched less often, or competes harder for the same limited attention, while the raw user count on the dashboard keeps climbing.
- **What it costs if nobody acts:** Growth stops meaning what management thinks it means, because the platform is adding users to the side that is already oversupplied and quietly degrading its own experience.
- **What the chapter says about it:** Surge pricing is the chapter's named response — a mechanism that helps manage excessive demand — and the general instruction is to evaluate the various same-side and cross-side effects carefully.

No answer to who pays · Negligible marginal cost is not a revenue model

- **The failure:** Platform business models are attractive partly because they often have negligible or zero marginal costs, but low costs are not income. The platform must still decide which participants, if any, are charged.
- **How it shows up on the platform:** The service is popular, usage grows, and the finance conversation keeps being postponed because nothing in the growth curve reveals where the money is supposed to come from.
- **What it costs if nobody acts:** The risk is that the charge arrives late and abruptly, or lands on whichever side is least able to leave — the same difficulty the chapter describes when a company reduces the service level of a free offering people have already built habits around.
- **What the chapter says about it:** The chapter puts two legitimate answers side by side: Airbnb charges a transaction fee to both parties, and YouTube charges neither side but sells to advertisers instead.

The rules and the rivals catch up · Disruption is not a permanent position

- **The failure:** The chapter says companies such as Uber and Airbnb try to gain a competitive advantage over traditional businesses by breaking, or at least bypassing, existing laws and regulations. An advantage held that way lasts only as long as the rules and the rivals stay still.
- **How it shows up on the platform:** Lawsuits, licensing fights, and legislative protection of the incumbent industry — Uber has been sued countless times, and many locales have attempted to protect their taxi systems in court or in statute.

- **What it costs if nobody acts:** A rival can copy the model and add reach the original lacks: DiDi Kuaidi merged the taxi-hailing services of Alibaba and Tencent, then widened into taxis, private cars, shared cars, shuttle vans, and buses.
- **What the chapter says about it:** The chapter's ending is blunt. Facing a hostile regulatory environment that tends to favor local businesses, Uber withdrew from China in mid-2016, selling its assets to its competitor for a minority stake.

The people doing the work are unprotected · The gig economy criticism

- **The failure:** In the gig economy, workers are not employed by the platform but hired for short-term, temporary jobs, so the protections that came with employment do not follow them onto the platform.
- **How it shows up on the platform:** No protection against discrimination, no social security, little job security, limited income opportunity, and few of the social interactions with coworkers that an ordinary workplace supplies.
- **What it costs if nobody acts:** The disruption delivers greater choice to end customers at a cost borne by humans and society, which is the specific criticism the chapter levels at these business models.
- **What the chapter says about it:** The chapter does not resolve it. It records that worker rights fought for since the Industrial Revolution are often neglected, and that bypassing existing law can reasonably be regarded as unfair competition.

When things go wrong: out-disrupting the disrupter

Uber built a successful business in many locations by using social, mobile, and Big Data to disrupt local transportation, and protected taxi monopolies struggled to react. Many locales fought back in legislatures or in court, and Uber has been sued countless times. In China a competitor tried the other approach: out-disrupting Uber.

The rival was itself a merger — DiDi Kuaidi combined the taxi-hailing services of Alibaba and Tencent, China's internet giants, so it began with scale.

It widened the platform — renamed DiDi Chuxing, it let users pick a taxi, private car, shared car, shuttle van, or bus, and in 2025 had more than 550 million users globally in the ride-sharing market.

Regulation settled it — as Amazon and Google learned before, regulation in China often favors local businesses, and Uber could not beat a state-backed rival.

In mid-2016 Uber withdrew from the Chinese market, selling all its assets to its competitor for a minority stake. DiDi Chuxing kept advancing and is on the cusp of launching a self-driving taxi.

Service-based business models and XaaS

Digital density enabled one more shift: selling the use of a product rather than the product, sometimes called **XaaS**, or X as a service. A manufacturer that used a pipe model to build and sell goods can instead offer two kinds of service.

Equipment services — offering the product itself as a service, or operations and optimization services, so the customer buys an outcome rather than a machine.

Information services — selling data or insights generated by the customer's usage of the product, which is revenue from something the product already produced.

Jet engines are the clearest case. Rolls-Royce and GE traditionally sold engines to aircraft owners; because the bargaining power of buyers is high, that market is competitive, and much of their revenue came from maintenance and spare parts.

Under a service-based model, GE is paid for guaranteeing continuous uptime — essentially thrust as a service. The customer buys neither engine nor spare parts and pays for hours of usage, turning fixed costs into variable ones, and the manufacturer earns only while engines run, so it gains a reason to improve the design and do preventive maintenance.

None of it works without the Internet of Things: sensors throughout the engines, connected to the internet, let GE monitor performance, predict failure, and schedule maintenance. Philips does the same on the ground, providing lighting as a service to dozens of parking garages of the Washington Metropolitan Area Transit Authority, which needs no upfront capital while Philips earns from the energy savings.

The transition is hard, and the difficulties are internal rather than technical.

The manufacturing goal inverts — rather than minimizing build cost, the firm focuses on life span and ease of repair, because it lives with the product.

Selling changes shape — a product sale negotiates price and features, while a service needs new pricing models, new business processes, and service level agreements.

Capacity gets harder — managing inventories and capacity is tougher, because demand for services fluctuates as a build-and-ship schedule does not.

The buyer is different — the segment targeted for products is often not the one buying services, forcing changes to marketing.

Across freemium, platforms, and XaaS the same move repeats: the firm stops charging once for an object and gets paid instead for continued access, participation, or performance.

CHECK YOURSELF

Platforms, effects, and services

Three questions on the harder half of this section, any unnamed company being a generic example written for practice; read every explanation, because the distractors are the confusions this material usually produces.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. An app store gains several hundred new apps this month, and the people who shop in it find it noticeably more useful as a result. Which effect is that?

- **A.** A same-side network effect, because the apps and the shoppers are all users of the same platform — Being on the same platform is not the same as being on the same side. Developers and shoppers occupy the two sides of the store, and the side an effect crosses is the only thing that names it.

- **B.** A same-side network effect, because the new apps now compete with each other for downloads — This describes a real same-side effect, but it is the wrong one for the question asked. Competition among developers is felt by developers, whereas the question describes a change in value felt by shoppers.
- **C.** A cross-side network effect, because the value changed for users on the opposite side from the ones who joined — **correct.** The chapter's own example is that the app store becomes more useful for users as more apps are added, which is the definition of a cross-side effect.
- **D.** Not a network effect at all, because the store simply widened its selection the way any retailer might — Widening a selection is genuinely not a network effect when a retailer buys the extra stock itself. What makes this one is that the value changed for one group because independent participants joined the other side, which is the chapter's own app store illustration.

2. As more riders use a ride-hailing platform, competition for the available rides increases. What is happening, and what is surge pricing for?

- **A.** A negative same-side effect, and surge pricing helps manage the excessive demand it creates — **correct.** Same-side effects can decrease value as well as increase it, and the chapter names surge pricing as the mechanism that helps manage that excessive demand.
- **B.** A cross-side effect, because riders and drivers sit on opposite sides of the platform — Riders and drivers do sit on opposite sides, so this would be the right label for an effect that crossed between them. The change described here happens among riders because of other riders, which keeps it on one side.
- **C.** A positive same-side effect, because a busier service is more valuable to the people using it — This assumes same-side effects can only add value, which is exactly what the definition rules out: the effect is an increase or a decrease as users join the same side.
- **D.** Ordinary supply and demand rather than a network effect, because too few cars for too many riders is a capacity problem — Capacity is genuinely part of the story, but the shortage is created by users arriving on one side and is felt by the users already standing on that same side. That is what the chapter calls a same-side network effect, and surge pricing is the response it names.

3. A jet-engine manufacturer stops selling engines and spare parts and is instead paid for guaranteeing continuous uptime. What actually changes?

- **A.** The customer finances the engine over time while the manufacturer keeps the spare-parts revenue it had before — Under the service model the customer does not have to purchase the engine or the spare parts at all, so this describes a financing plan for the old pipe model rather than the shift the chapter describes.
- **B.** The customer pays for hours of usage, turning fixed costs into variable ones, and the manufacturer now profits from durability and preventive maintenance — **correct.** The customer pays for hours of usage rather than buying the engine, which transforms fixed costs into variable costs, and the manufacturer earns only while engines operate, giving it an incentive to improve design and perform preventive maintenance.

- **C.** Nothing changes in the manufacturer's priorities, since minimizing manufacturing cost still decides how profitable each engine is — The priority inverts. Rather than focusing on minimizing manufacturing costs to increase profit margins, the manufacturer must focus on life span and ease of maintenance and repair, because it now bears the cost of failure.
- **D.** The arrangement needs no new technology, because maintenance intervals for engines are already well known — This model is only possible through rapid advances in Internet of Things technology: sensors throughout the engines, connected to the internet, let the manufacturer monitor performance, predict potential failure, and schedule maintenance.

Innovation and what it is worth

Ask most people what innovation means and they will describe a product — a better phone, a cleverer gadget, something you could hold up on a stage. That answer is not wrong, but it turns out to be one tenth of the picture, and this section is about the other nine tenths.

Inventing something is not the same as innovating

For organizations, **innovation** means creating new products, processes, or services that return value to the organization. The chapter is careful about that last phrase: in contrast to merely *inventing* new products, services, or processes, innovation involves realizing the value.

So a brilliant idea sitting in a notebook is an invention. It becomes an innovation at the moment it earns money, saves money, or wins a customer who would otherwise have gone elsewhere.

The stakes are not academic. Innovation is key for organizations attempting to gain or sustain a competitive advantage, whether through cost or through differentiation, and the chapter aims its warning squarely at the companies currently winning.

Being the leader is not protection. Companies that lead their markets or own strong brands often tend to find it difficult to react appropriately to new trends, and Cisco's former chief executive John Chambers predicted that almost half of today's leading businesses might not see the next decade. Read that as a statement about two different skills, since leading a market and reacting to a change in it are not the same job.

Two sizes of innovation

Not every innovation redraws an industry. The chapter sorts them by how new the underlying technology is and by what happens to whatever came before.

Incremental innovation — enhancing or upgrading an existing product, service, or process, which the chapter calls the most common form of innovation because it works with what a company already sells and already knows how to build.

Radical innovation — also called **disruptive innovation**, it uses a markedly new or different technology to reach new customer segments or to give existing customers significantly greater benefits, and it eventually marginalizes or replaces the products and services that were there before.

Telling them apart in a real case takes three questions, asked in this order.

- 1 Is the technology markedly new or different?** A faster version of the same machine is an upgrade; a machine that works on an entirely different principle is not, which is the first thing separating the two.
- 2 Who gets the benefit?** Radical innovation reaches customer segments the old product never served, or hands existing customers a significantly greater benefit rather than a slightly better one.
- 3 What happens to the incumbent?** If the older product or service is eventually marginalized or replaced, the change was radical; if the older product simply keeps selling in a better version, the change was incremental.

What radical innovation has already done

The chapter lists nineteen radical innovations beside the technology each one displaced or pushed to the margins, and all nineteen are reproduced below. Read them as pairs, and read the right-hand column as an industry that once looked permanent.

RADICAL INNOVATION	DISPLACED OR MARGINALIZED TECHNOLOGY
Digital photography	Chemical photography
Desktop publishing	Traditional publishing
Online stock brokerage	Full-service stock brokerage
Online retailing	Brick-and-mortar retailing
Free, downloadable greeting cards	Printed greeting cards
Distance education	Classroom education
Unmanned aircraft	Manned aircraft
Nurse practitioners	Medical doctors
Semiconductors	Vacuum tubes
Automobiles	Horses
Airplanes	Trains
Compact discs	Cassettes and records

RADICAL INNOVATION	DISPLACED OR MARGINALIZED TECHNOLOGY
MP3 players, music downloading, streaming	Compact discs and music stores
Smartphones	MP3 players, dedicated GPS navigation
Mobile telephony	Wire-line telephony
Tablets	Notebook computers
Xbox, PlayStation, smartphones	Desktop computers
3D printing	CNC milling
Camera drones	Helicopters for aerial filming and photography

Three things in that table are easy to read past, and each one changes how you should think about your own industry.

The chain does not stop — compact discs displaced cassettes and records, then music downloading and streaming displaced compact discs and music stores, and smartphones then displaced MP3 players and dedicated navigation units, so today's radical innovation can end up as a future entry in the right-hand column.

Displaced is not the same as deleted — the chapter's own heading says displaced or *marginalized*, and pairs such as distance education against classroom education, or nurse practitioners against medical doctors, describe a shift in share rather than an extinction.

Not every entry is a computer — automobiles displacing horses and airplanes displacing trains sit on the same list, because what makes an innovation radical is what it does to a market, not whether it happens to be digital.

SORT

Radical or incremental?

Each situation below is a generic, hypothetical example written for practice, with no real company named; drop it into the bucket the chapter's definitions call for, then read why it belongs there.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Radical innovation — markedly new technology, new segments or far greater benefits, and the old product ends up marginalized or replaced

- A camera maker stops using film and chemical developing altogether, and records the image on a sensor as data instead. — This is the chapter's own first pair: digital photography displaced chemical photography. The technology underneath is markedly different, not a better version of the same thing, and the older technology was pushed aside rather than improved.
- Sales of dedicated satellite navigation units collapse because one device already in the customer's pocket does the same job alongside everything else. — The chapter pairs smartphones with MP3 players and dedicated GPS navigation as displaced technologies. A general-purpose device absorbing a single-purpose one is the marginalize-or-replace pattern in its clearest form.
- Page layout that once required a print shop's specialist equipment and specialist operators is done on ordinary computers on ordinary desks. — Desktop publishing against traditional publishing is one of the chapter's pairs. The work reached people who were never going to buy typesetting equipment, which is exactly the access-new-customer-segments half of the definition.
- Aerial filming that used to require a helicopter, a pilot, and an airfield is done with a small flying camera launched from the ground. — The chapter lists camera drones as displacing helicopters for aerial filming and photography. The cost and skill barrier falls far enough to bring in customers who could never have chartered a helicopter.
- A manufacturer builds a part by adding material layer upon layer instead of cutting the shape out of a solid block. — 3D printing against CNC milling is one of the chapter's pairs. Adding material and removing material are opposite principles, so this is a markedly different technology rather than a faster version of the old one.
- A brokerage lets customers place their own trades through a website instead of telephoning a broker who advises them and charges a full commission. — Online stock brokerage against full-service stock brokerage is one of the chapter's pairs. The service reaches investors who were never worth a full-service broker's time, and it changes what the customer is paying for.

Incremental innovation — an existing product, service, or process is enhanced or upgraded

- A phone maker ships this year's model with a longer-lasting battery and a slightly brighter screen. — Enhancing or upgrading an existing product is the definition of incremental innovation, and the chapter notes this is the most common form. Nothing about the technology is markedly new, and last year's phone is not marginalized so much as superseded by its own successor.
- A grocery chain rewrites its checkout software so each scan completes a fraction of a second faster. — The process already exists and is being upgraded, so this is incremental. It may well be worth doing, because saved seconds at every lane add up, but no new technology and no new customer segment is involved.
- A notebook computer maker releases a model that is thinner than last year's and starts up more quickly. — Thinner and faster is an upgrade to an existing product, so it is incremental. Note the contrast with the chapter's own table, where tablets appear as a radical innovation that marginalized notebook computers as a category.

- A university records its existing lectures on video so students already enrolled in those classes can rewatch them later. — This upgrades a service that already exists for the same students, so it is incremental. The chapter’s radical entry is distance education against classroom education, which is different: it serves students who were never going to be in the room at all.

Why “we built a better product” is a fragile plan

Here is the trap the chapter is most concerned with. When people hear innovation, they often think only about innovative products or services — and products and services, even radical ones, can easily be copied.

Focusing only on them is therefore a dangerous path. Many leading companies, ranging from Apple to Dyson, all too frequently end up in lawsuits trying to protect their innovations from copycats, which is an expensive way to defend an advantage.

Successful organizations go beyond product and service innovation and add other kinds, such as new ways of earning revenue or entirely new business models. In their book *Ten Types of Innovation*, Keeley and colleagues set out ten distinct places a company can innovate to gain or sustain competitive advantage.

TYPE OF INNOVATION	WHAT IT CHANGES	EXAMPLES THE CHAPTER GIVES
Profit model	Finding novel ways of generating revenues from offerings	Dropbox using a freemium approach; Microsoft offering 365 on a subscription basis; GE selling “thrust as a service”
Network	Harnessing the capabilities and strengths of others	GlaxoSmithKline or Lego using open innovation for new product or service ideas; Netflix running contests for improving its movie recommendation algorithm; luxury hotels partnering with fashion designers
Structure	Using the company’s talent and assets in innovative ways	Southwest Airlines focusing on one aircraft type; Google allowing employees to use up to 20 percent of work time for personal projects
Process	Changing primary processes used to produce products or services	Toyota pioneering lean production; Zara moving fashion from initial design to stores in 3 weeks

TYPE OF INNOVATION	WHAT IT CHANGES	EXAMPLES THE CHAPTER GIVES
Product performance	Creating novel products or improving existing products through differentiation	Dyson's Airblade hand dryers; Corning's "unbreakable" Gorilla Glass; Coke's customizable Coke bottles
Product system	Creating bundles of complementary offerings	Microsoft bundling individual office programs into the 365 suite; Apple offering developer tools and an app store so developers can create novel apps; Marriott letting users test drive GoPro HERO action cams
Service	Supporting and enhancing the value of an offering	Zappos's WOW philosophy of delivering excellent customer service; Men's Wearhouse offering customers the ability to purchase free lifetime pressing
Channel	Using innovative ways to connect offerings with customers	Niketown offering immersive experiences; Nespresso partnering with hotels and airlines
Brand	Positioning the brand in innovative ways	The Virgin family of brands; German discount grocer Aldi's Trader Joe's markets
Customer engagement	Developing meaningful connections with customers	Swarm encouraging users to frequently "check in" to places; Apple tying customers to its ecosystem

Count the rows. Product performance — the new-and-better-product row that most people mean by the word innovation — is one entry out of ten. The other nine change how the money is made, whose capabilities are borrowed, how the firm is organized, how the work gets done, what the offering is bundled with, what happens after the sale, how the offering reaches you, what the name stands for, and what the relationship feels like.

EXPLORE**Six of the ten types up close**

None of these six changes the product itself, which is why they slip past a reader who hears the word innovation and pictures a better gadget; open each card and read all four panels.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Profit model innovation · How the money arrives

- **What it changes:** Finding novel ways of generating revenues from offerings, so what changes is the shape of the payment rather than the thing being paid for.
- **The chapter's examples:** Dropbox using a freemium approach, Microsoft offering 365 on a subscription basis, and GE selling "thrust as a service", a phrase that names the move from charging for a thing to charging for what the thing delivers.
- **Why it gets missed:** It gets missed because the product on the table looks identical before and after. A subscription version of a program is the same program; only the invoice has changed, and invoices are not what people photograph.
- **What a rival would have to copy:** Not the pricing page, which any competitor can rewrite in an afternoon, but the cash flow, the contracts, and the customer agreements that let a firm be paid in a different shape and at a different time than before.

Network innovation · Whose capabilities you borrow

- **What it changes:** Harnessing the capabilities and strengths of others, so the firm stops treating its own payroll as the boundary of what it can do.
- **The chapter's examples:** GlaxoSmithKline and Lego using open innovation to source new product or service ideas, Netflix running contests to improve its movie recommendation algorithm, and luxury hotels partnering with fashion designers.
- **Why it gets missed:** It gets missed because the credit lands on the finished product. When an outside contest produces a better recommendation algorithm, the customer experiences a better algorithm and never sees the contest.
- **What a rival would have to copy:** Not the announcement of an open call or a contest, which is easy to imitate, but the partners and outside contributors who choose to show up, because a competitor copying the announcement starts with an empty room.

Structure innovation · How talent and assets are arranged

- **What it changes:** Using the company's talent and assets in innovative ways, so the same people and the same equipment are organized to produce something the usual arrangement could not.
- **The chapter's examples:** Southwest Airlines focusing on one aircraft type, and Google allowing employees to use up to 20 percent of work time for personal projects.

- **Why it gets missed:** It gets missed because it is invisible from outside the building. A passenger boarding a flight cannot see that every aircraft in the fleet is the same model, even though that single decision reaches maintenance, spare parts, and crew training.
- **What a rival would have to copy:** Not a feature but an arrangement: matching a single-aircraft-type fleet would mean retiring the other models a rival already owns, has trained crews on, and stocks parts for.

Channel innovation · How the offering reaches you

- **What it changes:** Using innovative ways to connect offerings with customers, so the encounter happens somewhere the category has not traditionally been sold.
- **The chapter's examples:** Niketown offering immersive experiences, and Nespresso partnering with hotels and airlines, so that in both cases the customer meets the offering somewhere other than a conventional store.
- **Why it gets missed:** It gets missed because the product itself is unchanged, so nothing about it looks new. The innovation is the meeting place, and customers rarely think of a meeting place as an invention.
- **What a rival would have to copy:** Not the product but the route to the customer, and that route is assembled one negotiated partnership at a time, so a rival has to win each hotel and each airline rather than announce a policy.

Brand innovation · What the name stands for

- **What it changes:** Positioning the brand in innovative ways, so the name itself carries an expectation that competing names do not.
- **The chapter's examples:** The Virgin family of brands, and German discount grocer Aldi's Trader Joe's markets, which carry a name of their own rather than the parent grocer's.
- **Why it gets missed:** It gets missed because branding is often filed under advertising rather than innovation, and because the result is a feeling in the customer's head rather than an object you can put on a table.
- **What a rival would have to copy:** Not the name but what the name has come to mean, which is the accumulated result of everything done under it, so a competitor with a budget can buy attention faster than it can buy that association.

Customer engagement innovation · What the relationship feels like

- **What it changes:** Developing meaningful connections with customers, so the relationship continues between purchases instead of ending at the till.
- **The chapter's examples:** Swarm encouraging users to frequently "check in" to places, and Apple tying customers to its ecosystem.
- **Why it gets missed:** It gets missed because it produces no line on a product specification. Nothing on a specification sheet explains why a customer inside an ecosystem weighs one replacement against everything else already owned.

- **What a rival would have to copy:** Not the hardware, which the chapter says has already been imitated: the design and functionality of Apple's products have been widely copied, yet other companies find it difficult to copy its product system innovations or its customer engagement innovations.

Four pairs on that list get confused constantly, and keeping them apart is most of the skill.

Product performance against product system — product performance innovation creates novel products or differentiates existing ones, as Dyson did with its Airblade hand dryers, while product system innovation bundles complementary offerings, as Microsoft did by combining individual office programs into one suite.

Channel against network — channel innovation finds new ways to connect the offering with customers, as Nespresso did by partnering with hotels and airlines, while network innovation harnesses the capabilities and strengths of others, as Lego did by using open innovation for new product ideas.

Brand against customer engagement — brand innovation positions the name in a new way, as the Virgin family of brands does, while customer engagement innovation builds a meaningful ongoing connection, as Apple does by tying customers into its ecosystem.

Profit model against process — profit model innovation changes how revenue is generated, as Microsoft did by offering 365 on a subscription basis, while process innovation changes how the thing is produced, as Toyota did by pioneering lean production.

MATCH

Ten changes, ten types of innovation

Pair each type with the change it describes; every change here is one of the chapter's own examples, and the explanation names the company it came from.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Profit model innovation

A file-storage service gives a basic account away for nothing and charges for the extra capacity heavier users need This is Dropbox's freemium approach. The storage service is not what changed; the way revenue is generated from it is.

Network innovation

A toy company invites people outside the firm to submit ideas for new products instead of relying only on its own designers Lego and GlaxoSmithKline are the chapter's open innovation

examples, and the point is harnessing the capabilities and strengths of others.

Structure innovation

An airline flies a single type of aircraft, so crews, spare parts, and maintenance all serve one machine Southwest Airlines is the example, and it is structural because it changes how the company's talent and assets are arranged rather than what it sells.

Process innovation

A clothing retailer compresses the path from first design to clothes on the sales floor down to three weeks Zara is the example. The garments are ordinary garments; the primary process used to produce and deliver them is what was changed.

Product performance innovation

A glassmaker develops a cover glass tough enough to be marketed as unbreakable Corning's Gorilla Glass is the example, and it is the classic case of creating a novel product or differentiating an existing one.

Product system innovation

Separate word processing, spreadsheet, and presentation programs are sold together as a single suite Microsoft bundling individual office programs into 365 is the example, and bundling complementary offerings is what defines this type.

Service innovation

A menswear retailer lets customers buy free pressing of their garments for life Men's Wearhouse is the example. The suit is unchanged; what surrounds and supports the purchase is what enhances its value.

Channel innovation

A coffee-machine maker places its product with hotels and airlines so travellers meet it far from any store Nespresso is the example, and the innovation is the route to the customer rather than anything about the coffee or the machine.

Brand innovation

A German discount grocer owns a chain of markets that trade under a completely separate name Aldi's Trader Joe's markets are the chapter's example of positioning a brand in an innovative way.

Customer engagement innovation

A device maker ties customers into an ecosystem, so each purchase makes sense mainly in relation to the ones already made Apple is the example, and the chapter notes rivals find this far harder to copy than the design and functionality of the products themselves.

CHECK YOURSELF

Innovation, radical and incremental

Four options, one best answer; every situation here that names no company is a generic hypothetical written for practice, and every named company comes from the chapter, so read every explanation, including the ones for options you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A research team designs a working prototype of a new kind of battery, publishes the design, and the company then shelves it. By the chapter's definition, what does the company have?

- **A.** An innovation, because a new product was successfully created and demonstrated — Creating and demonstrating something new is real work, but the chapter draws the line elsewhere: innovation involves creating new products, processes, or services that return value to the organization. A shelved prototype has returned nothing yet.
- **B.** An invention, because innovation requires that the new thing return value to the organization — **correct.** The chapter contrasts innovation with merely inventing new products, services, or processes, and says innovation involves realizing the value. Until the design earns money, saves money, or wins a customer, the company has an invention.
- **C.** A radical innovation, because the underlying technology is markedly different from existing batteries — The technology may well be markedly different, which is half of the radical definition, but the other half is that it reaches new customer segments or delivers significantly greater benefits and eventually marginalizes what came before. A shelved design reaches nobody.
- **D.** An incremental innovation, because the design improves on batteries the company already sells — This would fit if the design were an upgrade the company shipped to its existing customers, since incremental innovation means enhancing or upgrading existing products, services, or processes. Nothing here was shipped or upgraded.

2. Which of the following best matches the chapter's definition of a radical innovation?

- **A.** A software vendor releases an annual update with a redesigned menu and faster file handling — A redesigned menu and faster file handling enhance an existing product, which the chapter defines as incremental innovation and calls the most common form. Useful, and genuinely innovation, but not radical.
- **B.** A manufacturer negotiates lower prices from its suppliers and passes some of the saving to customers — Cheaper supplies can improve competitiveness through cost, and the chapter does say advantage may come through cost or differentiation, but nothing here involves a markedly new or different technology or a displaced product.
- **C.** A device using a markedly different technology serves customers the old product never reached, and the old product is eventually marginalized — **correct.** Radical innovations use a markedly new or different technology to access new customer segments or provide significantly greater benefits to existing customers, and eventually marginalize or replace existing products or services.
- **D.** A company spends more on research than any of its competitors in the same industry — Research spending is an input, not an outcome. The chapter measures innovation by the value returned to the organization, so a large budget that produces nothing customers use has not produced innovation of any size.

3. The chapter's table pairs compact discs with cassettes and records, then pairs music downloading and streaming with compact discs and music stores. What does that sequence illustrate?

- **A.** That a technology which displaced an older one can itself be displaced later, so radical innovation is a chain rather than a finish line — **correct**. The compact disc appears in the table first as a radical innovation and later in the displaced column, and smartphones then displace the MP3 players that displaced it, which is why leading today guarantees nothing about the next round.
- **B.** That the chapter regards the compact disc as an incremental improvement on the cassette rather than a radical innovation — The table lists compact discs in the left-hand column, among the radical innovations, precisely because they used a markedly different technology and marginalized cassettes and records rather than improving them.
- **C.** That displacement moves slowly enough for the leading firm in each round to move into the technology that replaces it — The table records what replaced what and says nothing about the incumbent making the jump. The chapter points the other way, noting that market leaders and strong brands often find it difficult to react appropriately to new trends.
- **D.** That a radical innovation displaces a technology from its own industry, since one recorded-music format gave way to the next — The same table pairs smartphones with dedicated GPS navigation, automobiles with horses, and 3D printing with CNC milling, so the displaced technology frequently sits in a different industry from the innovation that displaced it.

Combining types is what makes an advantage stick

Because product performance innovations can often be copied easily, companies frequently combine several types at once to sustain a competitive advantage. The chapter offers two cases worth studying side by side.

Apple — typically known for its product innovations, and the design and functionality of those products have been widely imitated, yet other companies find it difficult to copy Apple's product system innovations or its customer engagement innovations.

Dell — challenged established computer manufacturers by combining profit model innovation, collecting money before building the product, with process innovation in the built-to-order model and channel innovation in selling computers online rather than in stores, as well as other types of innovation the chapter does not name.

A rival who copies your product has copied one row of ten; a rival who wants to copy several rows at once has to rebuild the company, which is why combinations outlast clever products.

Where information systems come in

Most, if not all, of these innovations are enabled by information systems, and several of them would not be possible without one. The chapter gives three examples that make the dependency concrete.

Airbnb — platform-based business models such as this one would not be possible without the internet, because the platform's entire job is to connect two groups of people who would otherwise never find each other.

Uber — its business model could not exist without the mobile devices carried by riders and drivers, and its dynamic pricing feature, called surge pricing, would not work without real-time analysis and prediction of traffic conditions and demand for services.

Manufacturers of physical goods — process innovations enabled by robotics and the Industrial Internet of Things offer improvements in efficiency, product quality, agility, and flexibility, allowing companies to mass-produce customized products.

That is why an organization often has to deploy new, state-of-the-art technologies, or redeploy existing ones in clever new ways, to gain or sustain a competitive advantage.

A firm can choose instead to keep continually upgrading its older systems rather than investing in new ones. The chapter's warning is about what those improvements buy: often, at best, only a short-lived competitive edge.

CHECK YOURSELF

Ten types, combinations, and the systems underneath

Three questions on where a company can innovate and on why information systems keep turning up in the answer; the manager quoted in the first one is hypothetical, while every company named is one the chapter names.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A manager says: "Our innovation strategy is simple. We build the best product in the category and defend the patents." What is the strongest objection the chapter supports?

- **A.** Competitive advantage comes from being the low-cost producer, so a strategy built on the best product aims at the wrong target — The chapter says the opposite about the target: innovation is key for organizations attempting to gain or sustain competitive advantage, be it through cost or differentiation. Building the best product is a legitimate aim; leaning on it alone is the problem.
- **B.** Product performance is one of ten types of innovation, and products, even radical ones, can easily be copied, so a product-only strategy is a dangerous path — **correct**. The chapter warns that innovative products or services, even radical innovations, can easily be copied and that focusing only on them is a dangerous path, which is why it presents ten types rather than one.

- **C.** Improving an existing product does not count as innovation, so defending a product line is not an innovation strategy at all — Enhancing or upgrading an existing product, service, or process is incremental innovation, which the chapter calls the most common form of innovation. The weakness in this plan is its narrowness, not that improvement fails to count.
- **D.** Innovation is the responsibility of the research department, so a manager outside it should not be setting an innovation strategy — Nothing in the chapter confines innovation to a research department. Profit model, structure, channel, and customer engagement innovations sit with the parts of the business that set pricing, organization, routes to market, and relationships.

2. The chapter describes Dell challenging established computer manufacturers by collecting money before building the product, building to order, and selling online rather than in stores. Which types of innovation are those three, in that order?

- **A.** Brand, service, and network innovation — Brand innovation positions the name in a new way, service innovation supports and enhances the offering, and network innovation harnesses the strengths of others. None of the three describes when cash is collected, how a machine is assembled, or where it is sold.
- **B.** Product performance, product system, and service innovation — These three are all about the offering itself: novel or differentiated products, bundles of complementary offerings, and support around the sale. Dell's changes were to its finances, its assembly, and its route to market.
- **C.** Profit model, process, and channel innovation — **correct.** Collecting money before building is a profit model innovation, the built-to-order model is a process innovation, and selling online rather than in stores is a channel innovation, and the chapter presents the combination as the reason it could challenge incumbents.
- **D.** Structure, customer engagement, and brand innovation — Structure innovation rearranges talent and assets, customer engagement builds an ongoing relationship, and brand innovation repositions the name. Dell may have done these too, but they are not the three changes described here.

3. Why does the chapter say Uber's surge pricing illustrates the link between innovation and information systems?

- **A.** Because riders and drivers carry mobile devices, which is the dependency the chapter attaches to surge pricing — Mobile devices are the chapter's condition for Uber's business model existing at all, not for surge pricing in particular. Surge pricing is tied to something narrower: real-time analysis and prediction of traffic and demand.
- **B.** Because the feature depends on real-time analysis and prediction of traffic conditions and demand, so with no system there is no feature — **correct.** The chapter states that Uber could not implement surge pricing without real-time analysis and prediction of traffic conditions and demand for services, just as its business model could not exist without the mobile devices riders and drivers carry.

- **C.** Because surge pricing raises revenue, which is the measure the chapter uses to judge whether an innovation succeeded — Revenue matters, since innovation is defined as returning value to the organization, but the chapter raises surge pricing to show a capability that only an information system can supply, not to score its profitability.
- **D.** Because a business model built on mobile devices is inherently harder for competitors to copy than one built on physical stores — The chapter argues nearly the reverse elsewhere: information systems are frequently bought from vendors and rivals can copy them, so a mobile business model carries no built-in protection.

Innovating in practice

Knowing that innovation matters is the easy part. This section is the hard part: why innovating with information systems so often fails, what a disciplined process for doing it looks like, and what an organization must change about itself first.

Nobody sees it coming

In 1999 Sir John Maddox, a physicist who edited the journal *Nature* for 22 years, was quoted in *Scientific American* saying that the most important discoveries of the next 50 years are likely to be ones we cannot now even conceive. The internet makes his point: in 1999 it was not on the radar screens of many business organizations, and firms with websites were mostly providing an electronic brochure.

The chapter's answer is structural. Organizations that want to stay relevant must become **bimodal**: one part exploits existing capabilities and technologies, a focus on **earning**, while another discovers and develops new interactions or service offerings enabled by increasing digital density, a focus on **learning**.

Four reasons innovating is hard

The chapter gives four separate reasons that innovating with information systems is harder than it looks, and they are genuinely different problems rather than four ways of saying one thing.

Successful innovation is difficult — systems are often bought from or built by someone else, so a firm typically cannot patent its use of one, the way a soft-drink company can patent a cola formula.

Innovation is often fleeting — even an in-house system uses hardware, software, and networking components rivals can also buy, so smart rivals quickly duplicate or improve on any use that visibly wins.

Innovation is often risky — choosing among innovative investments always entails risk, because the option most experts call superior is not always the one the market settles on.

Innovation choices are often difficult — picking a technology is like hitting one of several equally attractive fast-moving targets, because forecasting the growth, use, and importance of an emerging technology has defeated experts and organizations again and again.

The patent point is where intuition usually fails. The data in a system may be proprietary; the system itself typically is not, particularly when purchased from a vendor. One exception matters: a system

developed by a consultant or an outsourcing partner is proprietary to the organization.

Two things do resist copying. Amazon's patented "one-click" ordering process has been successfully defended in the courts, and it is often difficult to copy the "secret sauce" key to how a system functions or succeeds.

Risk is easiest to see in consumer electronics. Most experts agreed Sony's Betamax had better recording and playback quality than JVC's VHS, and VHS won the marketplace anyway; later, Sony's Blu-ray beat the HD DVD format Microsoft and Toshiba backed. For an organization the same choice is far riskier, given the size and mission-critical nature of the investment.

Forecasting is why the choices stay hard. The 1994 *Technology Forecast* from the consulting firm Price Waterhouse, now PwC, mentioned the word internet on five pages of a 750-page document; the next year more than 75 pages addressed it, and by the 1997 briefing it was pervasive throughout.

EXPLORE

The four reasons innovating with information systems is hard

Open each obstacle and read all four panels; the last panel is the one that changes what you would actually do.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Successful innovation is difficult · You usually cannot own it

- **What the difficulty is:** There are limits to using emerging information systems to innovate and gain or sustain a competitive advantage, because information systems are often bought from or built by someone else.
- **Why it happens:** If the systems are purchased from a vendor, they are usually not proprietary technologies owned by the organization. The data in the system may be proprietary, but the information system typically is not.
- **The chapter's example:** A soft-drink company can patent the formula of a cola and a pharmaceutical company can patent a new drug, yet an organization typically cannot patent its use of an information system.
- **What it means for a decision:** One exception is worth remembering: if the system is developed by a consultant or an outsourcing partner, it is proprietary to the organization. And because many business models have information systems at their very core, organizations are often forced to develop such systems regardless of what can be protected.

Innovation is often fleeting · Assume the copy is coming

- **What the difficulty is:** Given the pace of change in the digital world, advantages gained by innovations often have a limited life span, so the practical question is less whether the advantage erodes than how long it holds.

- **Why it happens:** Even where an organization developed an innovative system in-house, it usually did so with hardware, software, and networking components that others can also purchase, so rivals can copy innovative information systems.
- **The chapter's example:** The chapter's classic counterexample is Amazon's patented "one-click" ordering process, which has been successfully defended in the courts; it adds that the "secret sauce" key to the functioning or success of a system is often difficult to copy.
- **What it means for a decision:** Plan for duplication rather than against it. If use of a new system causes one organization to gain a significant advantage over others, smart rivals are quick to duplicate or improve on that use, and rivals can copy innovative products, services, and processes as well, so the question worth asking early is which part of the advantage is genuinely hard to copy.

Innovation is often risky · The better technology can lose

- **What the difficulty is:** Choosing among innovative investments in information systems, or among potential product, service, or process innovations, always entails risk that no amount of technical analysis removes.
- **Why it happens:** The risk does not come from bad analysis; it comes from the market, not the expert consensus, settling the outcome. In the early days of the videocassette recorder, most experts agreed that Sony's Betamax had superior recording and playback quality, yet JVC's VHS ultimately won the battle in the marketplace, so people who made the "smart" choice at the time chose badly.
- **The chapter's example:** The rematch came with high-definition discs: Microsoft, Toshiba, and many others backed the HD DVD format while Sony led the fight for Blu-ray and incorporated it into the PlayStation 3. Sony won, with the HD DVD Promotion Group dissolving in early 2008.
- **What it means for a decision:** For an organization the stakes are higher than for a shopper. Choosing among the plethora of available innovative technologies is far riskier given the size and often mission-critical nature of the investment, whereas a suboptimal video player is usually just disappointing.

Innovation choices are often difficult · Forecasts keep missing

- **What the difficulty is:** Choosing new technologies in the information systems area to develop innovative products, services, or processes is like trying to hit one of several equally attractive fast-moving targets.
- **Why it happens:** Given the pace of research and development in the information systems and components area, and the increasing digital density, staying current has been nearly impossible. Intel founder Gordon Moore predicted that the number of transistors squeezed onto a chip would double every 24 months, a number now often reduced to 18 months, and some computer hardware and software firms roll out new versions of their products every three months.

- **The chapter's example:** The 1994 Technology Forecast prepared by the major consulting firm Price Waterhouse, now PwC, mentioned the word internet on only five pages of a 750-page document; the next year more than 75 pages addressed it, and by the 1997 briefing it was a pervasive topic throughout.
- **What it means for a decision:** Back in 1994 it would have been difficult, perhaps even foolish, to forecast that growth. The chapter's table of predictions that were not quite correct makes the same point company by company, including Western Union's 1876 internal memo calling the telephone inherently of no value and Steve Ballmer saying in 2007 that there was no chance the iPhone would get any significant market share, so a technology choice is better treated as a hypothesis to test cheaply than as a forecast to commit to.

What actually lasts

So what survives contact with a competitor? If you use information systems for operating efficiencies, rivals can just as easily adopt the same types of systems and achieve the same gains. Two routes hold up longer.

Make the product or service unique — the Israeli tech startup Waze built a routing and navigation algorithm combining map data, real-time traffic information, and crowdsourced data, and Google purchased the app in 2013.

Raise the customer's switching costs — Salesforce.com stores the entire history of a firm's interactions with each customer, so switching to a rival would cost significant time and money and take years to rebuild.

Efficiency gains are borrowable; a customer's accumulated history is not, which is why the chapter treats uniqueness and switching costs as the durable forms of advantage.

The lean startup methodology

Traditionally companies would develop a product and then try to find a market for it, which the chapter calls inside-out thinking. Innovative companies reverse it, in outside-in thinking: they focus on the ends, meaning the customer's **job-to-be-done** and the friction points preventing the customer from accomplishing their goals.

That reversal matters more now because incumbents face competition from many small, nimble startups, not just a few traditional rivals. The **lean startup methodology**, pioneered by Eric Ries, gives the response a shape: a company first searches for a viable product and business model, and executes that model only after finding a fit between product and market.

Customer discovery, in search — you establish whether the perceived problem actually exists, whether a market for a solution exists, and whether your envisioned solution addresses it, which means “getting out of the building” before building anything.

Customer validation, in search — you test whether customers would be willing to pay, using a minimum viable product, because a finished product is an expensive way to learn you were wrong.

Customer creation, in execution — you test whether it is possible to establish a repeatable business model, now that the product has stopped being the open question.

Company building, in execution — the final step builds the company and scales operations and the organization around an already validated model.

A **minimum viable product** (MVP) contains just the features needed to demonstrate how your solution meets the problem, so its purpose is to test a hypothesis cheaply.

If a hypothesis fails during the search, it is time to **pivot** — change direction while staying grounded in what you learned. Instagram began as Burbn, an app of features from check-ins to gaming; when users did not value them, the founders stripped it to photo sharing and commenting.

Those iterations form the **build-measure-learn** cycle, which enables **validated learning**: empirically testing hypotheses through customer feedback to improve future iterations. You build the MVP quickly, measure by collecting data such as questionnaire responses, then learn by deciding whether to pivot or persevere. Established companies from Alaska Airlines to Philips to 3M use the method too.

SEQUENCE

The lean startup search, from first conversation to finished company

Put the six moves into the order the chapter describes, remembering that everything in the search phase happens before anything in the execution phase.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The lean startup methodology, pioneered by Eric Ries, splits the work into a search phase and an execution phase: a company first searches for a viable product and business model, and only executes that business model once it has found a fit between the product and the market.

1. **Get out of the building and talk to potential customers, to find out whether the perceived problem actually exists and whether there would be a market for a hypothetical solution.** — Customer discovery starts with the problem, not the product. The chapter is explicit that this entails talking to potential customers before starting to build something, which is why it comes first.
2. **Still within customer discovery, establish whether your envisioned solution would actually address the problem you have just confirmed is real.** — A real problem and a solution that fits it are two separate hypotheses. The second step of customer discovery tests the second one, and skipping it means building for a problem you cannot solve.
3. **Build a minimum viable product containing just the features needed to demonstrate how your solution meets the problem.** — The MVP is the instrument for the next test, so it comes after both discovery hypotheses hold and before anyone is asked to pay. Building it earlier means testing a solution to a problem nobody confirmed.
4. **Move into customer validation to test whether customers would be willing to pay, running build-measure-learn cycles and pivoting whenever a hypothesis does not hold.** — Validation is where willingness to pay is tested rather than assumed, and it is where the build-measure-learn cycle produces validated learning. A failing hypothesis here triggers a pivot back into discovery, not a push forward.
5. **Cross from search into execution with customer creation, testing the hypothesis of whether it is possible to establish a repeatable business model.** — The dividing line matters: the company only executes the business model once it has found product/market fit, so this step is meaningless before validation succeeds.
6. **Build the company, scaling operations and the organization around the model that has now been validated.** — The final step entails building the company and scaling operations and the organization, which is the only point at which heavy investment is justified by evidence rather than by hope.

Open innovation

Because innovating internally is this hard, organizations increasingly realize the potential for innovation often exists outside their boundaries, and they try to harness the creativity of external stakeholders.

Open innovation is the process of integrating external stakeholders into the innovation process.

Eli Lilly created InnoCentive, where companies post scientific problems, everybody can take a shot at them, and a reward is usually paid to a successful solver. P&G, Lego with Lego Ideas, and Heineken with Brewhouse have also used open innovation to solicit ideas for novel products, services, or processes.

What that arrangement changes is worth stating precisely.

The research network becomes ad hoc — companies rely less on a dedicated research-and-development department or on hiring specialists, because the problem is posted rather than staffed.

Contributors work on their own terms — people use spare time and expertise to solve problems and earn rewards, so effort arrives from outside the payroll.

The price is openness — the arrangement often involves making organizational data and knowledge accessible to outsiders, so benefits must be balanced against the potential drawbacks.

CHECK YOURSELF

Why innovating is hard, and what lasts

Four options, one best answer; read every explanation, including the ones for the options you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A manager says: “Our new logistics system gives us an edge, and because we built it in-house, competitors cannot copy it.” What is the accurate correction?

- **A.** The manager is right, because a system developed in-house is a proprietary technology the organization can patent — An organization typically cannot patent its use of an information system, unlike a soft-drink company patenting a cola formula or a pharmaceutical firm patenting a drug. Building it yourself changes who maintains it, not whether that use can be legally protected.
- **B.** The manager is right, because information systems developed in-house are harder for rivals to copy than innovative products, services, or processes are — The chapter puts them in the same position rather than ranking them. Rivals can copy innovative information systems, and rivals can copy innovative products, services, and processes as well, which is why the caution applies across the board.
- **C.** Even an in-house system is built with hardware, software, and networking components others can also purchase, so the advantage can be short-lived — **correct.** Even where an organization develops a system in-house it uses components others can buy, and smart rivals are quick to duplicate or improve on a use of a system that produces a significant advantage.
- **D.** The manager is wrong, because any advantage the system creates belongs to the vendors that supplied its hardware, software, and networking components — Those components are purchasable by anyone, which is exactly why the advantage tends to be short-lived, but the organization is still the party that gains it. The chapter also names cases where the gain held, such as Amazon’s defended one-click patent and a hard-to-copy secret sauce.

2. Which pair of facts does the chapter use specifically to show that forecasting emerging technologies is difficult?

- **A.** A 1994 consulting forecast mentioned the internet on five of 750 pages, and by the 1997 briefing the internet was a pervasive topic throughout — **correct.** The Price Waterhouse forecast went from five pages to more than 75 the following year to pervasive coverage three years later, which shows how fast an emerging technology can outrun expert expectation.
- **B.** Betamax had better recording quality than VHS, and Sony later led Blu-ray to victory over HD DVD — Both facts are accurate and both come from the chapter, but they illustrate the risk of choosing wrong between competing formats rather than the difficulty of forecasting which technologies will matter at all.
- **C.** Eli Lilly created InnoCentive, and P&G, Lego, and Heineken solicited ideas from outsiders — These are the open innovation examples. They describe one response to how hard innovation is, not evidence about anyone's ability to predict which technologies will grow.
- **D.** Instagram began as an app called Burbn, and its founders stripped it down to photo sharing and commenting — That is the chapter's illustration of a pivot inside the lean startup methodology, showing a company changing direction based on what it learned from users, not a claim about technology forecasting.

3. Your firm's information systems have sharply cut operating costs. Why is that unlikely to yield a lasting competitive advantage, and what would?

- **A.** Because savings in operating costs stay inside the firm, so they cannot influence how customers choose between one supplier and another — The chapter treats innovation as key to gaining or sustaining competitive advantage, be it through cost or differentiation, so a cost position is a legitimate route and customers do feel it in prices. What undermines this particular advantage is how easily rivals can imitate it, not the decision to chase efficiency.
- **B.** Because efficiency gains show up in published financial results, so rivals can read them there and respond in the same quarter — Rivals do not need to read a competitor's financial statements to catch up here. The chapter's mechanism is simpler and harder to hide from: rivals can just as easily adopt the same types of information systems and achieve the same gains.
- **C.** Because efficiency gains decay on their own over time, so the firm should relaunch the same system on a regular schedule — Nothing in the chapter says a working system decays by itself; what erodes the advantage is imitation by rivals. The chapter does warn that continually upgrading older systems rather than investing in new ones can at best give a short-lived competitive edge, so a relaunch schedule spends money without addressing the cause.
- **D.** Because rivals can just as easily adopt the same types of systems and achieve the same gains; making the product unique or raising switching costs lasts longer — **correct.** Rivals can adopt the same efficiency systems, so the durable routes are uniqueness and high switching costs — Waze's distinctive routing algorithm and Salesforce.com's accumulated record of every customer interaction are the chapter's two examples.

What an organization needs in place

Established organizations can benefit from radical approaches to innovation, but that sometimes requires fundamental changes to the organization itself: its processes, resources, and risk tolerance must be capable of adapting to and sustaining the development and implementation process.

Process requirements — organizational processes and incentive systems must change, because developing an MVP with agile approaches requires close collaboration of cross-functional teams, which is uncommon in established organizations.

Resource requirements — the firm needs enough employees with the relevant knowledge, skills, and time to quickly build MVPs, or able systems partners to outsource that development to.

Risk tolerance requirements — members must tolerate risk and uncertainty and be willing to test products that may not meet customers' requirements, and to fail fast if needed.

Two details are easy to skip and expensive to miss. Accountability and incentives belong at the team level rather than the individual level, because an experiment is a shared act. And the method needs a startup-like budgeting model, where funds are approved as a project moves through the stages, or can quickly be cut if the anticipated results do not appear.

Risk tolerance reaches further than products: open innovation typically requires sharing organizational knowledge with outside partners. Where people within the organization desire low risk, gambling on cutting-edge innovations with a lean startup approach will probably not be tolerable.

SELF-CHECK

Could this organization actually innovate?

Rate each statement honestly; anything you cannot do yet has a pointer to the exact place in this section to reread.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can name the three organizational requirements for innovation and say what each one actually demands of the company. The section on what an organization needs in place: process, resource, and risk tolerance requirements.
- I can explain why accountability and incentives have to sit at the team level rather than the individual level. The paragraph following the three requirements, where cross-functional collaboration and team-level incentives appear together.
- I can describe the budgeting model the lean startup methodology needs, and say why a single annual approval does not fit it. The startup-like model where funds are approved as a project moves through the stages, or can quickly be cut.

- I can state the two ways an organization can satisfy the resource requirement when it lacks the right people in-house. Enough employees with the relevant knowledge, skills, and time to build MVPs quickly, or able systems partners to outsource to.
- I can explain why risk tolerance applies to open innovation and not only to unproven products. The risk tolerance paragraph, on sharing organizational knowledge with outside partners.
- I can say what happens to a lean startup effort inside an organization whose members desire low risk. The closing line of the risk tolerance requirement, on gambling on cutting-edge innovations.
- I can explain what a bimodal organization is and why the chapter recommends becoming one. The opening of this section, on the part of the organization that earns and the part that learns.
- I can tell the difference between a requirement the organization must change and an obstacle it can only plan around. Read the three requirements against the four reasons innovating is hard, and notice which list the organization controls.

Three ways to think about radical innovations

The chapter closes by turning three familiar habits around, each uncomfortable on purpose.

Put technology ahead of strategy — with increasing digital density, technology is so important to strategy and success that you begin there, because retrofitting technology into an aging strategy is, many would argue, doomed.

Think in terms of problems before designing solutions — drop inside-out thinking for outside-in thinking, because customers understand their problems well but are not experts in solutions or in new technologies.

Innovation is continuous — innovative companies constantly look for the next new thing and keep getting out of the building to test new hypotheses, because the rate of technology evolution will not slow down.

Steve Jobs of Apple is quoted on the second point: you cannot just ask people what they want and then give it to them, because by the time you build it they will want something new. If your customers already know about a technology, your competitors probably do too, so it is not the one to rest your advantage on.

The third point is the troubling one. If you lead an industry you must continually learn to embrace and exploit radical innovations, potentially destroying your existing core business while building a new business around the innovation. If you fail to do this, your competition may do it for you.

DECIDE**Funding an innovation the disciplined way**

Work through four decisions in this clearly hypothetical situation, and read every outcome before moving on, including the ones you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

This is a hypothetical practice situation, not a report about any real company. You lead a small team at a regional appliance retailer whose leadership has set aside money for one innovation effort. The conditions below are supplied only so the reasoning has something to work on.

Decision 1. Two competing smart-home platforms are fighting to become the industry standard, and your engineers report that one of them is clearly the better technology. Leadership wants to commit the entire product line to it this quarter.

- Commit to the technically superior platform, on the reasoning that the better technology wins in the end — This is the Betamax mistake exactly. Most experts agreed Betamax had superior recording and playback quality, and VHS ultimately won the battle in the marketplace, so the people who made the “smart” choice at the time chose badly. Technical merit is evidence about the product, not a forecast of the market.
- Build a minimum viable product on one platform, designed so that moving to the other would be cheap — **the stronger call.** Right. An MVP carries just the features needed to demonstrate how your solution meets the problem, so it tests the idea without betting the product line on a format war. It also keeps the pivot affordable, and a pivot is a change of direction grounded in what you learned.
- Delay the entire effort until one of the two formats is formally declared the industry standard — Waiting trades format risk for a worse one. Some hardware and software firms roll out new versions every three months, so a company that only moves once uncertainty clears arrives after rivals have started accumulating the customer history that makes an advantage durable.

Decision 2. Your team proposes six months of development to finish the complete feature set before showing anything to a single customer.

- Approve the plan, because showing customers a half-finished product would damage the company’s reputation — This is inside-out thinking: develop a product, then look for a market for it. The chapter’s alternative is outside-in thinking, which starts from the customer’s job-to-be-done and the friction points in the way, and it requires getting out of the building before the building starts.
- Skip customer conversations and run a large advertising campaign to see whether interest exists — Marketing response is a legitimate measurement inside the build-measure-learn cycle, which names the effectiveness of search engine marketing campaigns as one kind of data. But it measures reactions to a message, and you have not yet confirmed that the underlying problem exists or that your solution addresses it.

- Send the team to talk to potential customers first, confirming that the problem is real and that your solution would address it — **the stronger call**. Right, and that is customer discovery in both of its steps. The chapter is blunt about the method: talk to potential customers, getting out of the building, before starting to build something. A six-month build tests nothing until the day it ships.

Decision 3. The MVP validates, but one hard technical problem inside it sits outside your team's expertise, and nobody currently on staff can solve it.

- Post the problem publicly and pay a reward to whoever solves it, accepting that some organizational knowledge becomes visible — **the stronger call**. Right. This is open innovation, the process of integrating external stakeholders into the innovation process, as Eli Lilly did with InnoCentive, where companies post problems, everybody can take a shot, and a reward is usually paid to a successful solver. You also named the cost correctly: such initiatives often involve making organizational data and knowledge accessible to outsiders.
- Hire and train a specialist team of your own over the next year — Resources genuinely are one of the three organizational requirements, and having employees with the relevant knowledge, skills, and time to quickly build MVPs is what that requirement asks for. The trouble is the clock: a year of recruiting is not quick, and the chapter offers routes designed for exactly this gap.
- Drop the feature, on the grounds that the organization cannot build what it cannot staff — The resource requirement offers an alternative before it offers a surrender: the organization must have resources and able systems partners available to outsource the development of such systems if necessary. Dropping the feature answers a staffing problem by abandoning a customer problem you already validated.

Decision 4. Finance will fund the work only through a single annual approval, and the company rewards individual employees for hitting their own personal targets.

- Accept the terms, because securing the funding matters more than the shape the funding takes — The shape is the part that decides whether the method can run at all. The iterative nature of the lean startup methodology necessitates a more startup-like budgeting model, where funds are approved as the project moves through the stages, or can quickly be cut if the project does not bring the anticipated results.
- Keep the annual budget and reassure leadership that none of the experiments will fail — A promise like that quietly converts a search phase into an execution phase. Validated learning means empirically testing assumptions or hypotheses through customer feedback, and a hypothesis nobody is permitted to disprove is not being tested, only announced.
- Ask for staged funding and team-level accountability, and say plainly that some experiments are expected to fail — **the stronger call**. Right on all three counts. The chapter asks for funds approved as the project moves through the stages, for accountability and incentives at the team level rather than the individual level, and for members with an appropriate tolerance for risk and uncertainty and the willingness to fail fast.

Startups and crowdfunding

Startups are typically technology-based new ventures with high potential for scalability and growth, often cofounded by entrepreneurs, developers, or designers with a promising idea. They were long built around services or platforms enabled by information systems, but 3D printing has enabled hardware startups built around physical objects, from Coin's electronic credit card to the Square Reader.

Prototyping quickly is not the whole job. Reaching manufactured stock also requires knowledge of the business environment, connections with the business community, and often substantial financial resources, and startups seek that support in four places.

Startup incubators — they provide various types of education and other resources, which is what an unproven idea usually needs first.

Startup accelerators — they typically provide education, investment, and intense mentorship for cohorts of startups, so money and guidance arrive together.

Angel investors — they can be a valuable source of funding, but often provide only limited education or mentorship, so the venture supplies its own direction.

Crowdfunding — through platforms such as Kickstarter or Indiegogo, it secures business financing from individuals in the marketplace, the "crowd," typically for a consumer-based product.

Crowdfunding has one rule that makes it unlike ordinary investment. Backers pledge a certain amount in return for certain benefits, generally one count of the item backed. If, and only if, the campaign reaches its funding goal are backers charged and the product produced; otherwise the campaign fails and nobody is charged. Kickstarter has helped fund more than 280,000 projects totaling more than US\$8 billion in pledges.

Equity crowdfunding, sometimes called crowdfinancing, secures financing the same way but changes what comes back: backers receive equity in return for their investment rather than a product.

Innovation, as this chapter treats it, is not a flash of cleverness; it is a search process with a budget model, a team structure, and a tolerance for being wrong in public.

CHECK YOURSELF

Lean startup, open innovation, and funding the venture

Three questions on the method itself and on where the money and the ideas come from.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What is a minimum viable product for?

- **A.** Releasing a cheap cut-down edition of a finished product to customers who cannot afford the full version — An MVP is not a budget edition aimed at a lower-priced segment. It belongs to the search phase, and its audience is the potential customers whose problem and willingness to pay you are still trying to confirm.
- **B.** Containing just the features needed to demonstrate how your solution meets the problem, so a hypothesis can be tested — **correct.** The MVP exists so that customer validation can test whether customers would be willing to pay, which is why it carries only the features that demonstrate the solution.
- **C.** Proving to investors that the engineering team can deliver a complete product on schedule — Delivery capability is a genuine concern, and the resource requirement does ask whether an organization has people who can build MVPs quickly. But the MVP itself is an instrument for learning about customers, not a demonstration of engineering throughput.
- **D.** Locking in the product design before customer feedback arrives, so the team avoids expensive changes later — This inverts the method. The build-measure-learn cycle exists precisely so that feedback changes the product, and discovering that a hypothesis does not hold is the signal to pivot rather than something to design around.

2. A company posts an unsolved technical problem on a public platform and pays a reward to whoever solves it. What has it gained, and what has it accepted?

- **A.** It has gained a permanent research department at no cost, and accepted that it must pay a reward to everyone who submits an attempt — The network created this way is ad hoc rather than a permanent department, and the reward is usually paid to a successful solver rather than to everyone who takes a shot at the problem. The cost the chapter names explicitly is a different one entirely.
- **B.** It has gained legal ownership of every idea submitted, and accepted a longer development timeline — Ownership of every submission is not what the chapter describes; it describes rewards paid to successful solvers. A longer timeline is not the point either, since open innovation is a response to how slow and hard internal innovation can be.
- **C.** It has gained a way to avoid hiring specialists, and accepted that it can no longer run a research department of its own — The first half is close, because companies can rely less on a dedicated research-and-development department or on hiring specialists. The second half does not follow, since relying less on an internal department is not the same as giving it up.
- **D.** It has gained an ad hoc research-and-development network, and accepted that organizational data and knowledge become accessible to outsiders — **correct.** An ad hoc research-and-development network is created, and integrating external stakeholders often involves making organizational data and knowledge accessible to them, so the benefits must be balanced against that exposure.

3. A hardware startup runs a Kickstarter campaign for a new gadget and falls short of its funding goal. What happens, and what would equity crowdfunding have changed?

- **A.** The campaign fails and no backers are charged any money; under equity crowdfunding, backers would receive equity rather than the product — **correct**. Backers are charged if and only if the campaign reaches its funding goal, and equity crowdfunding, sometimes called crowdfunding, is the version where backers receive equity in return for their investment.
- **B.** Backers are charged what they pledged and then receive a partial refund; equity crowdfunding would have removed the funding goal entirely — Backers are charged only when the goal is met, so there is nothing to refund. Equity crowdfunding changes what backers receive for their money rather than whether a campaign carries a target at all.
- **C.** The campaign proceeds at a reduced scale, and equity crowdfunding would have added mentorship alongside the money — A campaign that misses its goal fails rather than shrinking to fit. Mentorship belongs to incubators and accelerators, with accelerators typically providing education, investment, and intense mentorship for cohorts of startups.
- **D.** Backers automatically become part owners of the venture, and equity crowdfunding would have converted their pledges into ordinary pre-orders — This reverses the two models. Ordinary crowdfunding backers pledge in return for benefits, generally one count of the item being backed, while equity crowdfunding is the version in which backers receive equity instead.

AI in the strategy workflow

Imagine two regional chains subscribing to the same artificial intelligence service in the same week. One asks it what the company should do and believes the answer. The other uses it to assemble the evidence, then argues in a room full of people about what that evidence means. Same subscription, different **place in the workflow**. This supplemental section is about choosing that place deliberately.

How this supplement relates to the chapter. Chapter 2 is about how information systems create value and how they win advantage, and it names the conditions that make an advantage last or fade — the technology/strategy fit, the ten types of innovation, and the plain warning that innovation is often fleeting because rivals can copy it. Artificial intelligence appears in the chapter only in passing, monitoring equipment alongside IoT sensors and imagining a shopping assistant that has not arrived yet. What this section adds is **where AI fits inside that same analysis** — the analysis does not change just because the technology is new. That is why this appears as a labeled supplement rather than as a fourth chapter learning objective.

The question this section answers

Chapter 2 gives an information system three ways to add value. **Automating** is doing things faster, **organizational learning** is doing things better, and **supporting strategy** is doing things smarter. The chapter draws them as three bars of business value added, with automating the shortest and strategizing the tallest.

Artificial intelligence is not a fourth bar. It is a new means to those same three ends, so a proposal to buy it answers exactly the questions any other system answers.

Faster — the tool shortens work people already do, such as condensing a year of customer complaints into a readable list, and that is an automating benefit measured in time and cost like any other.

Better — the tool helps the organization use past behavior and data to improve a business process, such as noticing that those complaints cluster in one region, which is what the chapter calls organizational learning.

Smarter — the tool serves the plan the company chose for gaining or sustaining advantage over rivals, because a system that strengthens no chosen activity is not supporting a strategy at all.

The chapter is blunt about which of the three has to be present. A proposed system that will not clearly deliver strategic value, while also helping people work smarter and save money along the way, is not likely to be funded. Nothing about the letters A and I exempts a purchase from that test.

Where AI enters the strategy workflow

Strategy work has a shape. Someone gathers evidence about what is squeezing the business, someone turns that evidence into a claim about which force or which activity matters, and someone executes the initiative that follows. AI can enter all three of those places, and it enters each one differently.

- 1 Gathering the evidence** — it reads and condenses a great deal of material quickly, such as a stack of supplier contracts or a year of messages, which shortens the slowest part of a diagnosis without making any of that material true.
- 2 Analysing the evidence** — it can lay out how the same pattern reads as rivalry, as a new entrant, and as a substitute, which is useful precisely because one lost sale can honestly be told as a story about any of the five forces.
- 3 Executing the initiative** — it can carry the repetitive work inside whatever system the company decides to build, such as sorting arriving messages by topic, and this is where most of the measurable hours are actually saved.

There is a fourth step it does not enter. Choosing which force to answer, and what the company will therefore not do, is an executive-level decision, and the chapter calls those decisions **unstructured** because few or no procedures for them can be specified in advance.

INTERACTIVE DIAGRAM

Where AI enters, and where it does not

Pick a place in the strategy workflow to redraw it: what the step produces, what AI genuinely helps with there, what it cannot supply, and the control that keeps the step honest.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Gathering the evidence — Before the claim — assembling the material a diagnosis will rest on

The raw material (Complaints, contracts, prices, notes) → What AI does well (Reads a great deal, condenses fast) → What it cannot supply (Whether any of it is true) → The control (Someone who can check, checks)

- **What the step produces** — the observable facts a claim will stand on, such as sales falling only at the locations a new competitor reached. Evidence is what stops a strategy meeting from being two opinions.
- **What AI genuinely helps with** — volume. Condensing a year of messages or a shelf of contracts is the slowest part of a diagnosis and the part where a tool saves hours that can be counted.
- **What it cannot supply** — truth. A summary reads exactly the same whether every line came from the source or one line did not, so nothing in the output signals which sentences to distrust.
- **The control** — a person who already knows the material verifies the checkable parts, meaning figures, dates, names and terms, before any of it enters a recommendation that someone will fund.

Analysing the evidence — The claim — turning a pattern into a named force or a named activity

The pattern (Sales fell only where the club opened) → Competing readings (Rivalry, new entrant, or substitute) → What AI does well (Argues all three readings quickly) → What it cannot decide (Which reading the firm will act on)

- **What the step produces** — one named force from the five, or one named activity in the value chain, plus the reason that name beats the alternatives. Name the wrong one and the company buys the wrong system.
- **What AI genuinely helps with** — making the alternatives visible. Asked to argue the case for rivalry, for a new entrant, and for a substitute, it produces three readings faster than a team can, which is a real service when a lost sale fits all three.
- **What it cannot decide** — which reading the organization commits to. That commitment sets what the firm will not do, and the chapter classes decisions of that kind as unstructured, meaning no procedure for them can be written in advance.
- **The control** — the analyst states the chosen reading, the evidence for it, and the strongest argument against it, so the room is deciding between named alternatives rather than nodding at the only one presented.

Executing the initiative — After the decision — the repetitive work inside the system the firm builds

The initiative (One record of every supplier quote) → Work inside it (Sorting, matching, drafting, tagging) → Where the hours are (Volume, not judgment) → The limit (It runs the plan; it is not the plan)

- **What the step produces** — the working system itself, and the change in how people do the job around it. The chapter is firm that a significant implementation requires commensurate, significant organizational change rather than dropping software into an unchanged process.
- **What AI genuinely helps with** — the high-volume, low-stakes steps inside that system, such as tagging arriving messages by topic or matching a quote to the right contract, where being wrong is cheap and easily corrected.

- **Where the value shows up** — this is the automating rung, doing the same work faster and more cheaply. That is a genuine benefit, and on its own it is the shortest of the chapter's three bars of business value added.
- **The limit** — execution cannot rescue a badly chosen initiative. Automating a flawed process magnifies its errors, which is exactly the chapter's point about a loan department reviewing three times as many applications and approving three times as many bad ones.

Owning the recommendation — The step AI does not enter — deciding, and answering for the decision

The decision (Which force the firm will answer) → Its character (Unstructured, complex, long-term) → Who signs (A named person, before funding) → What they owe (An explanation at the review date)

- **What the step produces** — a named person who says this is what we should do and why, and who is still there when the measure is read. A recommendation with no owner is a document, not a decision.
- **Why it stays human** — executive-level decisions are unstructured, complex and nonroutine, with broad and long-term ramifications for the whole organization, and the chapter says few or no procedures for them can be specified in advance.
- **What is being owned** — not the words but the consequences. Funding one initiative means not funding another, and firms are constrained by time and money to build only the systems that add the most value.
- **The failure to avoid** — treating an output as the decision. If nobody in the room can explain why this force and not another, the organization cannot defend the spending and cannot learn from it when the numbers arrive.

Augment, automate, or leave it alone

The previous module offered a three-way decision for any piece of work. Strategy work is itself a set of tasks, so the same three words apply to it, and the vocabulary is kept identical on purpose so that the two modules agree rather than compete.

Augment — the system drafts, ranks or summarizes and a person decides, which suits work where a starting point saves real time but the judgment still matters, such as a first pass over three years of sales figures.

Automate — the system acts on its own inside limits written before launch, which suits high-volume, low-stakes, reversible work, such as pulling competitors' published prices into one weekly list.

Leave it alone — the task stays entirely with people because the stakes or the sensitivity are too high, such as choosing the generic strategy the firm will pursue, or anything resting on material that must not leave the company.

Most strategy work belongs in the first bucket, because an analysis is only worth as much as the argument a named person can make for it out loud, in a room, to people who disagree.

SORT Augment, automate, or keep it human?

Each item is a piece of strategy work at a hypothetical regional chain of bicycle shops. Place it in the mode it belongs to, judging by what one wrong answer would cost.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Augment — the system drafts, ranks or summarizes and a person decides before anything is acted on

- Condensing a year of customer complaint messages into a short list of recurring themes an analyst then interprets — Reading the pile is the slow part and interpreting it is the judgment, so the system supplies the themes and the analyst decides what they mean.
- Setting out how the same sales decline reads as rivalry, as a new entrant, and as a substitute, so an analyst can weigh all three — Producing the competing readings is genuinely useful, and choosing among them is the diagnosis itself, which is the shape of augmentation.
- Summarizing nine supplier contracts into one table of prices, terms and expiry dates that a buyer checks against the originals — A missed clause changes what the chain is bound to, so the summary saves the reading time while a person still verifies the terms that matter.
- Drafting the first version of the one-sentence recommendation, which the analyst rewrites before it reaches the board — A blank page is the expensive part of writing and the claim is the risky part, so the draft comes from the system and the wording stays with the analyst.

Automate — the system acts on its own inside limits written in advance — high volume, low stakes, reversible

- Collecting competitors' published prices into one weekly list that nobody assembles by hand — It is high volume, low stakes and reversible, since a price copied wrongly is corrected the moment anyone compares it with the published page.
- Tagging arriving customer messages by topic so a weekly count of each theme is ready on Monday morning — A message filed under the wrong topic is refiled in seconds, and no decision rests on any single message, so acting without a reviewer is worth it here.
- Reformatting last quarter's figures into the chart layout the board pack has always used — The numbers are already agreed and the task has one obvious right answer, so this is clerical work whose only possible error is visible at a glance.

Keep human-only — the stakes or the sensitivity are too high, so the task stays entirely with people

- Choosing whether the chain will compete as a low-cost leader or on differentiation — This is the unstructured executive decision every later system is judged against, and the chapter says no procedure for such a choice can be specified in advance.
- Deciding which of two underperforming stores closes when the analysis says one of them must — The consequences fall on named employees and a named community, it cannot be reversed, and the person accountable has to be able to explain it.
- Preparing the confidential file naming the smaller chain this company is considering acquiring — The document itself is the secret, so sending it to an outside service is the harm rather than a risk of one, whatever the quality of the output.

CHECK YOURSELF

Where it fits, and in which mode

Three questions on how AI relates to the chapter's three ways of adding value, and on which step of strategy work it can take.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A manager argues that artificial intelligence is a fourth way for an information system to add value, alongside automating, learning and supporting strategy. What is the most accurate response?

- **A.** It is a fourth way, because no earlier technology could produce written analysis without a person doing the writing — Novelty of the technique is not what the chapter's three categories measure; they describe the kind of value produced. A genuinely new fourth kind of value would have to be something other than faster, better or smarter.
- **B.** It is a new means to the same three ends, so a proposal to buy it is judged by the same three questions — **correct.** The three ways describe what an information system does for the firm, so a new technology enters through them rather than beside them, and the proposal still has to show which of the three it delivers.
- **C.** It belongs under automating only, because generating text faster than a person is by definition an efficiency gain — Automating is often where it lands first, and confining it there would miss the rest. The same tool can also reveal patterns in past behavior, which is organizational learning, and can serve the chosen strategy directly.
- **D.** It replaces the three, because a system that can analyse and recommend no longer needs a category to sit in — This overstates what any system does. The categories describe value delivered to the organization, not the sophistication of the software, and a system that analyses still has to be shown to help the firm compete.

2. In the strategy workflow, which step is the one AI does not take?

- **A.** Condensing a large volume of source material into a readable summary of what it contains — Condensing volume is the clearest thing it does help with, and it is the slowest part of gathering evidence, which is exactly why the verification control belongs immediately after it.
- **B.** Laying out how a single sales decline could be read as rivalry, as a new entrant, or as a substitute — Producing the competing readings is genuinely useful work, since one lost sale can be told honestly as a story about several forces, and seeing all of them is what makes a choice between them possible.
- **C.** Committing the organization to one reading and to what it will therefore not do — **correct.** Committing to one reading is an executive-level decision, and the chapter calls such decisions unstructured because few or no procedures for them can be specified in advance.
- **D.** Carrying repetitive work inside the system the organization has decided to build — Repetitive work inside a chosen system is the automating rung, which the chapter presents as real value, and it is where most of the countable hours are saved once the decision has already been made.

3. An analyst wants a tool to produce the first draft of the recommendation that will go to the board, which the analyst will then rewrite. Which mode is that, and is it defensible?

- **A.** Automate, and defensible, because drafting is repetitive work and the board will read the document anyway — Automate means acting without a reviewer, and here a person rewrites the document before anyone sees it, so the description does not match the arrangement being proposed.
- **B.** Augment, and defensible, because the draft saves time while the claim and the wording stay with the analyst — **correct.** A blank page is the expensive part of writing and the claim is the risky part, so the system supplies the first and the analyst keeps the second, which is what augmentation means.
- **C.** Augment, and indefensible, because any strategy document is too sensitive for a system to touch at all — Sensitivity is a real reason to keep some work human, and it applies to material that must not leave the company, such as a confidential acquisition file, rather than to every document about strategy.
- **D.** Keep human-only, and defensible, because a recommendation is an unstructured decision and no procedure exists for it — Deciding what to recommend is indeed unstructured and stays with the analyst. Writing a first draft of a decision the analyst has already made is a different task, and refusing help with it costs hours for no reduction in risk.

Why buying AI is not the same as gaining advantage

This is the heart of the section, and the chapter answers it directly. Innovation is often fleeting: even a system built in house is assembled from hardware, software and networking components that others can also purchase, so rivals can copy it and the advantage is short-lived.

A service any competitor can subscribe to the same afternoon is the clearest possible case of that warning. It is a commodity, and a commodity is where an advantage cannot live.

What a rival cannot buy is everything around the subscription. The chapter locates advantage in **resources** such as proprietary technology, brand equity and a loyal, established customer base, and in **capabilities**, meaning the organization's ability to leverage those resources in the marketplace. Three of those stay yours.

Your data — the record of what your customers actually did would take a competitor years to accumulate, which is why the chapter lists data among the sources of competitive advantage and gives Uber as its example.

Your processes — the chapter insists that any significant implementation demands commensurate, significant organizational change, so the firm that redesigns the work around a tool gets a result the firm that merely installs it never sees.

Your people — capabilities such as design quality or efficient operations exist in the people who exercise them, and resources plus capabilities are what produce the distinctive competencies a strategy is built on.

Then there is the fit. The chapter's **technology/strategy fit** says organizations do not want systems that fail to match their strategy, even when those systems offer real automating and learning benefits. Its own illustration is a firm pursuing overall low-cost leadership, which probably does not want a system that differentiates its products on high quality.

A purchase that does not serve the strategy the company chose is spending, not strategy — and the chapter's name for the narrow exception is **strategic necessity**, something an organization must do to survive whether or not it matches the plan.

The run below is a **hypothetical** practice situation, invented so the reasoning has something to bite on; no real company is described.

DECIDE Advising on an AI investment

Work the four decisions in this hypothetical company; after each one, read every outcome before continuing, including the ones you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

This is a hypothetical practice situation, not a report about any real company. A regional chain of bicycle shops has set money aside after two competitors publicly announced artificial intelligence projects. The chain has already chosen its strategy: overall low-cost leadership in its region, competing on price and on how cheaply it can run the repair counter. It also holds eight years of repair records that nobody has ever analysed. You are the analyst in the room.

Decision 1. First decision: what is the money actually for? Leadership wants to move this week, and three sentences are on the table.

- Buy the most capable general assistant available and give every employee an account, then see what people find to do with it — This is the tempting answer and it buys a commodity. The subscription is available to every rival at the same price on the same day, no activity has been named, so nothing can be measured and nothing has been made harder to copy. It is spending in the shape of a strategy.
- Name one activity the chain competes on, then ask what a tool would change inside it — **the stronger call**. Right, and in that order. The chain competes on running the repair counter cheaply, so the question becomes what happens to the cost of a repair, which is narrow enough to measure and specific enough to reject proposals that would not touch it.
- Buy whatever the two competitors announced, so the chain is not left behind on a technology the market clearly expects — This justifies parity and never advantage, and it hands the technology roadmap to rivals. It also assumes the competitors diagnosed their own business correctly, which is an assumption nobody in this room can check.

Decision 2. Second decision: a vendor offers a system that recommends premium components and custom fittings to each customer, promising a more differentiated, higher-quality buying experience. Another proposal would cut the time a mechanic spends diagnosing a common fault.

- Take the premium recommendation system, because a better customer experience helps any retailer — It probably would help some retailer, and it does not fit this one. The chapter's own illustration is exactly this mismatch: an organization pursuing overall low-cost leadership does not want a system that differentiates its products on high quality, however real the benefits are.
- Take the diagnosis proposal, because lowering the cost of the activity the chain competes on is what its chosen strategy values — **the stronger call**. Correct, and the reason is the technology/strategy fit. Under a low-cost leadership strategy, investments that drive costs down are valued over those that do not, so this proposal is judged against the plan the company already made rather than against how modern it sounds.
- Fund both, since each one is defensible on its own and the chain can decide later which mattered — Funding both is how a company ends up with neither. Organizations are constrained by time and money to build only what adds the most value, and running two initiatives at once means neither gets the organizational change the chapter says a significant implementation requires.

Decision 3. Third decision: the board asks where the advantage would come from, given that the same tools are for sale to everyone in the region.

- From the model itself, because the chain will subscribe to the most capable one on the market — This is the commodity trap stated out loud. Innovation is often fleeting precisely because a system is assembled from components others can also purchase, and a subscription a rival can start tomorrow is the least defensible asset the chain could name.

- From the eight years of repair records, plus rebuilding the counter process around what they show — **the stronger call**. Right on both halves. The records are a resource no rival holds and would take years to accumulate, which is why the chapter lists data among the sources of competitive advantage, and the process change is what turns the resource into a capability rather than a spreadsheet.
- From being the first chain in the region to announce it, which the other two have not yet done — Being first to announce is not a first-mover advantage in any market, and an announcement is the easiest thing in this entire plan for a competitor to copy. Nothing about it makes the chain harder to imitate next quarter.

Decision 4. Fourth decision: the board will fund the initiative and wants to know how the chain will judge it.

- Report how many employees use the tool each week, which shows the investment is being taken up — This measures activity rather than results. Usage rises simply because the tool exists and staff were asked to try it, and the number would keep rising even if no repair got cheaper and no customer noticed anything at all.
- Name the average cost of a common repair as the measure, fix the review date now, and name the manager answerable for the result — **the stronger call**. That is the full shape: a number tied to the activity the chain competes on, a date set before anyone has seen results, and a named person who has to explain the outcome. Keep, change and stop are all still genuinely available at that review.
- Use the vendor's dashboard, which reports the hours the tool saved across the business — The vendor is scoring its own product, and hours saved somewhere in the business is not the same as cost removed from the activity the strategy names. A measure the buyer cannot reconstruct is a claim rather than evidence.

AI as more than a product feature

Chapter 2 lists ten types of innovation, and two of them are worth setting side by side here. A **product performance innovation** creates a novel product or improves an existing one through differentiation. A **process innovation** changes the primary processes used to produce products or services.

In an established company, AI usually lands as the second kind. The tool rarely becomes the thing the customer buys; it changes how that thing is made, sold or supported. The chapter's examples of process innovation are Toyota pioneering lean production and Zara moving fashion from initial design to stores in three weeks.

That difference decides who has to be involved.

- **A product change** is owned by the people who design and sell the product, and customers deliver the verdict on it.
- **A process change** lands on the people doing the work, so their managers, their steps and their measures all have to change with it.

The chapter also explains why one type is rarely enough. Product performance innovations can often be easily copied, so companies frequently combine several types at once. Dell is the worked example, combining a profit model innovation in collecting money before building the product, a process innovation in the built-to-order model, and a channel innovation in selling computers online rather than in stores.

What goes wrong in strategy work specifically

These are not general complaints about the technology. Each failure damages a strategic analysis in particular, and each has a control that belongs inside the workflow rather than in a policy document nobody opens.

Evidence nobody verified — a confident summary of a market or a contract can contain a claim that is simply not in the source, so a person who can check has to check the figures, dates and terms before any of it reaches a recommendation.

Bias in the customer data underneath — an analysis of customer behavior can only describe the customers the company recorded, so a conclusion built on that record silently ignores whoever the company has never served.

Confidential strategy material leaving — a strategy document names what a firm intends to do next, and an outside service is still outside, which is the same balance the chapter asks companies to strike when open innovation makes organizational data and knowledge accessible to external stakeholders.

Accountability — a person still owns the recommendation, because a board asked to fund an initiative is entitled to ask who believes this and why, and no committee can act on the answer that the system suggested it.

The discipline this module keeps returning to

Whether or not a machine helped produce it, a recommendation is judged the same way it was judged before any of this existed. Name the force or the activity, give the evidence, state the initiative, then commit to a measure and a date.

Say it in one sentence. Because [force or activity] is [strong or weak], shown by [evidence], we should build [initiative], which we will judge by [measure] at [date]. AI can help you fill every slot in that sentence. It cannot be the person who says it, and it cannot be the person answerable when the date arrives.

CHECK YOURSELF**Commodity, fit, and the type of innovation**

Three questions on why a purchase is not an advantage, on matching a system to the chosen strategy, and on where AI usually lands among the ten types of innovation.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A board is told that subscribing to the most capable AI service available will give the company a lasting advantage. On this chapter's reasoning, what is wrong with the claim?

- **A.** Nothing is wrong, because a system built to gain or sustain advantage keeps producing it for as long as it runs — Sustaining an advantage is continuing work rather than a property a system acquires at purchase, and the chapter says the opposite of this option: advantages gained by innovations often have a limited life span.
- **B.** The claim fails because information systems are a commodity, so technology spending belongs on cost reduction only — This states a stronger position than the chapter takes. The chapter does not say technology spending must be limited to cost reduction; it says spending has to match the strategy and show clear, significant value.
- **C.** The advantage can be fleeting, because rivals can subscribe to the same service and copy the same use of it — **correct.** The chapter warns that innovation is often fleeting because systems are built from components others can also purchase, so smart rivals are quick to duplicate or improve on any use that visibly wins.
- **D.** The claim fails because the service was not developed in house, and only in-house systems create any advantage — In-house building is not the dividing line the chapter draws. It notes that even an in-house system is assembled from purchasable components, and separately that a system built by a consultant or outsourcing partner is proprietary to the organization.

2. A chain pursuing overall low-cost leadership is offered an AI system that would let it recommend premium, customized products to each customer. It genuinely automates work and reveals useful patterns. Should it buy?

- **A.** Yes, because the system delivers both automating and learning benefits, which are two of the chapter's three kinds of value — Automating and learning benefits are real and the chapter still says they are not enough on their own, since organizations do not want systems that fail to match the strategy even when those benefits are present.
- **B.** Probably not, because a system that differentiates on quality does not match a low-cost leadership strategy — **correct.** This is the chapter's own illustration of the technology/strategy fit: a firm pursuing low-cost leadership probably does not want a system differentiating its products on high quality.
- **C.** Yes, because pursuing differentiation as well as low cost gives the chain two sources of advantage instead of one — The chapter does describe a middle position, the best-cost provider strategy offering reasonably good quality at competitive prices, but that is a strategy a company chooses deliberately rather than a bonus acquired by buying a mismatched system.

- **D.** Probably not, because the chain should fix its cost position before it spends on anything customer-facing — Sequencing sounds prudent but is not the objection. The problem is not that this comes too early; it is that a system built to differentiate on quality pulls against a low-cost position no matter when it is bought, so waiting would not make it fit.

3. A manufacturer uses an AI system to change how it schedules and runs its production line. The product itself is unchanged. Using the chapter's ten types of innovation, what kind of innovation is this?

- **A.** A product performance innovation, because the output of the line will be more consistent for customers — Product performance innovation means creating novel products or improving existing ones through differentiation, as with an unbreakable glass or a distinctive hand dryer, and here the product has deliberately not changed.
- **B.** A profit model innovation, because lower production cost changes how the company generates its revenue — Profit model innovation means finding novel ways of generating revenues, such as offering software on a subscription or selling thrust as a service, and a cheaper line does not change what the company charges for.
- **C.** A process innovation, because it changes the primary processes used to produce products or services — **correct.** The chapter defines process innovation as changing the primary processes used to produce products or services, and gives Toyota's lean production and Zara's three-week design-to-store cycle as its examples.
- **D.** A channel innovation, because it changes how the finished product reaches the people who buy it — Channel innovation means using innovative ways to connect offerings with customers, such as selling computers online instead of through stores, and nothing in this scenario changes how the product reaches a buyer.

The analyst's toolkit: getting the evidence yourself

Every framework in this module ends in a claim — buyer power is our strongest force, inbound logistics is where the margin is leaking, this category is where our value is disappearing. A claim without evidence is a preference. This supplement is about producing the evidence yourself, with two tools that sit on almost every analyst's desk: a spreadsheet and a database query.

How this supplement relates to the chapter. Chapter 2 teaches the analysis — the competitive forces, the value chain, the business model. It also sets its own end-of-chapter exercises in a spreadsheet and a database, because the analysis has to be fed. This section takes the chapter's exercises and makes them runnable: you type a real formula or a real query on this page and watch it work on real rows. It is a labeled supplement because the tools serve the chapter's analysis rather than being a fourth learning objective of their own.

Why an analyst touches the data directly

It is tempting to think of data work as somebody else's job — that an analyst asks and a technical team answers. In practice the gap between the question and the answer is where most of the time goes, and the person who can close it themselves is the person whose recommendation arrives while the decision is still open.

Two skills close most of that gap, and both appear in this chapter's own exercises:

Cleaning and reshaping text — real data arrives inconsistent, with stray spaces, mixed capitalisation and invisible characters, and none of it can be grouped or matched until it is made uniform, which is why the chapter opens its spreadsheet exercise on exactly that problem.

Asking a database a precise question — a query turns a stored record into an answer about the business, and the discipline of writing one forces you to say exactly which rows and which columns your claim actually rests on.

The chapter puts the business case plainly: clean contact data matters because it makes communication, customer segmentation and personalized marketing possible, and those are the things a competitive advantage is built out of.

Part one: cleaning text in a spreadsheet

A spreadsheet function takes a value and returns a new one; it never changes the original. That is why cleaning is done in new columns rather than by editing the raw data in place — the original stays available when someone asks where a number came from.

The chapter names the functions this kind of work needs. These are the ones the exercises below use:

TRIM — removes extra spaces from a string, leaving only single spaces between words, which matters because a trailing space makes two otherwise identical addresses look like two different customers.

CLEAN — removes non-printable characters that may sit in raw data without being visible in the sheet, so a value that looks correct but refuses to match another one usually needs this.

LOWER — standardises capitalisation, and its companions UPPER and PROPER do the same job in the other directions, because a system that treats two capitalisations as two values will double-count a customer.

FIND — returns the position of one piece of text inside another, counting from one, and it is case sensitive; SEARCH does the same job without caring about case.

LEFT, RIGHT and LEN — take characters from the start of a string, from its end, and count how many there are, which together let you split a value at a marker you have located.

Start with the cleaning step. The raw column below is deliberately untidy, in the way exported contact data usually is. Fill each column with one formula and watch it run down every row — the formula is written once for row 2 and applied to the rest, which is exactly how it behaves in a real sheet.

SPREADSHEET

Make the addresses uniform

Type one formula for row 2 in each column, then run it; it is applied to every row the way a filled-down formula behaves in a real sheet.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Splitting a value at a marker

Once the addresses are uniform, they can be taken apart. The username is everything before the @ and the domain is everything after it, but neither has a fixed length, so you cannot simply take the first eight characters. You have to find the marker first and measure from it.

That is the whole idea behind the two formulas the chapter gives, and it is worth reading them slowly, because the same pattern splits a product code, a postcode or a file name:

- 1 **Locate the marker.** FIND returns the position of the @ counting from one, so in `priya.raman@northside.edu` it returns 12, because the twelfth character is the @ itself.
- 2 **Take everything before it.** The username is one character shorter than the marker's position, which is why the formula subtracts one: LEFT of eleven characters gives `priya.raman` and stops just before the @.
- 3 **Take everything after it.** The domain's length is the whole string minus the marker's position, so LEN minus FIND gives the number of characters to take from the right — no counting by hand, and it works on an address of any length.

Work the three columns below. The addresses have already been cleaned, so you can concentrate on the splitting.

SPREADSHEET

Split each address at the marker

The addresses are already cleaned. Find the @ first, then measure from it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

THE HABIT WORTH KEEPING

Notice that nothing in those formulas mentions a specific address. They describe a **rule** — find the marker, measure from it — which is why one formula handles six rows or sixty thousand. Writing rules rather than answers is the difference between a spreadsheet that scales and one that has to be redone every month.

Part two: asking a database a business question

A spreadsheet is the right tool while the data is small and you are still shaping it. Once records are shared, updated by several people and expected to stay consistent, they belong in a database, and you get at them with a query.

A query has a fixed shape, and reading it in the right order makes it far less mysterious than it looks:

SELECT names the columns you want back, and an expression such as `PurchasePrice - CurrentValue` can be one of them, given a name with AS.

FROM names the table the rows come from, and is the part that actually decides what you are counting.

WHERE keeps only the rows that satisfy a condition, so it runs before anything is totalled and quietly determines every number that follows.

GROUP BY collapses the surviving rows into one row per distinct value, which is what lets COUNT, SUM, AVG, MIN and MAX report per category rather than per record.

HAVING filters those grouped rows, and it exists because WHERE has already finished its work by the time the totals exist.

ORDER BY sorts the result, and LIMIT keeps only the first few rows, which together answer questions phrased as “the worst three” or “the largest.”

The table below is an asset register of the kind the chapter’s database exercise describes: an item, what it is, its condition, what it cost and what it is worth now. Everything on this page is a hypothetical practice situation, invented so the queries have something to bite on. Open the tables, then answer each question with a query — any query that returns the right answer is accepted, not just the one that was expected.

DATABASE

Question the asset register

Write a SELECT for each question and run it. Any query returning the right answer is accepted, so there is more than one correct way through.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

From a result set back to the analysis

A query result is not yet a finding, and a finding is not yet a recommendation. The last query above returns a number per department; on its own it is a fact nobody has to act on. It becomes useful only when it is attached to the analysis this module has been building.

The chain runs in one direction, and skipping a link is what makes a proposal easy to refuse:

- 1 **The result** is what the query returned — for instance that one department holds most of the assets recorded in poor or fair condition.

- 2 **The finding** is what that means in the business — equipment in that part of the operation is being run past the point where it works reliably.
- 3 **The framework** says where it bites — if that equipment sits in inbound logistics or operations, the value chain tells you the cost is landing on an activity the customer eventually pays for.
- 4 **The recommendation** names the initiative, the measure and the date, which is the same discipline the earlier sections asked for and the only form a decision-maker can actually approve or reject.

CHECK YOURSELF

Tools, evidence, and claims

Each question is about the reasoning behind the tool rather than the syntax; read every explanation.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. An analyst filters a sales table to this year's orders, groups by region, and totals the revenue. A colleague asks why one region looks so weak. What is the first thing worth checking?

- **A.** Whether the totals were sorted in the right direction before being read — Sorting changes the order rows are presented in, not their values. A region that looks weak will look equally weak wherever it appears in the list.
- **B.** Which rows the filter kept, because a condition applied before grouping silently decides every total that follows — **correct.** Right. The filter runs before the grouping, so anything it excluded is missing from every total. A region with orders recorded slightly outside the filtered range would look weak for a reason that has nothing to do with its sales.
- **C.** Whether a different aggregate such as AVG would make the region look stronger — Choosing the aggregate that flatters a result is the opposite of analysis. AVG and SUM answer genuinely different questions, and switching between them to get a nicer number is how an analysis stops being evidence.
- **D.** Whether the region column should have been selected after the grouping instead of before it — The order in which columns are named does not change which rows were counted. Grouping needs the column to be available, and it is, so this would not alter the result at all.

2. Two customer records read `R.0sei@Wexler-Foods.com` and `r.0sei@wexler-foods.com` . A segmentation report counts them as two customers. What has actually gone wrong?

- **A.** The database is missing a rule that would prevent two records for one person — A constraint would help future records, but it cannot explain this one: the system is behaving correctly by treating two different strings as different. The data was already inconsistent when it arrived.

- **B.** The values differ in capitalisation and trailing space, so they are two different strings until they are standardised — **correct.** Right. A trailing space and a capital letter make the strings genuinely different, which is exactly why the chapter puts TRIM and LOWER at the start of its exercise. Standardising first is what makes the match possible.
- **C.** The report should be counting domains rather than full addresses — Counting domains would answer a different question entirely, about which organizations the customers belong to. It would also merge everyone at the same company into one, which is worse.
- **D.** The addresses need to be split into username and domain before they can be compared — Splitting is useful for other purposes, but it does not fix this. Splitting these two values would produce two usernames that still differ in case and spacing.

3. A query shows that one department holds most of the assets recorded in poor condition. Which statement turns that result into something a decision-maker can act on?

- **A.** The register shows that department holds the most assets in poor condition — This is the query result restated. It is true and it is where the work starts, but it names no consequence, no initiative and no measure, so there is nothing to approve.
- **B.** Equipment reliability is a concern across the organization and deserves attention soon — This is a sentiment rather than a finding. It names no place, no evidence and no test, which means it can never be shown to have worked or failed.
- **C.** Ageing equipment in that department is slowing the operations activity, so we should replace the four worst units this quarter and judge it by unplanned downtime hours at the end of the quarter — **correct.** Right. It names the value chain activity affected, rests on the observable result, states a specific initiative, and commits to a measure and a date — the four parts that make a recommendation fundable rather than merely reasonable.
- **D.** That department should be given a larger equipment budget than the others next year — This jumps from evidence straight to a budget decision without saying what the money would buy or how anyone would know it worked. It also assumes spending is the answer before the cause has been established.

Strategy as code, and questions a database can answer

The analysis in this module produces claims: this force binds us, that activity is where the margin goes, this revenue model suits what we sell. Each of those claims is a calculation somebody is doing in their head, usually inconsistently. This supplement writes four of them down as functions, and then goes back to the database for the harder questions a single table could not answer.

How this supplement relates to the chapter. Nothing here adds to the chapter's material. The five forces, the value chain, the revenue models and the discipline of committing to a measure are all from the sections above; what this section adds is an executable form of each, so that a claim can be checked rather than asserted. The queries extend the previous supplement from one table to three.

Why write the analysis down

A framework applied in conversation drifts. The same analyst, given the same evidence a week apart, will weigh it differently, and two analysts will never agree on what they were even comparing. Writing the step as a function fixes three things at once:

The inputs become explicit — a function signature forces you to say precisely what evidence the conclusion rests on, which is the same discipline the recommendation format asks for.

The tie stops being invisible — when two forces score equally the honest answer is that the evidence does not yet separate them, and a function has to return something for that case rather than quietly picking one.

The result is reproducible — anyone can rerun it on the same inputs and get the same answer, which is what makes it evidence rather than an opinion with numbers attached.

Each exercise below gives you a stub, a contract and tests. Your code runs on this page with nothing installed and no network; a loop that never ends is stopped after two seconds. Ctrl-Enter runs the tests, Tab indents, and there is always more than one implementation that passes.

WRITE THE CODE**Four analysis steps as functions**

Write each function so every test passes; a failing test prints what your code returned beside what was expected.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Look back at what each function refuses to do. `bindingForce` will not name a winner it cannot justify, and `paybackMonths` will not report a payback that arrives after the customer has gone. Both refusals are the analysis being honest in a way prose lets you avoid.

Questions that need more than one table

The previous supplement queried a single asset register. Real questions rarely sit in one table, because the facts are deliberately kept apart: who the customer is, what they subscribe to, and what that plan costs are three different kinds of fact that change on three different schedules.

Joining is how they are brought back together for one question, and two ideas cover most of what you will need:

A join matches rows on a shared value — every subscription carries a customer identifier, so matching it against the customer table puts the name beside the subscription without ever storing the name twice.

HAVING filters groups rather than rows — WHERE has finished its work before any total exists, so a condition about a count or a sum has to wait until after the grouping, which is the distinction that catches most people out.

The three tables below describe a hypothetical subscription business; every company in them is invented. Answer each question with a query, and remember that any query returning the right answer is accepted.

DATABASE**Three tables, harder questions**

Each question needs a join, and some need grouping too. Any query that returns the right answer is accepted.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

From a result to a recommendation

The last two queries are worth sitting with, because together they say something neither says alone. One reports the revenue lost to churn by region; the other reports how long customers stayed before leaving, by plan.

- 1 **The results** are two tables, and neither is a finding. One region lost far more than the others, and one plan retained customers for far longer than another.
- 2 **The finding** is what they mean together: the losses are concentrated, and they are concentrated in a plan whose customers were expected to stay longest.
- 3 **The framework** says where it bites — if the customers leaving are the ones a competitor is courting, this is rivalry or buyer power rather than a service problem, and the two call for different responses.
- 4 **The recommendation** names the initiative and the number that should move, which is the only form that can be approved, rejected, or later shown to have worked.

THE HABIT THIS SECTION IS REALLY TEACHING

Every claim in a strategy document is either a calculation somebody can reproduce or an assertion somebody has to trust. Writing the calculation down — as a function, as a query, or as both — is what lets the argument be checked instead of merely believed. It is also, not incidentally, the difference between analysis that survives a challenge and analysis that does not.

CHECK YOURSELF

What the code and the queries settled

These ask about the reasoning, not the syntax; read every explanation.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Your `bindingForce` returns 'tie' when two forces score equally. A manager asks you to make it always name one force so the report reads decisively. What is the strongest objection?
 - **A.** Ties are rare enough in practice that the change would almost never matter — Rarity is not the issue and is not knowable in advance. Even a single manufactured winner sends real money at the wrong problem.
 - **B.** The function would need considerably more code to break ties fairly — Code length is not the objection, and a tie-breaking rule would be short. The problem is that any such rule invents a distinction the evidence did not supply.

- **C.** A tie means the evidence does not separate the two forces, so naming one manufactures a certainty the analysis does not have and points spending at a guess — **correct.** Right. The tie is information: it says go and get better evidence before committing. Converting it into a confident answer hides the one fact the decision-maker most needs.
- **D.** Naming one force would break the tests, and the tests are the specification — The tests would indeed change, but tests follow the specification rather than justify it. This answer describes a consequence and skips the reason.

2. `paybackMonths(100, 25, 0.5)` returns `-1`: four months to repay, but a two-month expected lifetime. A colleague proposes returning `4` anyway, since that is the arithmetic. What is lost?

- **A.** Nothing meaningful, because the payback period and the customer lifetime are separate measures — They are separate measures whose whole point is to be compared. Reporting one without the other is what makes a losing proposition look profitable.
- **B.** The single fact that decides the case — that the customer leaves before the cost is recovered, so the acquisition loses money every time it succeeds — **correct.** Right. A payback that arrives after the customer has gone never arrives at all. Returning `4` describes a repayment that does not happen, which is precisely the error the horizon was added to catch.
- **C.** Precision, since four months is a rounded figure and the true value has a fractional part — Rounding up is deliberate, since a partial month still costs a whole one, and it is not what the objection turns on.
- **D.** Consistency with the other functions, which all return a number rather than a sentinel value — Return-type consistency is a style question. Here the sentinel is carrying the finding, and a uniform number would carry none.

3. A query totals revenue per plan by joining subscriptions to plans and summing `Months * MonthlyFee`. Why is the fee stored on the plan rather than copied onto each subscription?

- **A.** Because a join is the faster way to reach a value than reading it from the row itself — This has the trade-off backwards: reaching a value through a join generally costs more at query time than reading it from the row. Normalising is done for correctness, and the join is the price paid for it.
- **B.** Because SQL does not permit an arithmetic expression inside SUM — SQL is perfectly happy aggregating an expression, which is exactly what this query does. Nothing here is a limitation of the language.
- **C.** Because the fee is a fact about the plan, so storing it once means a price change is one edit rather than one per subscriber, with no chance of some rows disagreeing — **correct.** Right. One fact, one place. Copying the fee onto every subscription creates as many opportunities to disagree as there are rows, and the disagreement surfaces later as totals nobody can reconcile.
- **D.** Because storing it on the subscription would make the table too large to query — A few more bytes per row would not trouble any database. The cost of duplication is inconsistency, not size.

4. Churned revenue is concentrated in one region, and the plan with the longest average tenure is also the one losing the most money. Which statement is a finding rather than a result or a recommendation?

- **A.** Two queries returned totals grouped by region and by plan — That describes what was run rather than what was learned, and no decision follows from it. It is the result, the step before a finding.
- **B.** The customers we expected to keep longest are the ones leaving, and their departures account for most of the revenue lost — **correct.** Right. It states what the numbers mean together, in business terms, without yet proposing an action — which is exactly what a finding is and what a recommendation must be built on.
- **C.** We should introduce a retention programme for premium customers next quarter — This is a recommendation, and a thin one: it names an initiative but no measure and no date, so nobody could later say whether it worked.
- **D.** Churn is a problem the business needs to take more seriously — This is a sentiment. It names no evidence, no place and no test, so it can be agreed with indefinitely without anything changing.

REFERENCE

Chapter vocabulary

Every term the chapter defines, in one place, written so it makes sense on its own. Search by word, or filter by the objective the term belongs to. Use this while you read rather than after — a good deal of the difficulty in this chapter is two terms that sound alike meaning genuinely different things, such as efficiency and effectiveness, or a resource and a capability.

affiliate marketing

2.2

A revenue model in which a business pays other businesses that bring or refer customers to it; revenue sharing is typically used. It is one of the revenue models the internet has enabled or enhanced, alongside the sales, transaction fee, and advertising models already common offline.

For example: A creator posts a link to a pair of headphones and earns a cut of the price when a follower buys through that link, the way Amazon's Associates program works.

augmented reality

2.3

Using information systems to enhance what a person perceives of the real world by supplying relevant information about their surroundings. Special glasses or another device layer computer-generated content on top of the actual view rather than replacing it.

For example: Google Glass, eyeglasses with a tiny built-in screen, could show the wearer weather, transit schedules, and reviews of the restaurant they were looking at.

automating

2.1

A way of thinking about technology that asks how to get an existing task done faster, more cheaply, and perhaps more accurately and consistently — in other words, how to raise efficiency. It speeds up the work that is already being done rather than questioning whether the work is being done well.

For example: A bank that once had staff key in paper loan applications lets customers fill the same form out online, so the checking and data entry happen the instant it is submitted.

best-cost provider strategy

2.1

A middle-of-the-road position in which a company offers products or services of reasonably good quality at competitive prices, rather than being the absolute cheapest or the most distinctive. The chapter gives Dell as an example.

For example: Buying the laptop that is not the cheapest on the shelf and not the most expensive, because it is decent quality for what it costs.

business/IT alignment

2.1

Making sure the information systems an organization builds or buys actually match the strategy it is pursuing. Companies try to maximize this, and generally do not want a system that fights the strategy even if that system would save time or money.

For example: A store competing purely on low prices funds a system that squeezes cost out of its warehouse rather than one that adds a luxury personal-shopper service.

business model

2.2

A summary of a company's strategic direction that lays out how its objectives will be reached: what it does, how it uniquely does it, what resources and activities that takes, how it gets paid, and what it all costs. It names the value proposition along with the customer segments, channels, customer relationships, revenue streams, key resources, key activities, key partners, and cost structure.

For example: Writing out, on one page, who would buy your food-truck tacos, how they would hear about you, what you charge, and what the truck and ingredients cost you.

business process

2.1

The activities an organization performs to reach its business goals. These include core activities that take inputs and turn them into outputs, plus supporting activities that make those core activities possible.

For example: Everything a coffee shop does between a customer ordering and walking out with a drink — taking the order, pulling the shot, handing it over — along with the ordering of beans that keeps it possible.

capabilities

2.1

An organization's ability to actually put its assets to work in the marketplace. Owning something valuable is not enough; capabilities are what let a firm turn it into cost savings or a better product.

For example: Two restaurants buy the same ingredients from the same supplier, but only one has cooks who can turn them into a dish people line up for.

collaborative economy

2.2

Another name the chapter uses for the sharing economy: an economic system in which assets or services are shared between private individuals, either free or for a fee, typically by means of the internet. What matters is access to the asset rather than owning it.

For example: Borrowing a neighbor's drill through an app instead of buying a drill you would use twice a year.

cross-side network effects

2.2

The change in value a user gets when more people join the other side of a platform. Platforms have two sides — those who produce and those who consume — and growth on one side pulls the other side along.

For example: An app store gets more useful to you as more apps are added, and developers get more interested in building for it as more people use it.

crowdfunding

2.3

Securing business financing from individuals in the marketplace — the crowd — to fund an initiative, typically some consumer-based product. The supporters, called backers, pledge an amount in return for certain benefits, and on a platform such as Kickstarter or Indiegogo they are charged only if the campaign reaches its funding goal.

For example: Pledging money toward a board game on Kickstarter and paying nothing at all, because too few backers were interested and the campaign never reached its goal.

customer creation

2.3

The first step of the execution phase in the lean startup approach, where a company tests whether it can establish a business model that repeats reliably rather than working just once. Building and scaling the company itself follows after this.

For example: After a few strangers pay for your tutoring service, checking whether you can keep finding paying students week after week instead of only the first three friends of friends.

customer discovery

2.3

The first search step in the lean startup approach: finding out whether the problem you think exists actually does, whether there would be a market for a solution, and then whether the solution you have in mind would really address it. It means talking to potential customers before you build anything.

For example: Asking twenty students whether finding a study partner is genuinely a hassle for them, before writing a single line of an app to fix it.

customer validation

2.3

The second search step in the lean startup approach, where a company puts its minimum viable product in front of people to test whether they would actually pay for the solution. Saying a problem is annoying and handing over money are two different things.

For example: Putting a price on the study-partner app and seeing how many of the students who said they liked the idea will actually subscribe.

differentiation strategy

2.1

Competing by providing better products or services than rivals rather than cheaper ones. A company can aim that broadly at many kinds of buyers or focus it on one segment, as Apple did for years with high-quality computers for home and educational buyers.

For example: Paying more for a pair of shoes because that brand fits and lasts better, not because it was the lowest price on the rack.

digital twin

2.1

A digital representation of a physical object, used to model how that object behaves and interacts with its environment. Feeding it live sensor data lets people troubleshoot or plan maintenance on equipment that is hard or impossible to reach.

For example: NASA kept physical replicas of space modules such as Apollo 13 on the ground to work out how to operate or repair the real one; a digital twin does the same job in software.

disruptive innovation

2.3

Another name the chapter uses for a radical innovation: something built on a markedly new or different technology that reaches new customers or gives existing ones far greater benefits, and eventually pushes aside the product it competes with.

For example: Streaming music pushed compact discs and music stores to the margins.

distinctive competency

2.1

What an organization's resources and capabilities add up to in the marketplace — things like innovation, agility, quality, or low cost. These are what make its product genuinely valuable to customers compared with what rivals offer.

For example: Toyota being known for quality and Nintendo for value are the kinds of competencies the chapter lists as sources of competitive advantage.

effectiveness

2.1

The extent to which goals or tasks are accomplished well. It is about doing the right things properly, which is not the same as doing them quickly.

For example: A tutor who actually gets you to understand the material, even if the session runs long.

efficiency

2.1

The extent to which goals are accomplished faster, at lower cost, or with relatively little time and effort. Systems at the operational level are usually aimed at this.

For example: A self-checkout lane that gets you out of the store in two minutes instead of ten.

executive level

2.1

The strategic level of an organization, where the president and chief executive officer, chief information officer, vice presidents, and possibly the board handle long-term questions such as which products to make and which countries to compete in. Their decisions are unstructured and carry broad, long-term consequences.

For example: The people who decide whether a company should stop making a product line entirely, a call that changes employment and profit for years.

first-mover advantage

2.1

The edge a company can get from being the first to enter a market. Most organizations do not have this luxury and must instead compete inside industries that are already established.

For example: The first food-delivery app in a college town gets known as the one everybody uses before a second one shows up.

freemium

2.2

A blend of free and premium: giving away a limited version of a digital product or service at no charge to build a large user base, then charging for the unrestricted version, usually on a subscription. It works because the cost of serving one more user of a digital product has fallen so low.

For example: Dropbox and iCloud hand out free accounts with a small amount of storage and charge you once you want more space.

functional area information system

2.1

A system built to support the particular business processes of one discrete part of an organization, such as marketing, accounting and finance, human resources, or production and operations. Each area has its own activities and therefore needs its own kind of system.

For example: The payroll system that pays employees is separate from the system marketing uses to plan advertising.

gig economy

2.2

An arrangement in which workers are not employed by the company at all but are hired for short-term, temporary jobs. The chapter notes it often leaves workers without protection against discrimination, without social security, with little job security, and with limited contact with coworkers.

For example: Driving for a ride-hailing app on weekends with no set schedule, no benefits, and no manager or coworkers you ever meet.

in-app purchase

2.2

Buying extra features, content, or in-game currency from inside an app you downloaded for free. It lets a product build a huge user base quickly while still producing a continuous stream of revenue.

For example: Downloading a free-to-play game such as Candy Crush and then spending real money inside it on extra content or in-game currency.

innovation

2.3

Creating new products, processes, or services that return value to the organization. The chapter is careful to separate it from mere invention: innovation means actually realizing the value, not just having the new idea.

For example: Plenty of people sketch a better shopping cart; innovation is the company that builds it, sells it, and makes money on it.

key performance indicator (KPI)

2.1

The handful of metrics judged most critical for assessing progress toward a particular goal. They are usually shown on digital dashboards so managers can see at a glance where things stand.

For example: A store manager's dashboard showing sales per hour, items per transaction, and how much inventory is left.

lean startup methodology

2.3

An approach pioneered by Eric Ries in which a company first searches for a viable product and business model and only executes that model once it has found a fit between the product and the market. The older habit was to build a product first and then hunt for customers.

For example: Testing whether people want your idea by talking to them and shipping a rough version, instead of spending a year building the polished thing nobody asked for.

low-cost leadership strategy

2.1

Competing by offering the best prices in the industry on your goods or services. The chapter gives Walmart as the example.

For example: A store that wins your business simply because the same box of cereal costs less there than anywhere else.

managerial level

2.1

Also called the tactical level, where functional managers in areas like marketing, finance, manufacturing, and human resources monitor and control operational activities and pass information up to higher levels. Their decisions are moderately complex, usually stay inside one business function, and look days to months ahead.

For example: A marketing manager deciding how to split the advertising budget across the coming three months.

minimum viable product (MVP)

2.3

A product built with just the features needed to demonstrate how your solution meets the problem, and nothing more. It exists to test a hypothesis quickly, not to impress anyone.

For example: A single working screen that lets someone match with a study partner, with no profile pictures, messaging, or settings yet.

open innovation

2.3

Bringing outsiders into the innovation process because good ideas often exist beyond the organization's own walls. It usually means opening some organizational data and knowledge to those outsiders, which is a tradeoff a company has to weigh.

For example: Eli Lilly created a site called InnoCentive where companies post scientific problems, anyone can attempt a solution, and a reward goes to whoever solves it.

operational level

2.1

The bottom and widest level of an organization, where routine day-to-day business processes and customer interactions happen. Foremen and supervisors here make highly structured, recurring decisions, and planning covers only a few hours or days.

For example: The supervisor on a store floor confirming that the system's automatic reorder of running shoes should go through.

organizational learning

2.1

An organization's ability to use its past behavior and its own data to improve its business processes. It builds on automating, because the data a system creates while automating work can reveal how that work should change.

For example: A bank studying which approved loans later went bad, then tightening the rules it uses to approve the next batch.

organizational strategy

2.1

A firm's plan for accomplishing its mission and goals and for gaining or sustaining a competitive advantage over rivals. Information systems are judged by whether they help carry that plan out.

For example: A company deciding it will win on speed of delivery, and then funding every project that makes deliveries faster.

platform-based business model

2.2

A model in which the company does not supply the product or service itself but coordinates others — other businesses and its own users — so they create value for each other. That makes it a two-sided market whose success depends on the network effect.

For example: Airbnb owns none of the rooms and Uber owns none of the cars; both depend on users to offer the service and other users to consume it.

radical innovation

2.3

An innovation built on a markedly new or different technology that opens up new customer segments or gives existing customers significantly greater benefits, and eventually marginalizes or replaces what came before. The chapter contrasts it with incremental innovation, which just enhances or upgrades what already exists.

For example: Digital photography displaced chemical photography, and smartphones displaced MP3 players and dedicated GPS units.

resources

2.1

The specific assets an organization uses to achieve lower cost or a product different from its competitors'. Proprietary technology, brand equity, and a loyal, established customer base are the examples the chapter gives.

For example: The chapter names a proprietary asset at Coca-Cola, a user base at Facebook, and a brand at Nike among its sources of competitive advantage — assets a rival cannot simply go out and buy.

revenue model

2.2

A description of how a firm will earn revenue, generate profits, and produce a superior return on the money invested in it. Even nonprofit organizations need one.

For example: Netflix charging a recurring monthly fee, while Google gives its service away and charges advertisers instead.

same-side network effects

2.2

The rise or fall in value that happens when more users join your own side of a network. More users on your side can help you, but they can also crowd you out.

For example: As more people switch to iOS, everyone on it benefits from wider compatibility for the Keynote presentation format; but as more riders open Uber at once, competition for available rides pushes prices up.

semistructured decision

2.1

A decision where some of the procedure can be laid out in advance but not enough to produce a specific recommendation, so judgment and expertise are still required. This is the typical shape of managerial-level decision making.

For example: A production manager looking at sales forecasts, inventory levels, and capacity, then deciding which product to build first.

sharing economy

2.2

An economic system in which assets or services are shared between private individuals, either free or for a fee, typically by means of the internet. Platforms make it work, and the point is being able to access an asset rather than owning it.

For example: Using a car-sharing service instead of buying a car, since a shared vehicle sits idle far less of the day than a private one.

startup

2.3

A new venture, usually technology based, with high potential to scale up and grow quickly. They are often cofounded by entrepreneurs, developers, designers, or others with a promising idea, and may turn to incubators, accelerators, angel investors, or crowdfunding for support.

For example: Two people with an idea and a prototype working out of a shared office, hoping to reach millions of users rather than one town.

strategic necessity

2.1

An investment an organization has no real choice about — something it must do simply to survive, whether or not it lines up neatly with the overall strategy. Not every system a company funds is chosen freely.

For example: A store that has to accept contactless payment because customers now expect it, even though it wins the store no advantage over rivals who all accept it too.

strategic planning

2.1

The work senior managers do to form a vision of where the organization needs to head, turn that vision into measurable objectives and performance targets, and craft a strategy for reaching them. It happens at the executive level.

For example: Leadership setting a three-year target for entering two new countries and deciding what has to be built to get there.

structured decision

2.1

A decision where the procedure to follow in a given situation can be specified ahead of time. Because they are so straightforward, they can be programmed right into operational systems and made with little or no human involvement.

For example: An inventory system that watches stock levels and automatically orders more shoes whenever the count drops below a set number.

transaction

2.1

Anything that happens as part of a firm's daily business that it must keep a record of. These pile up at the operational level and become the raw material higher levels analyze later.

For example: Every single sale rung up at a register, along with the timestamp and the items in the basket.

unstructured decision

2.1

A decision for which few or no procedures can be specified in advance, because the problem is complex and nonroutine. Executives face these, and must weigh the effects on the whole organization.

For example: Top managers deciding whether to launch an entirely new product or kill an existing one, knowing either choice reshapes employment and profit for years.

validated learning

2.3

Testing your assumptions empirically through real customer feedback so the next version is better informed. It is what the build-measure-learn cycle produces: you build the smallest thing that tests a hypothesis, measure how people respond, then decide whether to change direction or keep going.

For example: Instagram began as an app called Burbn crammed with features from check-ins to gaming; once the founders learned users did not value all that, they stripped it down to sharing photos and commenting.

value chain

2.1

The whole set of activities inside an organization that add value as supplies come in, get turned into products and services, and are marketed, sold, distributed, and supported after the sale. Employees can use information systems at any point along it to add more value.

For example: For a hotel, the chain runs from receiving supplies and taking reservations, through checking guests in and cleaning rooms, to resolving a complaint after the guest has gone home.

value chain analysis

2.1

The process of examining an organization's activities to determine where value is added to its products or services and what it costs to add that value. You map out each activity, work out the costs and what drives them, and then decide which activities to optimize to cut costs and gain an advantage.

For example: A pizza shop laying out every step from buying flour to delivering the box, pricing each step, and finding that delivery routing is what is quietly eating the profit.

value creation

2.1

What happens when an organization can offer its products at a lower cost or with better, more differentiated benefits than customers can get elsewhere. Distinctive competencies are what make it possible, and it is how a company gains a competitive advantage.

For example: A phone that costs the same as its rivals but has a camera nobody else can match gives you more for your money.

value proposition

2.2

The usefulness a product or service offers to its customers — the reason they need it, the problem it solves, and why they would pick it over a competitor's. It is one of the core components of a business model.

For example: A meal-delivery service's promise that you get a hot dinner without cooking, shopping, or leaving your apartment.

virtual reality (VR) headset

2.3

A head-mounted device that creates an immersive three-dimensional experience, replacing what you see with a computer-generated environment. Companies redeploy this kind of technology in new ways to gain an advantage.

For example: Architects and interior designers let clients walk through a planned building with a headset such as the Apple Vision Pro or Meta Quest before construction has even started.

XaaS

2.2

Short for X as a service: a business model built on providing something as an ongoing service instead of selling it as a product. The customer pays for use rather than buying the equipment, which turns their fixed costs into variable ones and gives the provider a reason to make the equipment last.

For example: Rather than selling an airline an engine and spare parts, GE is paid for guaranteeing continuous uptime — essentially selling thrust as a service.

PUT IT TOGETHER

Final challenge

Thirty situations rather than thirty definitions. Each describes something happening in an organization and asks what is actually going on, which is the form this material takes once you are the person being asked. Answer all of them, then score: the result breaks out by objective, so a weak area points at the section to reread instead of disappearing into a single number.

FINAL CHALLENGE**Final challenge**

Answer key — the interactive, self-scoring version loads when scripting is available.

1. **2.1** At a hypothetical parts distributor, three decisions are made on the same morning. A shift supervisor confirms a reorder the inventory system proposed when a bin fell below its set level. A marketing manager allocates next quarter's advertising budget across four product lines. The executive committee debates discontinuing a product line altogether. Which decision does the chapter call **structured**, and what follows from that?

- **A.** The supervisor's, because the procedure for this situation can be specified in advance, so it can be programmed straight into an operational system — **correct**. Structured decisions are those in which the procedures to follow for a given situation can be specified in advance, and because they are relatively straightforward they can be programmed directly into operational information systems and made with little or no human intervention. The chapter's own example is an inventory system that issues an order when levels drop below a set point and the manager simply confirms it.
- **B.** The manager's, because a quarterly budget allocation follows an established annual procedure even though the amounts change every time — That is a semistructured decision at the managerial level, where some procedures can be specified in advance but not to the extent that a specific recommendation can be made. The chapter uses a marketing manager allocating a quarter's advertising budget as its illustration of exactly this level.
- **C.** The executive committee's, because discontinuing a line follows the formal governance rules the company has already written down — That is an unstructured decision at the executive level, where few or no procedures for the situation can be specified in advance, and the chapter's example is top managers deciding to develop a new product or discontinue an existing one. Governance rules settle who decides, not what the decision should be.
- **D.** None of the three, because any decision that commits money needs human judgment and so cannot be fully specified beforehand — Money is committed at all three levels, so spending cannot be what separates them. The chapter's dividing line is whether the procedure can be written down beforehand, and an automatic reorder shows that some decisions involving money can be.

2. **2.1** The chief executive of a hypothetical hospital group asks for a system that will show her how the group is trending, where it stands against comparable groups, and what the next two years look like under different assumptions. A vendor proposes a screen showing yesterday's ward-by-ward patient throughput hour by hour. What is the accurate objection?

- **A.** There is no objection to make, because operational detail is the raw material every level works from, so a system that reports it serves an executive as well as a supervisor — Operational data does feed the higher levels, and the chapter says so: systems at the operational level help automate repetitive activities but also gather valuable data for higher decision-making levels. Passing that detail upward unaggregated is not the same as summarizing it, and the chief executive asked for trends, projections, and comparisons.

- **B.** The proposal belongs at the managerial level, since monitoring and controlling operational activity is what midlevel managers do, so the executive should receive the same screen weekly rather than daily — Monitoring and controlling operational activities is indeed the managerial level's work, and this screen would be a reasonable managerial tool. Changing how often an executive receives operational detail does not turn it into the aggregate summaries and projections the chapter says executive decisions require.
- **C.** The proposal fails because executive systems work from unstructured external data such as economic conditions and changing customer tastes, so anything built from the group's own records cannot serve that level — Executives do need unstructured data such as global economic factors, demographic changes, and shifting customer tastes, which is why this option sounds right. The chapter also has executive systems providing key performance indicators that balance performance across the organization and benchmark it against competitors, and those come from the organization's own records.
- **D.** The proposal serves the operational level, where systems automate routine activity to improve efficiency, while the executive level needs aggregate summaries, projections of the future, and benchmarks against competitors — **correct.** Information systems at the operational level automate routine and repetitive activities to improve efficiency, while at the executive level they provide aggregate summaries of past data, projections of the future, and indicators that benchmark the organization against its competitors. The vendor has answered a supervisor's question with an executive's budget.

3. **2.1** A hypothetical lender's approval criteria let weak applications through: reviewers handled about four applications a day and accepted roughly two poor ones a week. The lender then automated the same criteria, reviewers now handle about twelve a day, and poor approvals have climbed. Nothing in the software is malfunctioning. Which reading matches the chapter?

- **A.** The system was implemented badly, because a correctly built approval system would have identified the weak applications as it processed them — A system that screened applications on better criteria would indeed catch them, and that is the point: the criteria are the thing that is wrong. Blaming the build implies the specification was right, when the chapter's example is a specification that approves a borrower carrying heavy debt merely because no payment was recently late.
- **B.** Volume is the whole story: more applications processed produces proportionally more mistakes, and the rate of bad approvals per application is unchanged — More decisions do produce more errors, and this reading stops one step early. Figure 2.9 shows cumulative errors rising at an accelerating rate as automation increases, and treating the increase as arithmetic leaves the faulty acceptance criteria untouched and unnoticed.
- **C.** Automating a flawed process magnifies the existing business problem and can mask it, which is why the chapter pairs automating with learning from the data the system creates — **correct.** The chapter's worked example moves from four applications a day with about two bad approvals a week to twelve a day with up to six, and says the technology would serve only to magnify the existing business problems. Without learning it is more difficult to uncover the bad business process underlying the system, which is why an automating use can mask the problem rather than reveal it.

- **D.** The criteria were sound and the reviewers stopped attending to them once the work sped up, so the remedy is retraining people rather than changing the process — Attention does fall when work speeds up, and retraining is a common first response. Nothing here suggests reviewers are departing from the criteria; they are applying criteria that were producing bad approvals before the system arrived.

4. **2.1** Three proposals sit on the table at a hypothetical building-supply firm. **A:** scan delivery tickets at the dock so nobody retypes them into the system. **B:** analyze two years of delivery exceptions by route, season, and crew, and change how routes are planned as a result. **C:** build reordering from each customer's own consumption history, which no competitor offers. Which proposal is the clearest example of an information system used for **organizational learning**?

- **A.** A, because capturing the ticket at the dock removes handling time and errors, and the chapter treats faster and more accurate work as the first form of value a system adds — A is an automating use: it helps complete a task faster, more cheaply, and with greater accuracy. That is genuine value and it is the base of the chapter's ladder, which is why this would be the right answer to a question about improving efficiency at the operational level.
- **B.** B, because organizational learning is using past behavior and data to improve business processes, and this proposal changes how the work is done on the strength of what the data showed — **correct.** Organizational learning is the ability of an organization to use past behavior and data to improve its business processes, and the chapter's illustration is a loan system whose accumulated records let a manager see patterns and change how the work is planned. Analyzing two years of exceptions and rewriting the routing rules is the same move.
- **C.** C, because the system that produces a competitive advantage is the one the organization has learned the most from building and running — C is a strategizing use, the top of the chapter's ladder, because it looks for competitive advantage rather than for understanding of the process. It would be the answer to a question about which proposal serves the organization's strategy.
- **D.** None of them, because learning happens only when a system adjusts its own rules without a person intervening — Learning in this chapter is something the organization does, not something the software does unattended. The manager who reads the trend and changes the criteria is where the learning happens, and the chapter's loan example works exactly that way.

5. **2.1** A hypothetical commercial printing firm is in trouble. Its two long-standing competitors match every price cut within days, and margins at all three are thin. Its own market study shows something else as well: total pages printed by all three firms together has fallen by about a third over three years, as clients moved documents into shared online workspaces. Which force is the strongest here?

- **A.** Traditional rivals, because the competitors matching every price cut are the visible cause of the lost margin, so the answer is to compete on service rather than on price — Rivalry is genuinely present and it explains why nobody in the industry earns much on the pages still being printed. It cannot explain why a third of the pages disappeared, because a rival taking work would move share between the three firms rather than shrink the total.

- **B.** Threat of new entrants, because the workspace providers have entered a market the printing firms used to hold and found the barriers low enough to walk through — A new entrant is a firm that starts selling what you sell. A shared online workspace prints nothing and sells no pages, and that difference is exactly what separates an entrant from another industry's answer to the same need.
- **C.** Customers' bargaining power, because clients are refusing to pay printing prices and are holding up the online alternative as leverage in every negotiation — Buyer power is customers forcing better terms while still buying from your industry, which shows up as pressure on price, quality, and service. These clients are not negotiating over a page rate; they have stopped needing the pages.
- **D.** Threat of substitute products or services, because the same need is being met from outside the industry and the whole category is shrinking rather than one firm's share of it — **correct.** A substitute meets the same underlying need by a different route, so the customer never enters your industry, and the chapter's implications for this force are product returns, decreased market share, and losing customers for life. The tell is the category-level decline, which nothing happening inside the industry could produce.

6. 2.1 A hypothetical chain of fourteen bakeries buys its flour blend from one of only two mills that make it. Both raised prices this year and lengthened lead times, and the chain has switched to a coarser blend that customers have noticed and mentioned. Each shop still phones its own order in, so nobody at head office can state what the chain buys in total. Which force is at work, and what does the chapter say it implies?

- **A.** Suppliers' bargaining power, whose implications the chapter gives as increased costs and reduced quality, both of which have already reached the customer here — **correct.** Suppliers' bargaining power is the ability of the firms you buy from to set terms because you have few alternatives, and Table 2.3 gives its implications as increased costs and reduced quality. Both landed in the product the customer eats, and a chain that cannot state its own total volume has nothing to negotiate with even when an alternative exists.
- **B.** Traditional rivals, because competing bakeries buying from the same two mills face the same increase, and none of them can undercut the others while it lasts — Shared exposure is a real and useful observation, since a rival facing the same cost will not undercut you easily. It still does not name where the pressure originates, which is upstream of the bakeries rather than between them.
- **C.** Customers' bargaining power, because the customers who noticed the coarser product are the ones who will now press for a lower price or leave — Buyer power is customers forcing your price down or your quality up because leaving costs them little. Customers here are reacting to a change that was made to them; the terms were set two steps earlier, by the mills.
- **D.** Threat of new entrants, because two mills serving a whole region shows the barriers to entry in that market are low enough to attract others — The threat of new entrants concerns firms beginning to compete in your industry, which for this chain would mean a new bakery. Two mills supplying a whole region is evidence of high barriers in milling rather than low ones.

7. **2.1** A hypothetical furniture manufacturer builds a system that lets its designers assemble and stress a chair frame virtually before any tooling is cut, so weak joints are found in the model rather than on the production line. Where does this initiative sit on the value chain?

- **A.** Operations, because the chair is what the customer buys and this system exists so that it is built correctly — Operations integrates supplies into the products and services the customer buys, and a better frame certainly reaches operations eventually. Naming the activity a system actually changes is what keeps the analysis usable, and no one is cutting or assembling anything at the point this system is used.
- **B.** Inbound logistics, because the timber and fittings the model specifies determine what must be received before production can begin — Inbound logistics is receiving what has been purchased and getting it inside before any of it is sellable. A design decision changes what will later be ordered and received, which puts it upstream of this activity rather than inside it.
- **C.** Product research and development, technology, and systems development, a support activity, because it improves the product and the systems the whole chain runs on — **correct.** The support band covering product research and development, technology, and systems development is where the product and the processes themselves are improved, and Figure 2.15 places computer-aided design systems among the systems feeding that supporting layer. The payoff is spread across the whole chain, which is the signature of a support activity.
- **D.** Procurement, because deciding which joint and which fitting to use is a buying decision taken before anything is manufactured — Procurement is the decision about whom to buy from and on what terms, and it is a support activity in its own right. Specifying a joint is a design choice; choosing the supplier who will provide the fitting is the procurement decision that follows it.

8. **2.1** A hypothetical regional insurer competes on the speed and care of its claims handling. Its industry then adopts a shared electronic claims exchange: connecting confers no distinction because every carrier will be on it, but brokers will have no way to submit claims to a carrier that is not. The investment committee is split. What does the chapter's framework support?

- **A.** Decline it, because organizations try to maximize business/IT alignment and a system that offers no differentiation fails that test — Alignment is the general rule and the chapter states it plainly: in most cases organizations do not want systems that fail to match the strategy. This is the recognized exception to that rule, and applying the rule without the exception would disconnect the insurer from the brokers who bring it work.
- **B.** Fund it as a strategic necessity, which the chapter describes as an investment an organization must make to survive and which may or may not coincide with its overall strategy — **correct.** The chapter says organizations sometimes have no choice about investments that may or may not coincide with their overall strategy, and calls such an investment a strategic necessity, something the organization must do to survive. Survival, not differentiation, is the test being applied.

- **C.** Fund it and present it to the board as a competitive advantage, since being connected to the exchange is something the insurer will be able to offer its brokers — Every carrier will be connected, so this buys parity rather than advantage, and the chapter reserves competitive advantage for what rivals have not matched. Calling a necessity an advantage also hides the real reason for spending, which makes the next proposal harder to judge.
- **D.** Defer it until the exchange has proved itself, because a system every rival will also hold cannot repay what it costs to build — Waiting is a reasonable posture toward an optional system whose value is unproven, and it would be the right instinct if the only question were payback. Here brokers cannot submit claims to a carrier that is not connected, so deferring is a decision to lose the work in the meantime.

9. **2.2** A hypothetical home-repair booking company employs nobody who holds a wrench.

Repairs are carried out by independent tradespeople it recruits, rates, and pays per job, and its own scheduling and dispatch software is the thing rivals have not been able to reproduce. Using the chapter's business model components, how should those two be classified?

- **A.** The tradespeople are key resources and the software is a key activity, because the people are what customers actually pay for and dispatching is what the company spends its day doing — This reverses both definitions. Key resources are the most important assets needed to make the business model work, which fits the software rather than the tradespeople, and key activities are the most important things the company does rather than a piece of software it owns.
- **B.** Both are channels, because between them they are how the service reaches the customer who booked it — Channels are the ways the product or service offering reaches the target customers, such as an app or a counter. The people and the software are how the service is produced and coordinated, which the model files under partners and resources instead.
- **C.** The tradespeople are a customer segment, because the company recruits them, rates them, and has to keep them satisfied to stay in business — Producers on a two-sided market do have to be attracted and kept, which is why this reading is tempting. Customer segments names the customers targeted with the offering, and the component for the outside people and firms the model depends on is key partners.
- **D.** The tradespeople are key partners and the software is a key resource, because partners are the network needed to make the model work and resources are its most important assets — **correct.** Key partners are the network of partners and suppliers needed to make the business model work, which is what independent tradespeople are here, and key resources are the most important assets needed to make it work, which is what proprietary scheduling and dispatch software is.

10. **2.2** A hypothetical language-learning app has three sources of income. Anyone can use it free with advertisements. Learners who want the advertisements removed and extra practice sets pay a monthly fee. When a learner books a session with an independent tutor through the app, the app keeps a commission. How should its revenue model be described?

- **A.** It runs three at once: advertising on the free tier, a freemium subscription for the paid tier, and a transaction fee on tutor bookings, because the revenue streams component can hold more than one — **correct**. A revenue model describes how the firm will earn revenue, and the business model component is revenue streams, plural: the chapter's own examples show firms combining approaches, with Dropbox freemium and subscription, Amazon selling and running an affiliate program. Naming only one here would leave two thirds of the income unexplained.
- **B.** Freemium, because the entire design is a limited free version alongside a paid version with the restrictions removed — Freemium is accurately described in this option, and it does cover the free tier and the paid tier: limited versions given away to build a base, with a premium charged for the unrestricted version. It says nothing about the advertisements or the commission on bookings.
- **C.** Advertising, because the free tier is what brings in the users that everything else depends on — Advertising is free services provided to customers and paid for by a third party, which is exactly what the free tier does. It is one of the three streams rather than the whole answer, and it would be the right answer if the paid tier and the tutor commission did not exist.
- **D.** Transaction fees, because the commission is the only income tied to something the app actually arranges for the learner — Transaction fees or brokerage is a commission paid to the business for aiding in a transaction, which the tutor booking fits precisely. Being the income most visibly attached to a service does not make it the only income the model earns.

11. **2.2** A hypothetical kitchenware manufacturer has always made its own pans and sold them through its website. It now adds two things: reviews written by owners, and a marketplace where independent restorers list refurbished pans and pay the manufacturer a commission on each sale. Which description is accurate?

- **A.** It is now a platform business rather than a manufacturer, because a platform coordinates others instead of producing anything itself — A platform-based business model does coordinate others rather than providing the product itself, which is why this sounds right. The manufacturer has not stopped making pans, and the chapter's point is that the two can sit in one business rather than that one replaces the other.
- **B.** It is still purely a pipe, because it produces goods and pushes them to customers, and reviews and third-party listings are marketing features rather than parts of a business model — The pipe description is accurate as far as it goes, since products are still produced and pushed to customers. The chapter says explicitly that many pipeline-based businesses now include platform elements such as reviews created by users, and a commissioned third-party marketplace is a good deal more than a marketing feature.
- **C.** It has added platform elements to a pipe: the marketplace coordinates others to co-create value, creating a two-sided market whose success depends on network effects — **correct**. A platform-based business model coordinates others, both businesses and users, to co-create value, and the chapter notes that many pipeline businesses include platform elements and that every business must consider how far its model will include them. The restorers produce value that buyers consume, which is the two-sided market the chapter describes.

- **D.** It has entered the sharing economy, because assets that already exist are being passed between private individuals rather than newly manufactured — The sharing economy is an economic system in which assets or services are shared between private individuals, free or for a fee, typically over the internet. A refurbished pan sold outright through a commissioned listing is a sale, not shared access to an asset somebody keeps.

12. 2.2 On a hypothetical marketplace for freelance editors, two things are observed over the same quarter. First, as more editors join, each editor faces more competition for the same jobs. Second, as more editors join, the publishers who hire them find the site more useful because more specialists are available. Which classification is correct?

- **A.** The first is a cross-side effect and the second a same-side effect, because the harm lands on the editors themselves while the benefit crosses over to the publishers — This has the two labels the wrong way round. A same-side effect is the change in value as additional users join on the same side, and editors crowding other editors is that; a cross-side effect is the change in value for a user when users join on the other side, which is what the publishers experience.
- **B.** The first is a same-side effect and the second a cross-side effect, because value falls for users on the same side in one case and rises for users on the other side in the other — **correct.** Same-side network effects relate to the increase or decrease in value as additional users join on the same side, and the chapter's own negative example is riders competing for available rides. Cross-side network effects relate to the change in value for a user when users join on the other side, as when an app store becomes more useful to users as more apps are added.
- **C.** Both are same-side effects, because in each case the change was caused by editors joining rather than by publishers joining — The cause is the same in both observations, which is what makes this reading tempting. Effects are classified by whose value changes rather than by who joined, so the same arrival of editors produces a same-side effect for editors and a cross-side effect for publishers.
- **D.** Neither is a network effect: the first is ordinary competition between sellers and the second is simply a bigger catalogue for buyers to choose from — Competition among sellers and a larger catalogue are exactly what these effects look like from inside; the terms name the mechanism rather than replace the description. A platform only has value if users participate on both sides, which is why the chapter asks platform businesses to evaluate these effects deliberately.

13. 2.2 On a hypothetical grocery-delivery platform, couriers accept jobs one at a time, are not employed by the company, have no protection against discrimination, no social security contributions, and little job security. The company answers criticism by saying it merely removes friction from transactions people were already free to make. How does the chapter frame this?

- **A.** This is the gig economy, where workers are hired only for short-term temporary jobs, and the chapter names precisely these losses to workers as a criticism of such business models — **correct.** In the gig economy workers are not employed by the company but are hired for short-term, temporary jobs, and the chapter lists the consequences directly: rights fought for since the Industrial Revolution neglected, no protection against discrimination, no social security, little job security, and limited income opportunities and social contact.

- **B.** This is disintermediation, because the platform has removed the traditional middleman standing between the shopper and the courier — Disintermediation is real here, since many platforms match providers and customers without traditional middlemen, and that is roughly the argument the company is making. It names what happened to the intermediary rather than what happened to the people doing the work.
- **C.** This is a same-side network effect, because couriers competing with each other for the same jobs is what determines what the work pays — A same-side effect is a genuine feature of this market, and more couriers competing for the same jobs is one. It describes how value shifts among users of a platform, not the employment arrangement the criticism is about.
- **D.** This is unfair competition, which the chapter defines as operating with no employees while established rivals carry the cost of a workforce — The chapter does raise unfair competition, and it defines it as gaining advantage over traditional businesses by breaking or at least bypassing existing laws and regulations. Employing nobody is a feature of the gig arrangement rather than the definition of unfair competition.

14. **2.2** A hypothetical maker of commercial laundry machines is considering a change: instead of selling machines to hotels and then selling spare parts and repairs, it would keep ownership and charge per completed wash cycle. Which consequence does the chapter say follows from a move like this?

- **A.** Revenue becomes more predictable and little else changes, because the same machines go to the same hotels under a different payment schedule — Smoother revenue is a fair expectation and one reason firms find these models attractive. The chapter is explicit that a great deal else changes, including pricing models, business processes, service level agreements, capacity management, and the customer segment being sold to.
- **B.** The maker should now press harder on minimizing manufacturing cost, since it keeps every machine it builds and margin per unit matters more than it did — Minimizing manufacturing cost is the instinct a pipe-based manufacturer brings with it, which is exactly why the chapter calls it out. Under a service-based model the manufacturer has to focus instead on life span and ease of maintenance and repair, because it now pays for every failure.
- **C.** The change is essentially a financing arrangement, so the important new work is assessing each hotel's credit rather than altering the product or the processes — Money does move differently and credit exposure is real, so this concern belongs somewhere on the list. Treating the change as financing misses the operational half: the maker only earns while the equipment runs, which rewrites what it should design for and monitor.
- **D.** The hotel's fixed costs become variable and the maker earns only while machines run, so its incentive moves to life span and preventive maintenance, which requires connected sensors and analytics — **correct.** The chapter's jet-engine example has the customer paying for hours of usage rather than buying the engine, transforming fixed costs into variable costs, while the manufacturer earns only for operating equipment and therefore has an incentive to improve design and perform preventive maintenance. It adds that such models are possible only through sensors connected to the internet and advanced analytics that predict failure and schedule maintenance.

15. **2.3** A hypothetical veterinary group made two changes last year. **A**: it upgraded its booking software to a version that sends automatic reminders, and missed appointments fell. **B**: it began remote video consultations, which brought in rural clients who had never used any practice regularly, and which are steadily taking over routine follow-up visits. Which change is a **radical innovation** as the chapter defines it?

- **A**. A, because the software version is new to this practice and the change produced a measurable business result — A is an incremental innovation, which the chapter calls the most common form: enhancing or upgrading existing products, services, or processes. Being new to the adopter is not the test, otherwise every purchase a firm had not made before would count as radical.
- **B**. Neither, because both run on software the practice bought rather than technology it developed itself — The chapter's own table of radical innovations lists online retailing, distance education, and online stock brokerage, none of which required the adopting organization to invent the technology. Who built it is a separate question from what it displaced.
- **C**. B, because a markedly different technology has reached a customer segment the practice never served and is marginalizing an existing service — **correct**. Radical innovations use a markedly new or different technology to access new customer segments and/or provide significantly greater benefits to existing customers, and eventually marginalize or replace existing products or services. Both halves are present here: a segment that was never served, and routine visits being displaced.
- **D**. Both, because the chapter treats any innovation that returns value to the organization as radical — Returning value to the organization is what separates innovation from mere invention, so both changes qualify as innovations. That test does not distinguish the two sizes, and treating every valuable change as radical empties the word of the meaning the chapter gives it.

16. **2.3** A hypothetical grocery chain stops charging a fee for each home delivery and offers a flat monthly delivery pass instead. The products, the stores, the vans, and the delivery routes are unchanged; what changed is how the customer pays. Which of the ten types of innovation is that?

- **A**. Service innovation, because the customer experiences delivery differently now and their experience of the offering is what changed — Service innovations support and enhance the value of an offering, and the chapter's examples are a retailer's philosophy of delivering excellent customer service and a clothier offering free lifetime pressing. Nothing about the standard of service changed here, only the basis on which it is billed.
- **B**. Profit model innovation, because it is a novel way of generating revenue from an offering that is otherwise the same — **correct**. Profit model innovation is finding novel ways of generating revenues from offerings, and the chapter's examples are a storage firm using a freemium approach, a software firm offering its suite on subscription, and an engine maker selling thrust as a service. A pass replacing per-delivery fees is the same kind of change.
- **C**. Channel innovation, because delivery is how the offering reaches the customer and delivery is what the change is about — Channel innovations use innovative ways to connect offerings with customers, such as an immersive flagship store or a coffee brand partnering with hotels and airlines. The van reaching the same doorstep is the same channel; what moved is the money.

- **D.** Process innovation, because the way the chain plans and staffs its delivery rounds will have to change to serve pass holders — Process innovations change the primary processes used to produce products or services, such as pioneering lean production or moving fashion from design to store in three weeks. Scheduling may well adapt, but adapting to a pricing change is not the same as changing how the work is produced.

17. **2.3** A hypothetical team building scheduling software for salons ships a minimum viable product with six features. Testing shows that owners use only the waitlist feature, will not pay for the other five, and say they would happily pay for a waitlist tool on its own. What does the lean startup methodology say the team should do?

- **A.** Persevere and add more features, because a single-feature product cannot support a business and the other five need longer to be understood — Adding features is the inside-out instinct the chapter warns against, and it is what the evidence contradicts. The chapter's example runs the other way: an app combining check-ins, gaming, and photo sharing was stripped down to the features users valued and became successful afterwards.
- **B.** Abandon the venture, because a hypothesis that failed in customer validation shows there is no market for the product — Abandoning is the response to learning nothing, and this test taught the team a great deal. When a hypothesis does not hold the methodology asks for a pivot, a change of direction grounded in what was learned, rather than the end of the venture.
- **C.** Move into the execution phase, because customers have said they would pay and willingness to pay is what the search phase exists to settle — Willingness to pay is indeed what customer validation tests, which makes this tempting. What customers said they would pay for is not the product that was built, so the search phase has not yet produced a product/market fit to execute on.
- **D.** Pivot: change direction while staying grounded in what was learned, which is what the build-measure-learn cycle exists to produce — **correct.** If at any stage of the search process a hypothesis does not hold, it is time to pivot, changing direction while remaining grounded in what was learned, and teams typically go through several iterations and pivots before finding product/market fit. The measure step here has produced exactly the learning a pivot is made from.

18. **2.3** A hypothetical equipment-rental firm surveys its regular customers about which emerging technology it should adopt. The customers name one they already encounter elsewhere, and the firm decides to build its differentiation on it. What does the chapter's account of innovation say about this reasoning?

- **A.** It is weak, because customers understand their own problems very well but are not experts in solutions, and a technology they already know is one competitors know as well — **correct.** The chapter says customers are likely to understand their problems very well but are not experts in solutions and are not likely to know about new technologies, and adds that if they do know about a technology then competitors probably do too, which makes it the wrong one to rest a competitive advantage on.

- **B.** It is sound, because the customers are the people who will use it, and outside-in thinking means building what customers ask for — Outside-in thinking is the chapter's recommendation, so the label is right and the application is not. Outside-in means starting from the customer's job-to-be-done and the friction points blocking it, which is a question about problems rather than a shopping list of technologies.
- **C.** It is sound, because a technology customers already recognise carries less adoption risk than one they have never seen in use — Familiarity does lower adoption risk, and that is a real benefit worth weighing. The chapter's objection is about advantage rather than adoption: a technology already visible to everyone will not distinguish this firm from the rivals who can see it too.
- **D.** It is weak, because customer surveys are inside-out thinking, and outside-in thinking means starting from the firm's own capabilities and finding a market for them — This reverses the two terms. Inside-out thinking is developing a solution and then trying to find a market for it, while outside-in thinking starts from the customer's need, so a survey of customers is not itself inside-out.

19. **2.3** A hypothetical logistics firm must choose between two incompatible shipment-tracking standards. One is clearly better engineered; the other is already used by more of the firms it trades with. A manager argues that the choice is obvious, because the better technology will win in the end. What does the chapter's treatment of innovation support?

- **A.** Agree, because the superior technology is the safer investment and markets correct toward quality given enough time — Quality does matter and it often wins, which is why this is the intuitive answer. The chapter's videocassette example exists precisely to show that most experts agreed one format recorded and played back better and that the other format won the marketplace anyway.
- **B.** Agree, because a technically superior system is proprietary technology, which is the kind of resource rivals cannot copy — This misapplies the word proprietary. The chapter says an organization typically cannot patent its use of an information system, particularly one purchased from a vendor: the data in the system may be proprietary, but the system usually is not.
- **C.** Disagree: choosing among innovative investments always entails risk, and the technically superior option can still lose the market, as Betamax did to VHS — **correct**. Choosing among innovative investments always entails risk, and the chapter's classic example is the videocassette format contest in which the technically superior choice turned out to be the unfortunate one. It adds that for organizations the risk is greater still, given the size and often mission-critical nature of the investment.
- **D.** Disagree, and wait until one standard has plainly won, because a firm that never commits early cannot make an expensive mistake — Waiting removes one risk and creates another, and the chapter's account of innovation is that it has to be continuous, with organizations constantly on the lookout and never resting. It also leaves the manager's reasoning untouched, since the eventual winner is exactly what nobody can read off engineering quality.

20. **2.3** A hypothetical insurer announces it will adopt the lean startup approach for new services. Its funding is approved once a year for a whole project, individual bonuses depend on delivering the plan as written, and a cancelled experiment counts against the person who proposed it. What stands in the way?

- **A.** Nothing serious, provided the teams are made cross-functional, because the methodology's requirements concern how teams are composed rather than how work is funded and rewarded — Cross-functional teams are one of the requirements the chapter names, so this is half right. It stops there, and the chapter also calls for incentives at the team level rather than the individual level and for a startup-like budgeting model, both of which this firm currently contradicts.
- **B.** A resource requirement, because what the firm needs is enough people with the knowledge, skills, and time to build a minimum viable product quickly — Having enough people with the relevant knowledge, skills, and time is a genuine resource requirement the chapter lists, and it may well bind here too. Nothing in the situation says the insurer lacks people; what it describes is how money is released and how behavior is rewarded.
- **C.** Nothing, because the methodology was developed for startups, so an established insurer adopting it needs no change to how it funds and rewards work — The chapter answers the first half directly, saying the approach does not only work for startups and naming established companies that have used it successfully. The second half is where this fails: the chapter treats process, resource and risk-tolerance requirements as conditions the organization has to meet, not as things a team can work around while funding and rewards stay unchanged.
- **D.** Process and risk-tolerance requirements: accountability and incentives need to sit at the team level, funding must be released in stages and be cuttable, and failure has to be tolerable — **correct.** The chapter's process requirements call for cross-functional collaboration, incentives and accountability at the team level rather than the individual level, and a startup-like budgeting model where funds are approved as the project moves through stages or cut quickly. Its risk-tolerance requirement adds that members must be willing to test products that may not meet requirements and to fail fast.

21. **2.4** An analyst at a hypothetical distributor asks an AI assistant which competitive force is pressing hardest on the firm. It returns a fluent paragraph naming customers' bargaining power, citing an industry switching-cost figure and a named competitor's loyalty program. Neither the figure nor the program can be found anywhere. What is the right next move?

- **A.** Keep the diagnosis and drop the two citations, because the force it named is the part the analysis actually needs — This is the tempting shortcut, and it treats the conclusion as separable from the support that came with it. The same process produced both, and the module's own test for a recommendation is a named force together with the observable evidence that it is the strong one.

- **B.** Treat the paragraph as a hypothesis and go and collect the firm's own evidence, because a confident output can be invented and a recommendation has to rest on something observable — **correct.** A confident, fluent output that turns out to be untrue is the failure this module has been guarding against, and the remedy is a person between the output and the decision. Naming a force is a hypothesis worth testing against renewal history, account concentration, and win-loss records, which are things only this firm holds.
- **C.** Ask the assistant to rate its own confidence in each claim and keep only the claims it rates highly — A self-assessment is produced the same way as the rest of the passage, and the paragraph in question already read as confident while being wrong. Asking a text generator to judge its own text adds a second output to verify rather than a check on the first.
- **D.** Stop using the assistant for work of this kind, because a tool that can invent a citation cannot contribute to a strategy discussion — Refusing the tool entirely is the over-correction, and it gives up real assistance in framing a question and listing candidates. The lesson the module draws is to keep the person in control of the decision, which is a role for the tool inside the work rather than a reason to bar it.

22. **2.4** The strategy team at a hypothetical manufacturer wants a summary of its unreleased pricing model and its supplier contract terms. Someone proposes pasting both documents into a public AI assistant the firm holds no agreement with, whose provider states that submitted content may be retained and used to improve the service. What should be raised first?

- **A.** That submitting the documents moves the firm's own proprietary material outside its control to a provider that says it may keep it, and that material is the resource the firm's position rests on — **correct.** The chapter treats resources such as proprietary technology and accumulated commercial knowledge as the assets an advantage is built from, and unreleased pricing and negotiated supplier terms are exactly that. Once the file is submitted it has left the firm's control, and the provider has said it may retain the content.
- **B.** That the summary may contain an invented figure, so every number in it will have to be checked against the source documents before anyone relies on it — Checking figures against the source is right and would be the first concern if the documents held nothing sensitive. It is a concern about what comes back rather than about what goes out, and the damage in this case is done at the moment of submission.
- **C.** That an assistant trained largely on freely available public content is unlikely to read commercial supplier contracts well enough to summarize them accurately — How well an assistant reads an unfamiliar commercial document does affect the quality of the summary, so this is a fair question about output. It is a separate question from whether the document should be sent to an outside service at all.
- **D.** That summarizing these documents is work the strategy team is there to do, so handing it over makes the team less valuable to the firm — Drafting help is not the same as surrendering the decision, and a summary the team then reads and argues with is assistance of the kind this module endorses. What is wrong here is what the assistant is being handed, not that it is being asked.

23. **2.4** A hypothetical retailer trains a model on eight years of its own promotion results to propose next season's promotions. The chain now wants to win a customer group it has never served. The model keeps proposing variations of what has worked for the customers it already has and proposes nothing for the new group. Nothing is malfunctioning. What explains the result?

- **A.** The output is a hallucination, because the proposals do not correspond to what the chain intends to do next season — A hallucination is a confident statement that is not true, such as a fabricated figure or a citation to something that does not exist. These proposals invent nothing; they repeat faithfully what the chain actually did, which is a different fault needing a different fix.
- **B.** The result shows the new customer group is not worth pursuing, since eight years of the chain's own data turned up nothing that would reach them — This reads silence as evidence, which is the reasoning error the situation is built to expose. Records of customers the chain never sold to do not exist, so the data cannot speak for or against that group and the absence of a proposal says nothing about its value.
- **C.** The model reflects the data it was given: the record holds the chain's history rather than its strategy, so the remedy lies in the inputs and in human judgment rather than in better prompting — **correct.** Output quality rests on input quality, and a dataset of eight years of one chain's own promotions describes the customers it had rather than the customers it wants. No amount of phrasing will produce evidence about a group the record never contained, so the work belongs with the analyst and with new data collection.
- **D.** The problem is how the request was phrased, and a prompt naming the new customer group explicitly would produce a proposal the chain could act on — Sharper requests do produce better responses, and naming the group would certainly change the wording of the answer. It would not change what the model has to draw on, so the reply would be assembled from patterns belonging to the existing customers and presented as though it were about the new ones.

24. **2.4** Four uses of an AI assistant are proposed during a strategy review at a hypothetical wholesaler. Which one fits the discipline this module has been teaching?

- **A.** Let it choose which of the three competing proposals to fund, because it can weigh them without the internal politics that distort the room — Hard judgment is where analysis feels most valuable, and the wish to take politics out of a funding decision is a real one. Handing over the decision itself is what the companion rule refuses; having it compare the three proposals against stated criteria while the committee decides would fit.
- **B.** Let it write the final recommendation, complete with the measure and the review date, and take that draft to the executive committee as written, since the analyst supplied the evidence it was built from — Supplying the evidence is a real safeguard, and drafting is genuinely the kind of help this module endorses, which is what makes this the hardest of the four to refuse. The verification step belongs after the drafting rather than before it, because a figure, a date or a term can still be restated wrongly in the draft, and whoever carries it into the room is the person who will be asked to defend it.

- **C.** Let it generate the evidence for the force we have already agreed is strongest, so that the recommendation reads convincingly when it reaches the committee — This inverts the whole sequence the module teaches, which runs from evidence to a named force to an initiative. Assembling support for a conclusion already reached is the opposite of a diagnosis, and it would survive the meeting only until somebody checked the numbers.
- **D.** Let it list candidate substitutes and possible cost drivers for the analyst to test against the firm's own sales and purchasing records — **correct.** Candidates and cost drivers are the kind of breadth an assistant supplies well, and every one of them is then checked against records only this firm holds. The analyst keeps control of what the evidence supports, which is the arrangement that makes the person more valuable rather than less.

25. **2.4** For several months a hypothetical wholesaler's team has received a weekly competitive-scan summary drafted by an AI assistant from public news and vendor sites. Nobody now opens the underlying sources. A manager proposes making the summary the firm's main strategic input, since it covers more ground than the analysts ever did and arrives days sooner. How should that proposal be judged?

- **A.** Sound, because the summary reaches more sources than the analysts could read and speed is what makes a diagnosis useful while a decision is still open — Coverage and speed are genuine gains, and the module does argue that the analyst who closes the gap fastest is the one whose recommendation arrives while the decision is open. Speed does not settle where an advantage comes from, which is the claim this proposal is actually making.
- **B.** Weak, because the summary rests on content any rival's assistant can read too, so it can support parity at best, while the firm's own records are the input rivals do not have — **correct.** The chapter's account of advantage turns on resources and capabilities superior to those of competitors, and on the observation that rivals can copy what they can also obtain. Public sources are available to every competitor, while this firm's own purchasing, renewal, and margin records are not, which is why the summary belongs alongside the internal evidence rather than in place of it.
- **C.** Sound as long as the assistant is required to cite each source, because a summary with citations attached is evidence in the sense this module has been using the word — Citations do make verification possible and they are worth insisting on. A cited public article is still a statement about the industry rather than observable evidence about this firm's buyers, suppliers, or costs, which is what a named force has to rest on.
- **D.** Weak, because the real loss is that nobody on the team reads a source before repeating what the summary says about it — This names a real cost, since a team repeating an unchecked summary cannot say where a claim came from. It is an argument for keeping the reading habit rather than an answer to whether a public-source summary can carry the firm's strategy.

26. **2.5** An analyst at a hypothetical wholesaler wants a list of departments whose total spending for the year exceeded fifty thousand. Working from a single spending table, she groups the rows by department and totals the amounts, but places the condition on the total in the clause that filters rows before any grouping happens. The query will not give her what she wants. Why?

- **A.** The row filter runs before the rows are grouped, so the totals it is being asked to test do not exist yet; a condition on a total belongs in the clause that filters groups — **correct**. The row filter keeps only the rows that satisfy a condition and it runs before anything is totalled, which is why it quietly determines every number that follows. Filtering on a total requires the clause that filters grouped rows, and that clause exists precisely because the row filter has finished its work by the time totals exist.
- **B.** The total was never named among the columns the query returns, and a condition can only test a column that appears in the returned result — This is a common belief and it is worth testing: conditions and returned columns are separate decisions, and a query can perfectly well filter on something it does not display. The trouble here is not visibility but timing, since the total has not been calculated at the moment the row filter runs.
- **C.** The result was never sorted, and a condition on a total is applied as the rows are put in order, so nothing is compared until a sort is added — Sorting arranges the rows that survive; it neither creates nor removes any of them. Adding a sort here would order a result that is still wrong, which is why an ordering clause never repairs a filtering mistake.
- **D.** The department names and the amounts have not been brought together, so a join is needed before either column can be used in a condition — Everything the analyst needs sits in one table, so nothing has to be joined before the condition can be written. A join brings related rows from two tables together on a matching column, which would matter if the department names lived somewhere else.

27. **2.5** A hypothetical service desk holds six tickets: **A** north, 5 days; **B** north, 2 days; **C** south, 9 days; **D** north, 4 days; **E** east, 3 days; **F** north, 3 days. An analyst keeps only the tickets from the north region that took more than three days, counts what survives, and reports that the north region resolves most of its tickets slowly. What is wrong with the report?

- **A.** Nothing: two of the four northern tickets took more than three days, which is a fair basis for calling the region slow — The arithmetic is nearly right and the conclusion is not: two of four is half rather than most. Even that comparison is unavailable from this result, because the surviving rows no longer include the fast northern tickets the share would be measured against.
- **B.** The condition should have been three days or more rather than more than three, which would have kept ticket F and changed the count — Boundary conditions genuinely matter, and ticket F at exactly three days is excluded by the wording as written. Moving the boundary changes the count by one and leaves the report's two real problems exactly where they were.
- **C.** The filter removed every other region as well as the fast northern tickets, so the result can neither compare north with anywhere else nor say what share of northern tickets is slow — **correct**. A filter applied before anything is counted decides every number that follows, and this one discarded both halves of the comparison the claim depends on: the northern tickets resolved quickly, and every ticket from the south and east. The result supports only the statement that two northern tickets took more than three days.
- **D.** The count is right but the rows should have been grouped by region first, because a count means nothing until the rows are grouped — Grouping would produce a count for each region present in the surviving rows, and only one region survived the filter. The group would therefore be north on its own, which is the same number reported in a different shape.

28. **2.5** A hypothetical retailer is matching a loyalty file against a web-store file on the customer email address. Around forty per cent of the records fail to match even though staff recognise the same customers in both. A spot check shows differences of capitalisation, trailing spaces, and in a few rows an invisible character. What is the right first move?

- **A.** Correct the raw values in both files by hand until the two columns agree, since matching them is what the work is for — Editing in place does eventually produce values that agree, and it destroys the originals, so nobody can afterwards show where a matched record came from. It also has to be redone by hand every time either system exports again.
- **B.** Build standardised columns beside the raw ones in each file and match on those, leaving the original values in place so anyone can later check where a match came from — **correct.** A spreadsheet function takes a value and returns a new one without changing the original, which is why cleaning is done in new columns: the raw values stay available when someone asks where a number came from. Removing stray spaces, stripping non-printable characters, and standardising capitalisation is what makes two records for one customer match at all.
- **C.** Match on the part of each address after the marker instead, because the domain is stable and does not carry the stray spacing — Domains are stable, which is what makes this idea appealing. Matching on the domain merges every customer at the same organization into a single record, which replaces a matching problem with a larger counting problem.
- **D.** Ask for fresh exports from both systems, because data with invisible characters in it cannot support a customer analysis — A fresh export is worth asking for when a file is genuinely corrupt, and here the inconsistency is in how the values were entered rather than in the transfer. The same untidy values would arrive again, and the cleaning step would still be waiting.

29. **2.5** An analyst at a hypothetical wholesaler pulls the username out of each email address by taking the first eight characters of the address. It looks right for a few rows and is wrong for many others. What has gone wrong, and what is the rule that fixes it?

- **A.** The addresses still carry stray spaces, and once those are removed a fixed count of eight characters will land in the right place — Trimming is a real first step and it does shift every character along by however many spaces came off. A fixed count of eight would still be wrong for every address whose username is not eight characters long, which is most of them.
- **B.** The function being used is case sensitive, so the addresses containing capital letters were cut at the wrong point — One of the position-finding functions is case sensitive and its companion is not, and that distinction matters when the marker itself is a letter. The marker here is the same character in every row, so capitalisation cannot be what moved it.
- **C.** The values vary in length, so the split has to be done by looking each address up against a list of known usernames — Varying length is exactly the problem, and a lookup list is the wrong remedy for it: the list would need maintaining and would fail on the first customer nobody had entered yet. The point of a formula is to state a rule that works on rows nobody has seen.

- **D.** The marker sits in a different position in every address, so the rule has to locate the marker first and measure from it rather than count a fixed number of characters — **correct**. Neither the username nor the domain has a fixed length, so the position of the marker has to be found for each row and the length measured from it. That is why the pattern is find the marker, take everything up to one character before it, and take the remainder by subtracting the marker's position from the total length.

30. **2.5** A query against a hypothetical wholesaler's sales table returns revenue by account and shows that four accounts produced roughly two thirds of the year's revenue. Four sentences are written about it. Which one is the **finding** — what the numbers mean in the business — rather than the result itself or a recommendation?

- **A.** Four accounts produced about two thirds of revenue in the period the query covered — This is the result restated, and it is true and it is where the work starts. It names no consequence for the business, which is why on its own it is a fact nobody has to act on.
- **B.** We should build a purchase history for every account and make offers from it, judged by the share of revenue coming from accounts outside the top four at the next review — This is a recommendation, and a well-formed one: it states a concrete initiative and commits to a measure and a date. It is the step after the finding rather than the finding itself, and it only makes sense once the finding has explained why account concentration matters.
- **C.** A few buyers hold enough of our revenue to set terms at renewal, which is customers' bargaining power pressing on us — **correct**. A finding says what the result means in the business, and this one attaches the concentration of revenue to a named force: buyer power grows when a few accounts hold most of your revenue and can use that leverage at renewal. It is the link between the query result and any initiative anyone might propose.
- **D.** We should be spending more time with our largest accounts than we currently do — This is a preference. It offers no evidence, nothing anyone could be assigned to build, and no measure by which it could later be judged to have worked.

31. **2.6** A scoring function for competitive forces returns the single highest-scoring force. Two forces tie at the top and it returns the first one it happened to see. What is the practical consequence?

- **A.** None, since either force is equally worth addressing when they score the same — Equal scores do not make the responses interchangeable; relieving buyer power and relieving supplier power call for entirely different systems and budgets.
- **B.** Spending is aimed at whichever force the data happened to list first, and the report gives no sign that the evidence never separated them — **correct**. Right. The ordering of the input decides the recommendation, and nothing in the output warns the reader of that. A confident answer resting on row order is worse than an admitted tie.
- **C.** The score itself becomes unreliable and the analysis must be redone from the beginning — A tie does not make the scores wrong. The scoring is fine; the reporting of what it found is what fails.

- **D.** The tie will resolve itself once more evidence is gathered, so the output is provisional — It may well resolve later, but the report as issued presents a winner as settled, and decisions get taken from it now.

32. **2.6** A subscription business calculates that a customer costing 240 to acquire returns 30 a month, and reports a payback of eight months. Customers on that plan churn at 20 percent a month. What has the calculation missed?

- **A.** Nothing; eight months is the correct payback period for those figures — The division is right and the conclusion is wrong, which is what makes this the common error. Eight months only matters if the customer is still there in month eight.
- **B.** That the expected lifetime is about five months, so the customer leaves roughly three months before the acquisition cost is recovered — **correct.** Right. A churn rate of twenty percent a month implies an average life near five months, so the eight-month payback never arrives and each acquisition loses money.
- **C.** That the monthly return should have been discounted for the time value of money — Discounting would refine the figure slightly and would not change the verdict. The horizon, not the discount rate, is what decides this case.
- **D.** That acquisition cost should be spread across the whole customer base rather than charged to one customer — Spreading the cost differently changes the presentation, not the fact that this customer leaves before repaying what winning them cost.

33. **2.6** An analyst needs total revenue per plan and writes a query that filters with WHERE on the summed revenue. The database rejects it. Why?

- **A.** Sums cannot be filtered at all; they can only be sorted — Sums are filtered routinely; the question is only where in the statement that filter is allowed to sit.
- **B.** WHERE is evaluated before rows are grouped, so no total exists yet for it to test; a condition on an aggregate belongs in HAVING — **correct.** Right. WHERE decides which rows enter the grouping and has finished before any total is computed. HAVING runs afterwards, which is why the condition belongs there.
- **C.** The query is missing an index on the column being summed — An index changes how quickly a query runs, never whether it is a legal statement.
- **D.** SUM may only appear in the SELECT clause and nowhere else in a query — Aggregates appear in HAVING and ORDER BY as well, which is precisely the point this question turns on.

34. **2.6** A customer's name is stored once in a customer table and joined in wherever it is needed, rather than copied onto each of their subscriptions. Which reason best justifies that design?

- **A.** Joining is faster than reading a column that is already on the row — Joining generally costs more at query time than reading a local column. Speed is the argument against this design, not for it.

- **B.** The name is a fact about the customer, so keeping one copy means a correction is made once and no two rows can disagree — **correct.** Right. One fact in one place. Copying the name onto every subscription creates as many chances to disagree as there are rows, and reconciling them later is far more expensive than the join.
- **C.** It reduces the total storage the database consumes to a meaningful degree — The saving is real but trivial at this scale, and nobody would restructure a schema for it. Correctness is the reason that holds.
- **D.** It is required before a GROUP BY can be used on any of the joined tables — Grouping works perfectly well on duplicated columns. Nothing about GROUP BY requires the data to be split across tables.

What to do next. Take one organization you already deal with and run the whole chain on it once: name the force that squeezes it hardest, find the value chain activity where that force lands, say what system would relieve it, and state the number that should move. Then ask what evidence would settle whether you are right, and how you would pull it. That last question is the one that separates an opinion from an analysis.